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EC-Council CND (Certified Network Defender)

Technical Network Security – Segmentation, Protocol Hardening, PKI, VPN & IDS

Executive Summary

Module 3 of the Certified Network Defender course implements technical network security controls on the existing Active Directory environment (crack.local / corp.local) from Module 2.

It introduces three new virtual machines (VPN01, IDS01, ATTACK01) and delivers the following core controls:

- **Network segmentation with Windows Firewall rules on DC01**
- **Disable of insecure protocols (SMBv1, LLMNR, NetBIOS, mDNS)**
- **TLS 1.2 enforcement and weak-cipher blocking**
- **Enterprise Certificate Authority (AD CS) with autoenrollment**
- **SoftEther L2TP/IPsec VPN server on VPN01**
- **Suricata IDS with custom detection rules on IDS01**
- **Sysmon enhanced logging on DC01 and ADMIN01**
- **Windows Admin Center for secure remote management**

All controls are validated through live attack simulations from ATTACK01 (reconnaissance, EternalBlue checks, brute-force, LLMNR poisoning, VPN access, and TLS verification).

The lab demonstrates that effective technical security requires consistent application across every endpoint and works in tandem with the administrative foundations built in Module 2.

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1. INTRODUCTION

This lab implements technical network security controls as part of Module 3 of the Certified Network Defender course. It builds directly upon Module 2 and requires Module 2 to be fully completed before proceeding. The Active Directory domain, users, group policies and client machines are configured in Module 2 form the foundation that Module 3 builds upon.

Prerequisites

Module 2 must be fully completed and working before starting this lab. This includes the Active Directory domain crack.local, all virtual machines, user accounts, group policies and monitoring tools.

Lab Environment

All machines operate on the internal network 192.168.100.0/24 under the domain crack.local. Module 3 adds three new virtual machines to the existing Module 2 setup.

VM	**IP Address**	**Operating System**	**Role**
DC01	192.168.100.100	Windows Server 2019	Domain Controller — from Module 2
ADMIN01	192.168.100.110	Windows 10 Pro	Management Workstation — from Module 2
CL01	192.168.100.120	Windows 10 Pro	Client Workstation — from Module 2
VPN01	192.168.100.130	Windows Server 2019	VPN Server — new in Module 3
IDS01	192.168.100.140	Kali Linux	Intrusion Detection — new in Module 3
ATTACK01	192.168.100.150	Kali Linux	Attack Simulation — new in Module 3

Security Objectives

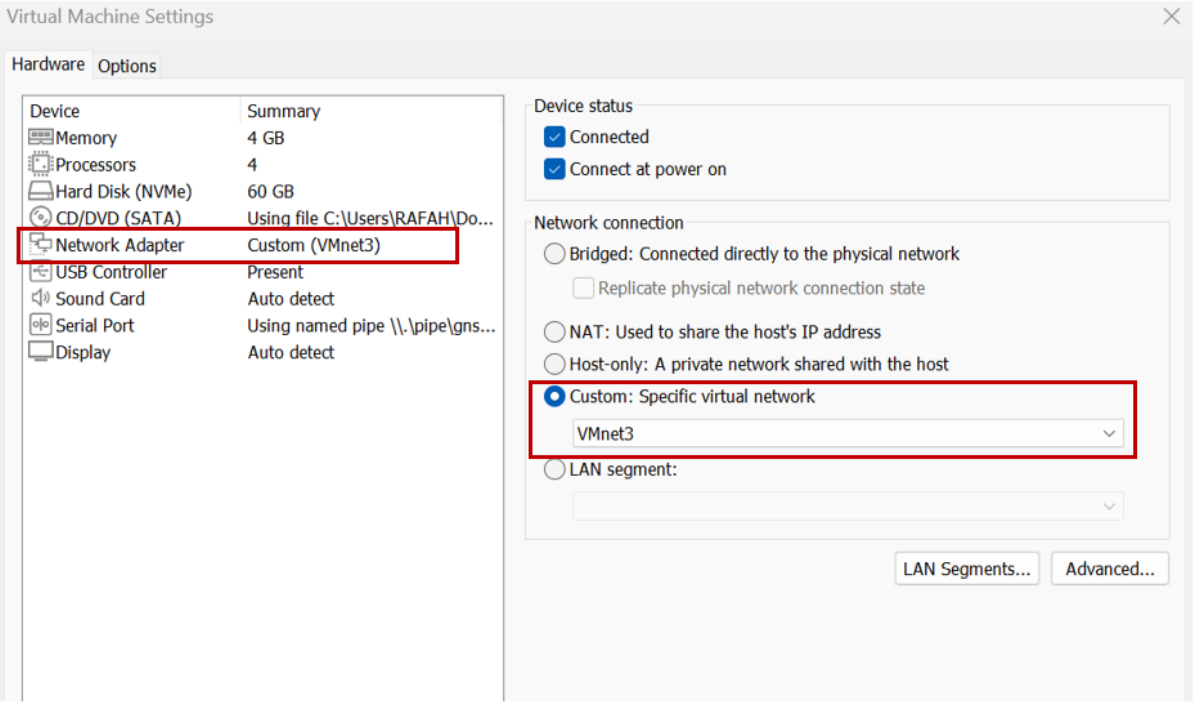
Module 3 implements the following technical security controls:

- Network segmentation using Windows Firewall rules
- Disabling insecure protocols including SMBv1, LLMNR, NetBIOS, mDNS, Telnet and FTP
- TLS hardening by enforcing TLS 1.2 and blocking weak ciphers
- Deploying an Enterprise Certificate Authority on DC01
- Configuring a SoftEther L2TP/IPSec VPN Server on VPN01
- Installing Suricata IDS with custom detection rules on IDS01
- Installing Sysmon for enhanced logging on DC01 and ADMIN01
- Setting up Windows Admin Center for secure remote management
- Validating all controls through live attack simulation from ATTACK01

2. Create VPN01 Virtual Machine

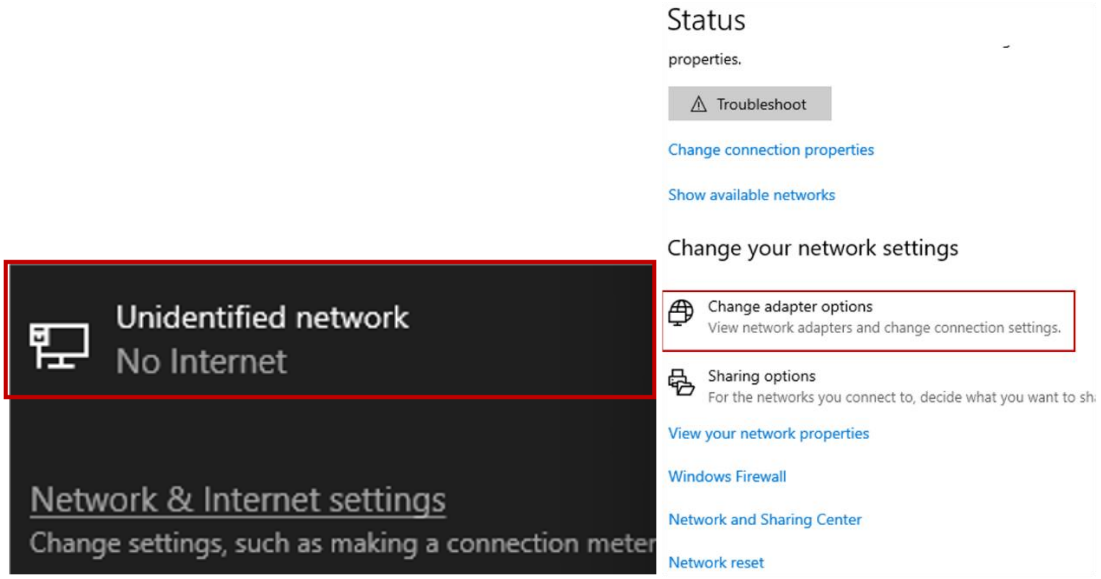
Create a new VPN Windows Server Virtual Machine. Set Network Adapter to host only. Use your existing adapter that is being used in the DC01 Virtual Machine.

- Go to the VPN virtual machine settings and set the Network Adapter

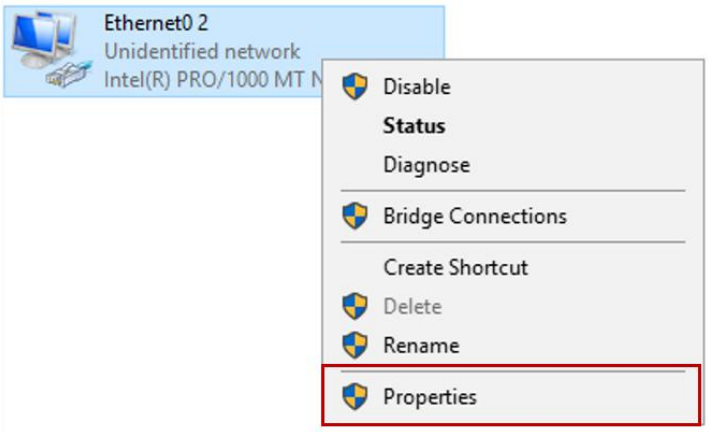


Click OK and Run the Virtual Machine

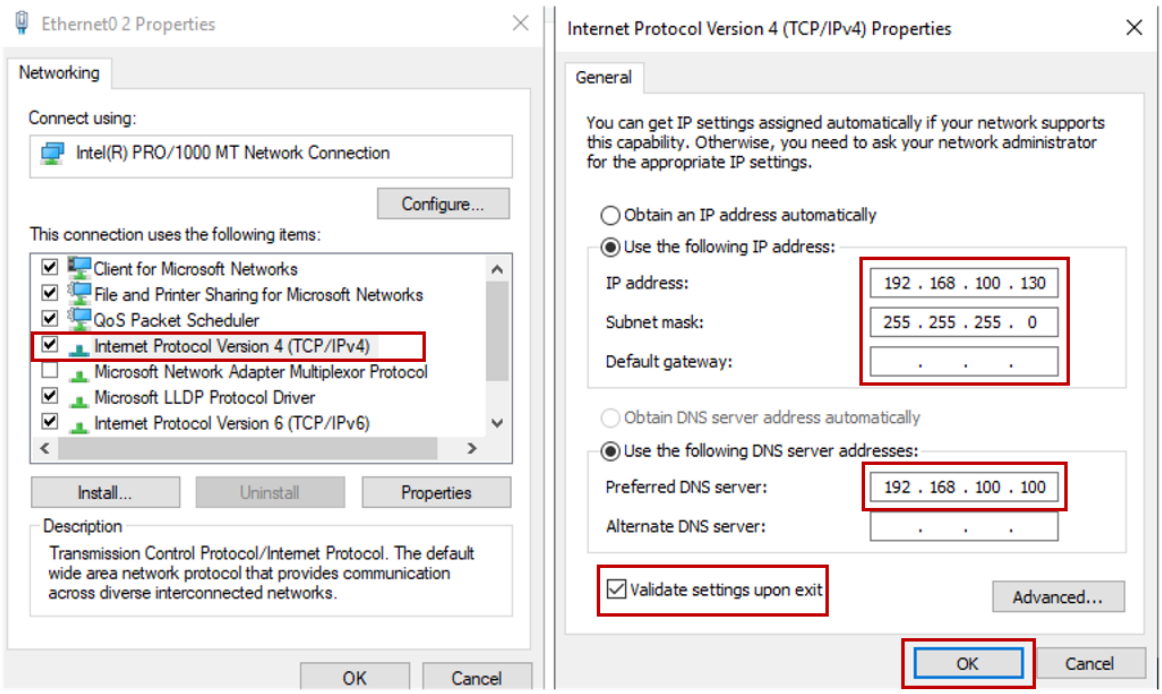
- Go to adapter settings and click on Change Adapter options



- Double click your Adapter and go to properties



- Click on IPv4 settings
- Set static ip address and set DNS to DC01 ip address

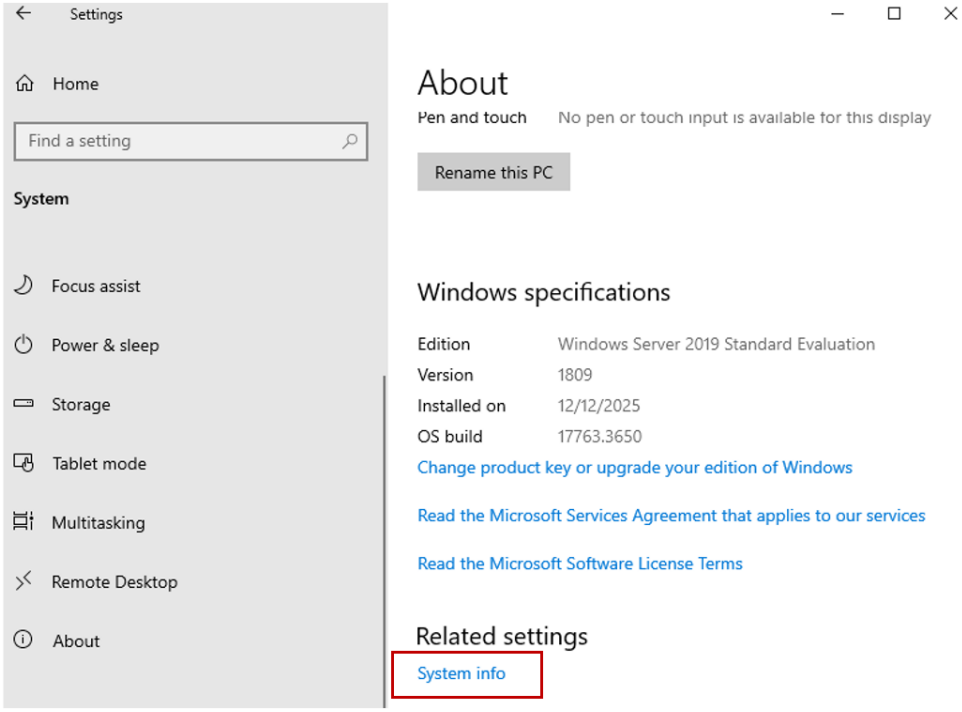


Click OK for the changes to take place.

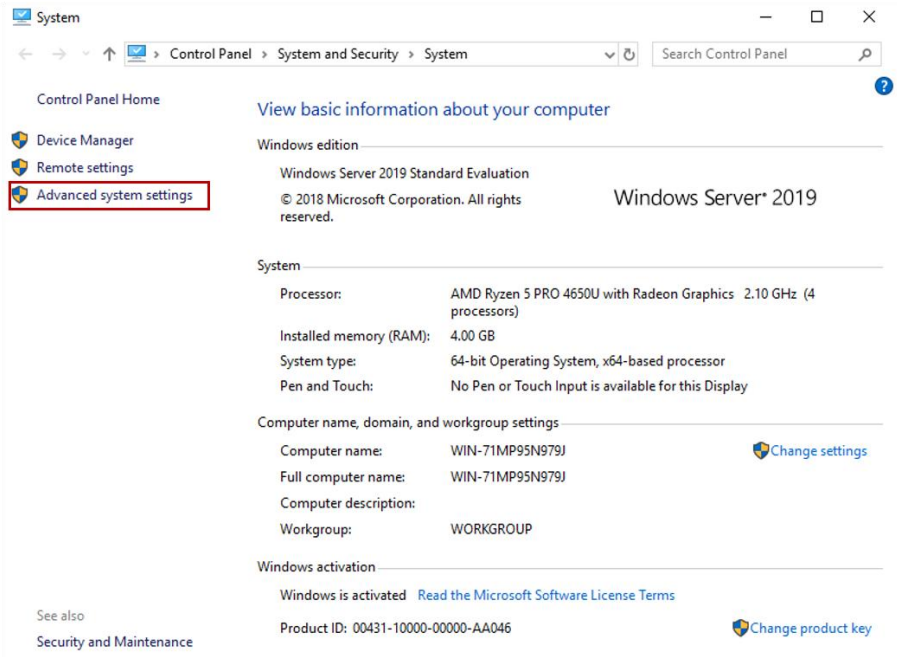
3. Join VPN01 to domain

Go to System settings and click on About.

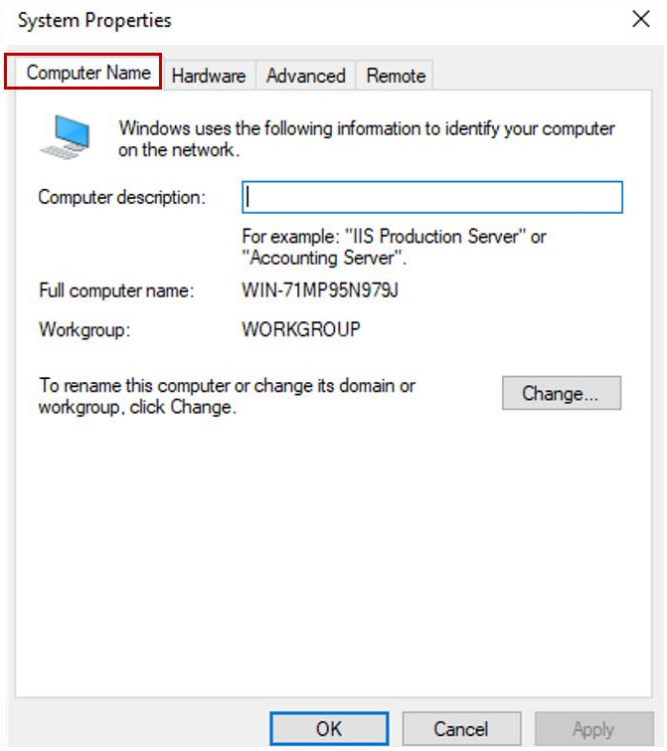
- Click on System info



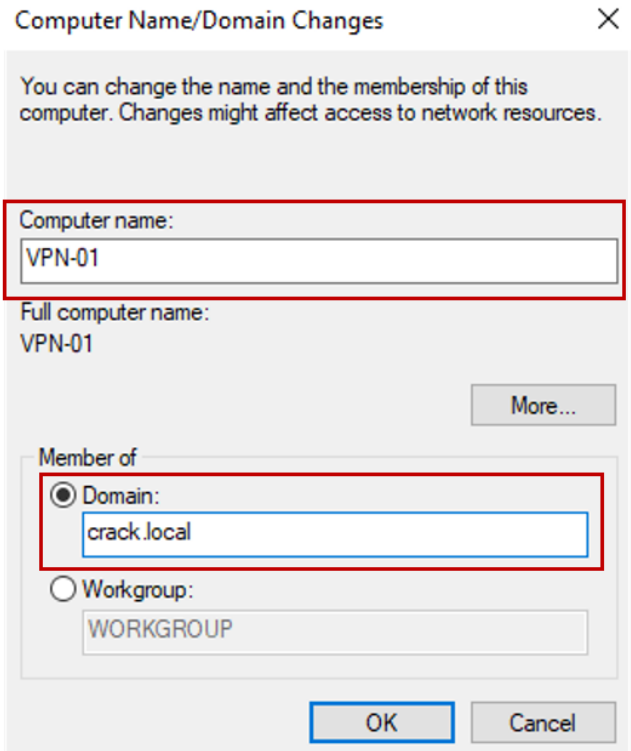
- Select Advanced System Settings



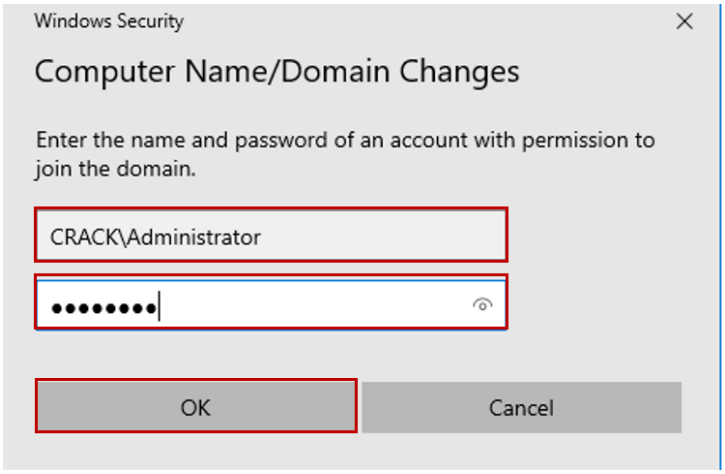
- Go to Computer Name and click on Change



- Enter the Computer name and join the domain: **crack.local**



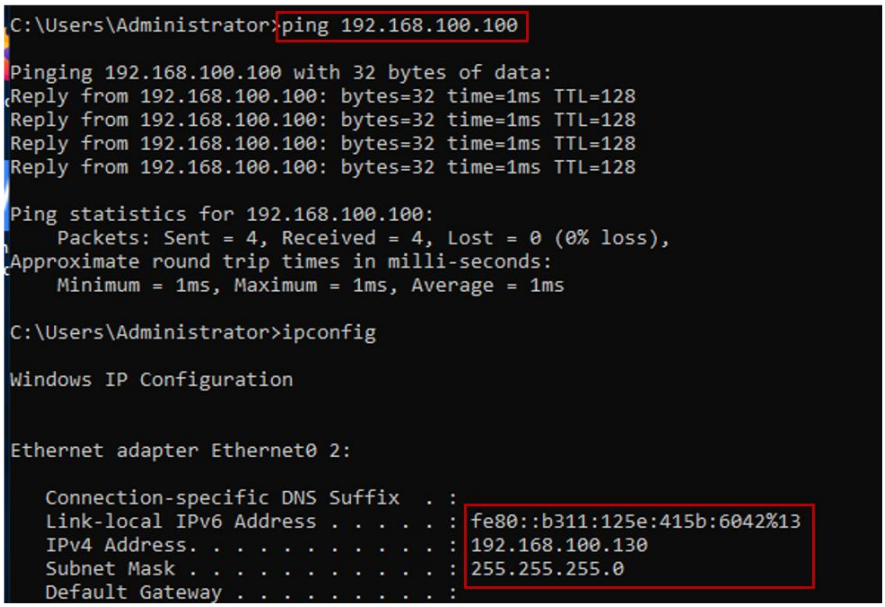
- Enter you crack domain credentials



Restart your windows server for changes to take effect.

Login as:

- CRACK\Administrator
- Checking Connection & IP

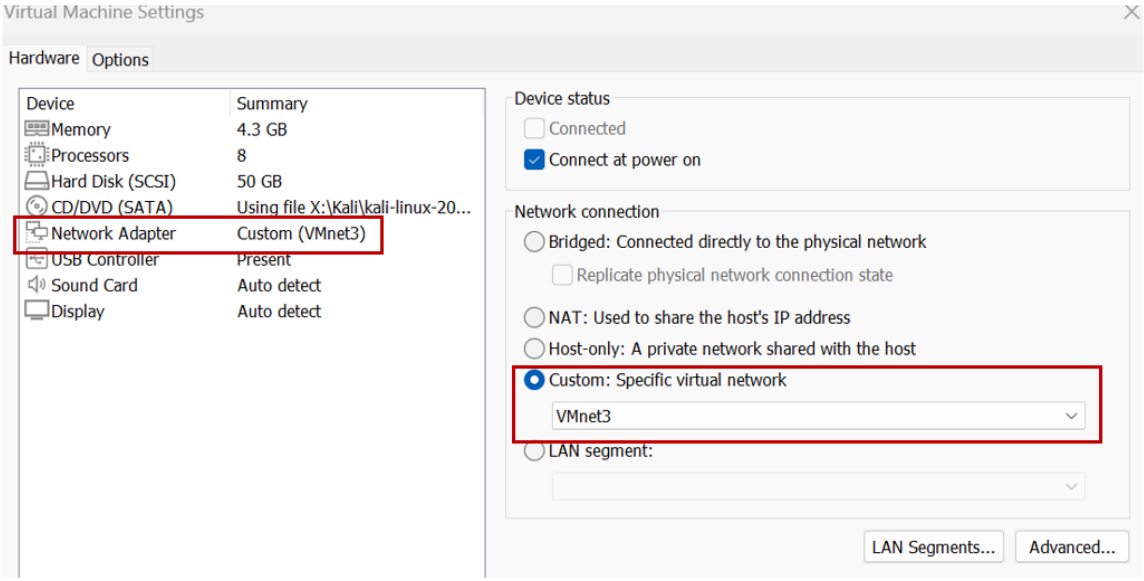


Now domain join is complete

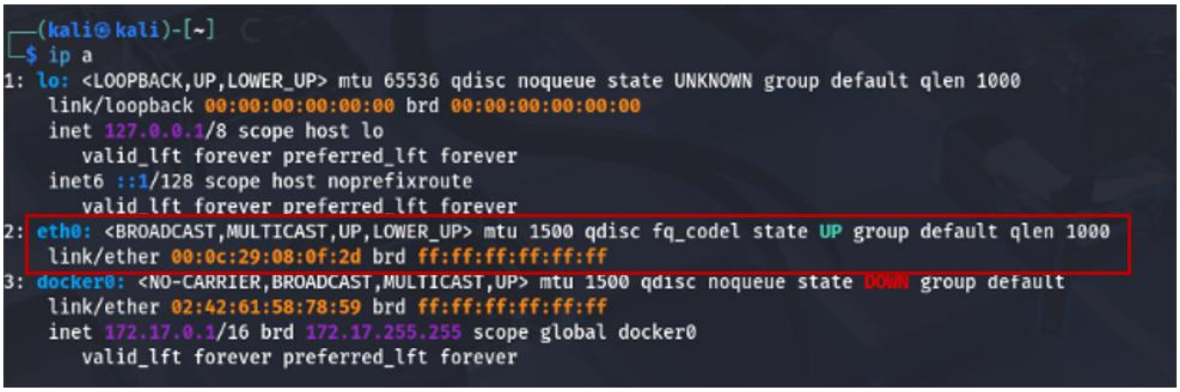
4. Create IDS01 Virtual Machine

Create a new kali Linux Virtual Machine for intrusion detection.

Network: **same VMnet as your other VMs (VMnet3)**



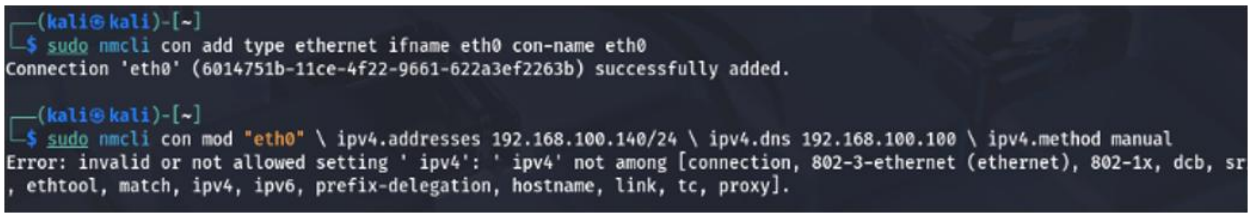
Static IP Configuration



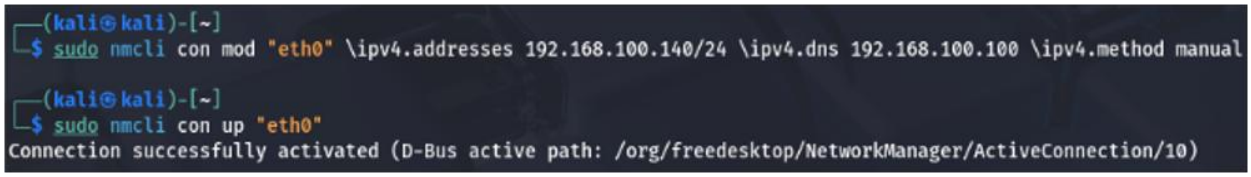
Use nmcli:

Enter these commands to change the IDS ip address

```
sudo nmcli con mod "eth0" \
  ipv4.addresses 192.168.100.140/24 \
  ipv4.dns Enter your DC01 Ip Address \
  ipv4.method manual
sudo nmcli con up "eth0"
```




```
sudo nmcli con up "eth0"
```

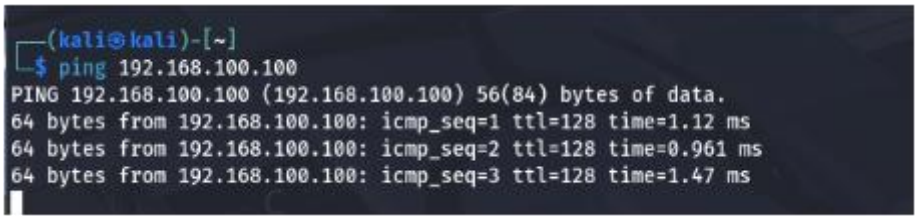



```
ip a
```

ping Enter your DC01 Ip Address Ping the DC01

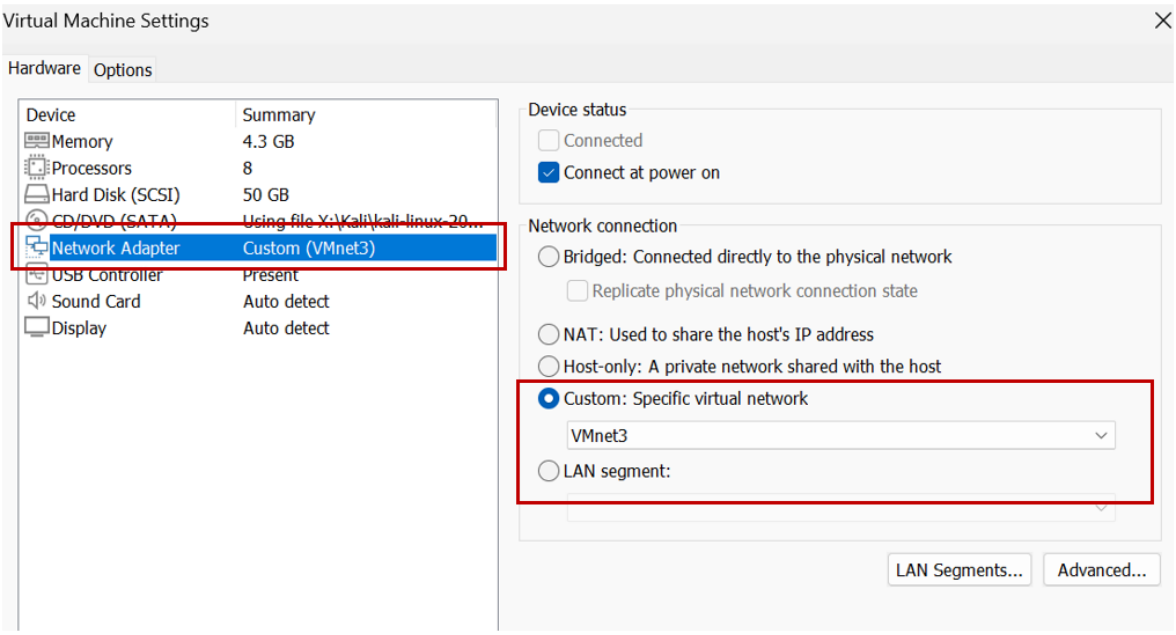


Ping Successful



5. Create ATTACK01 Virtual Machine (Kali Linux)

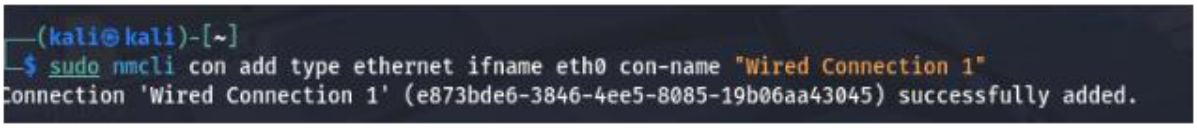
Network Adapter 1: **same VMnet as your lab (VMnet3)**

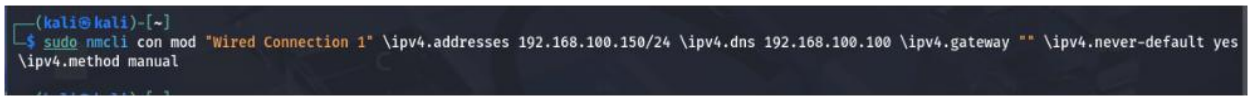


Set Static IP on ATTACK01:

Enter these commands to change the IDS ip address

```
sudo nmcli con mod "Wired connection 1" \
  ipv4.addresses 192.168.10.150/24 \
  ipv4.dns Enter your DC01 Ip Address \
  ipv4.gateway "" \
  ipv4.never-default yes \
  ipv4.method manual
```

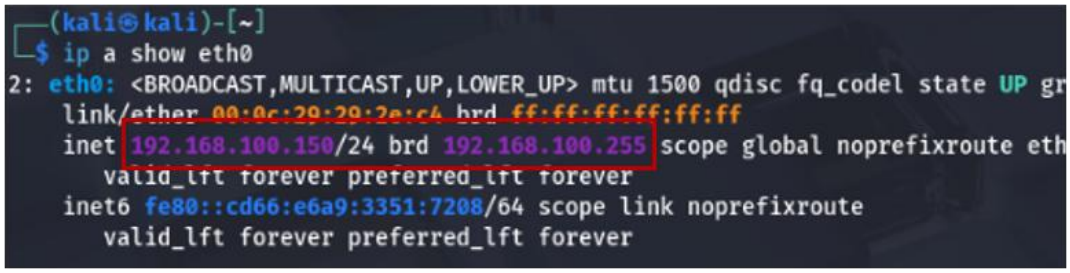





```
sudo nmcli con up "Wired connection 1"
```




```
ip a
```



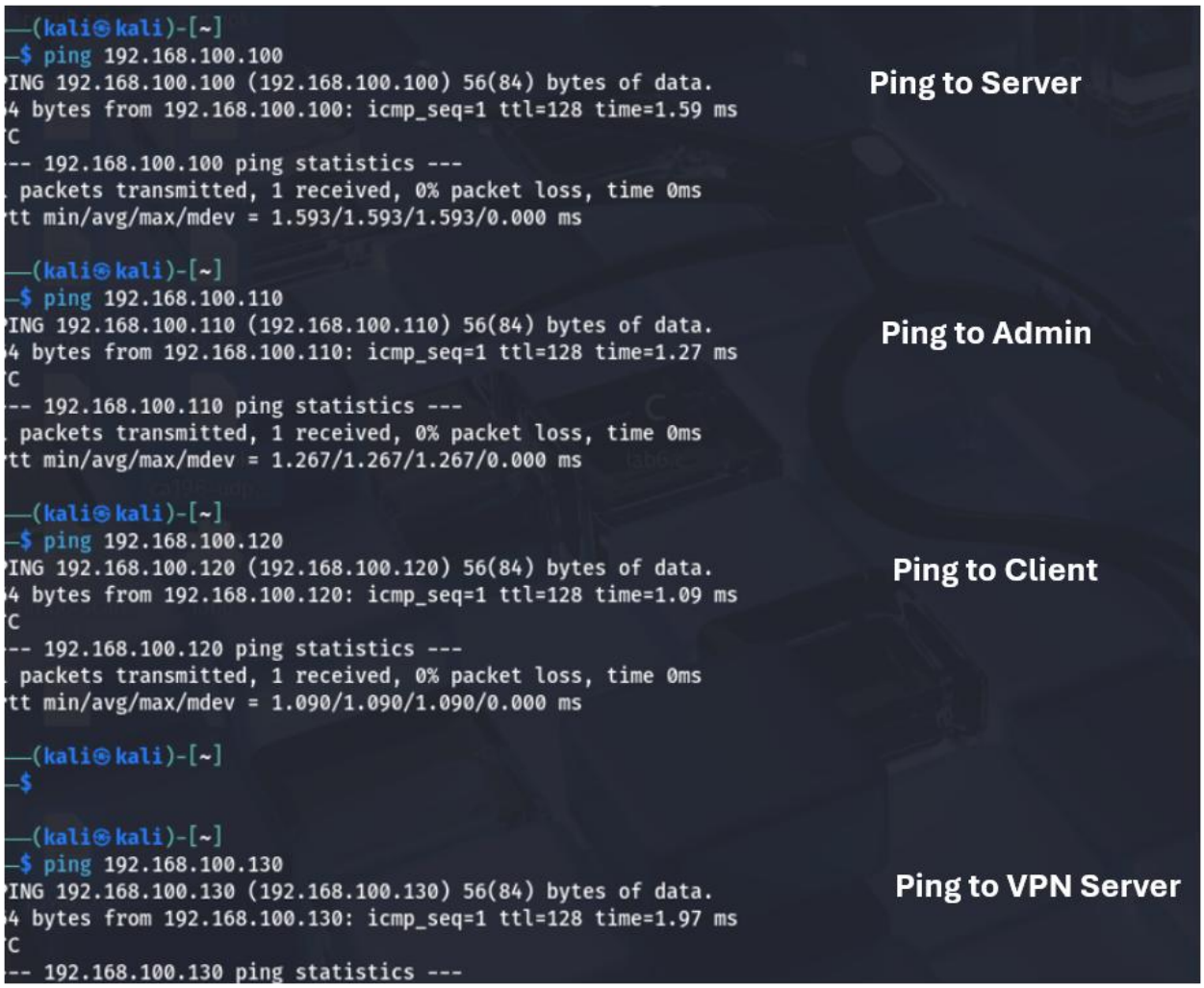

```
sudo systemctl restart networking
```

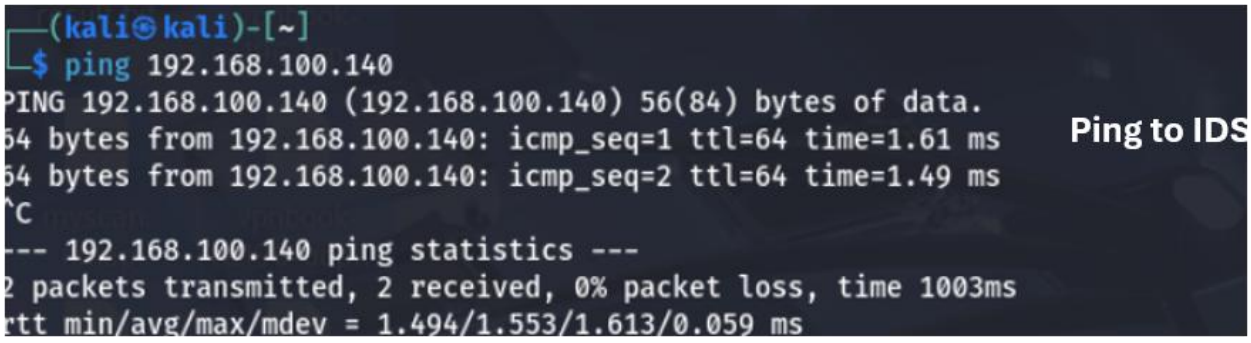


Lesson: Always run sudo nmcli con show first on each machine — never assume the connection name is the same across different VMs.

Verify All VMs Can See Each Other

Make sure your windows firewall should be off while ping from 1 network to another



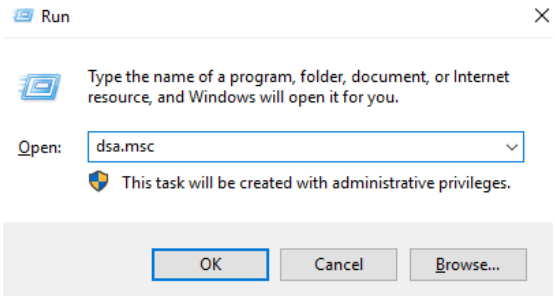


6. FIX YOUR EXISTING AD (ADD MISSING ITEMS)

Your Module 2 has users but is missing some OUs and groups needed for Module 3. Do this on **DC01**.

Open Active Directory Users and Computers:

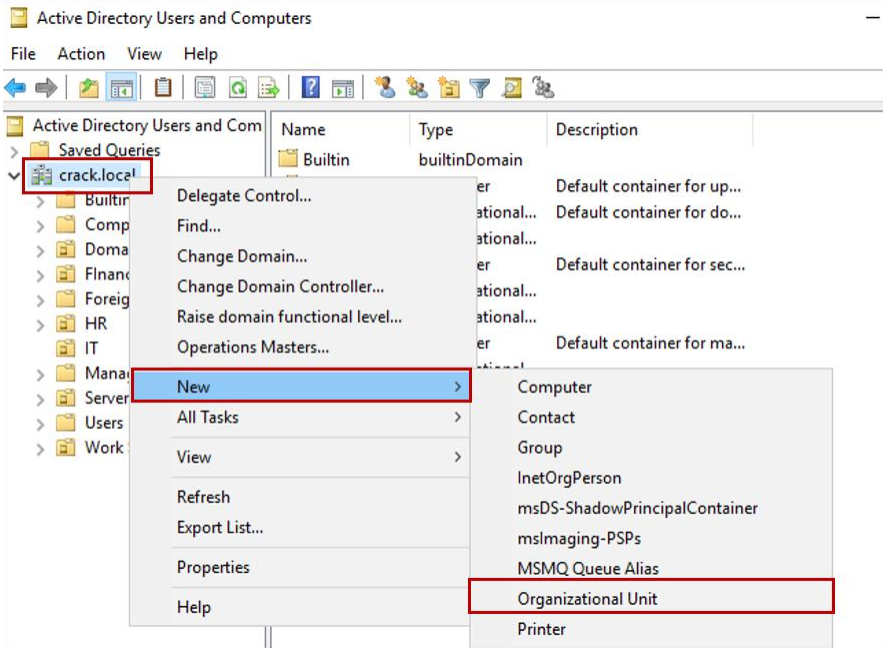
- Win+R → dsa.msc



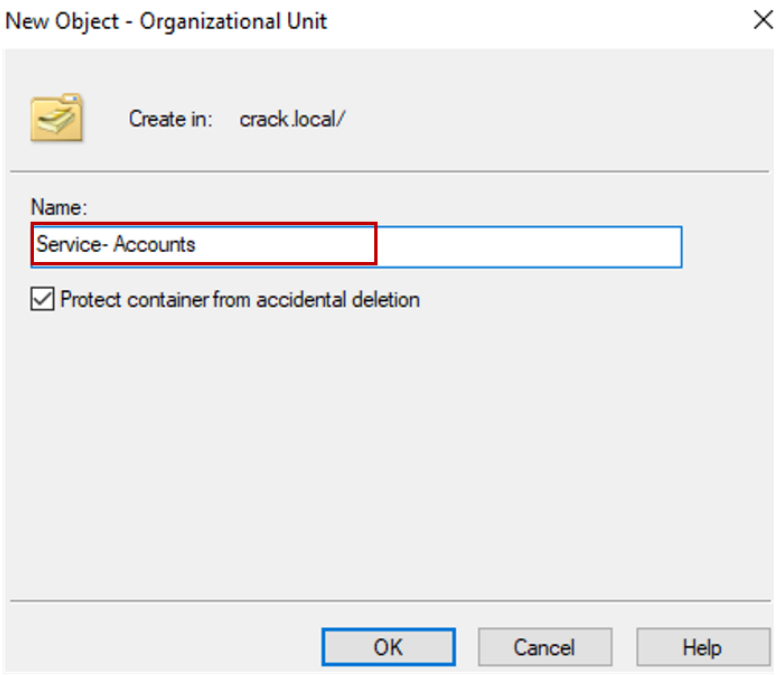
Add Missing OUs:

Right-click corp.local → New → Organizational Unit

- Create: ServiceAccounts (you already have IT, HR, Finance, Workstations, Servers)

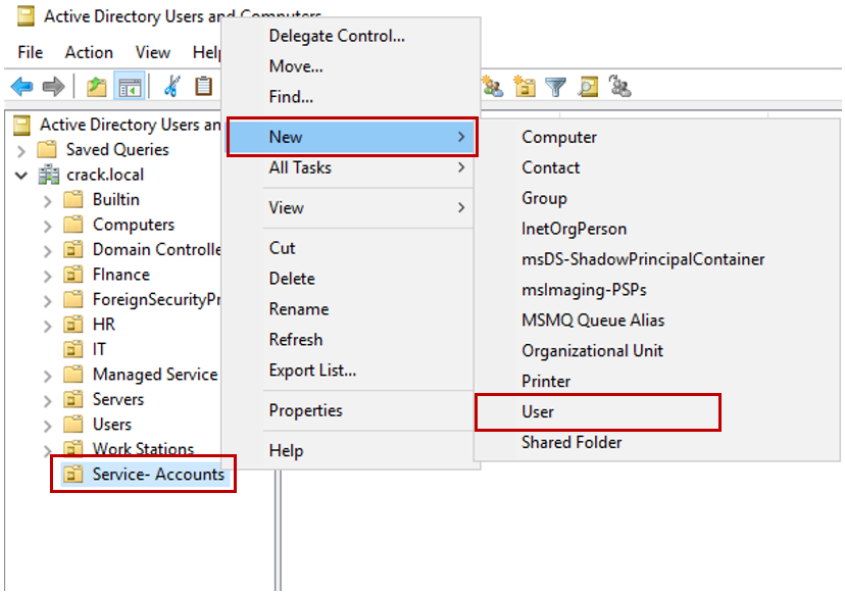


Enter the Name of the Organizational unit and click OK

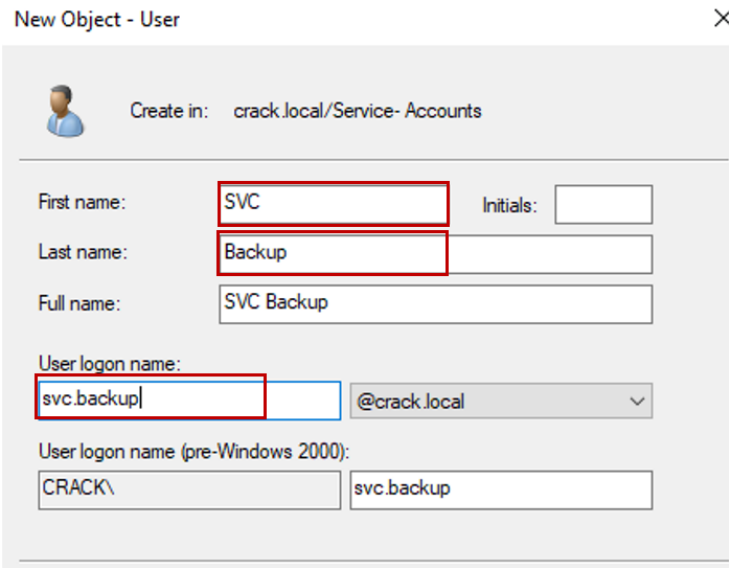


Create a Service Account:

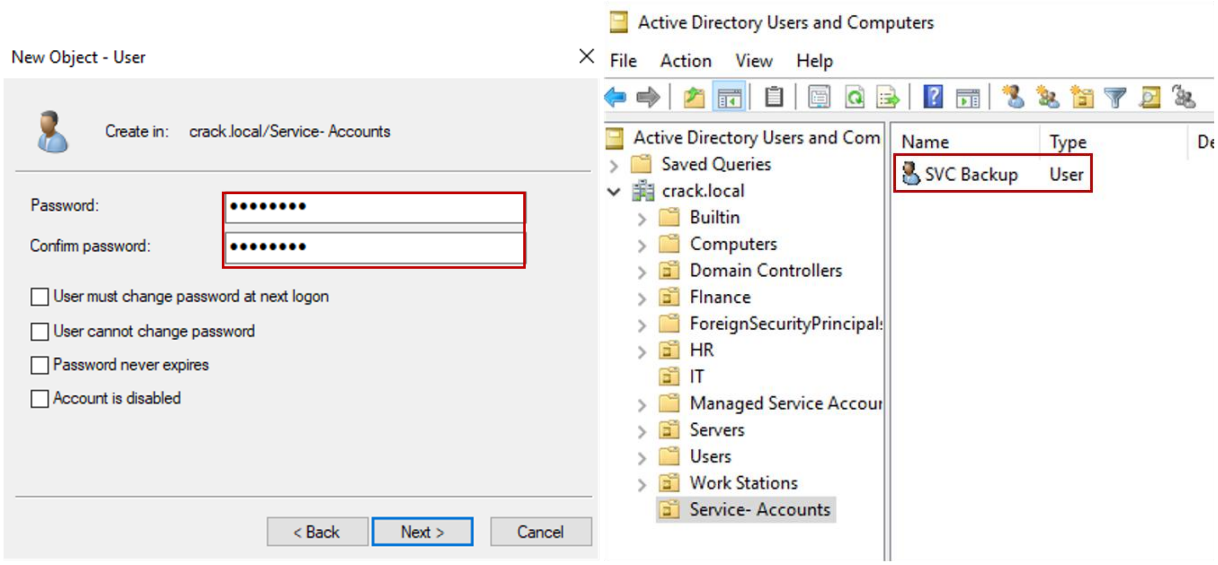
1. Click ServiceAccounts OU → Right-click → New → User



2. First name: SVC Last name: Backup. User logon name: svc.backup

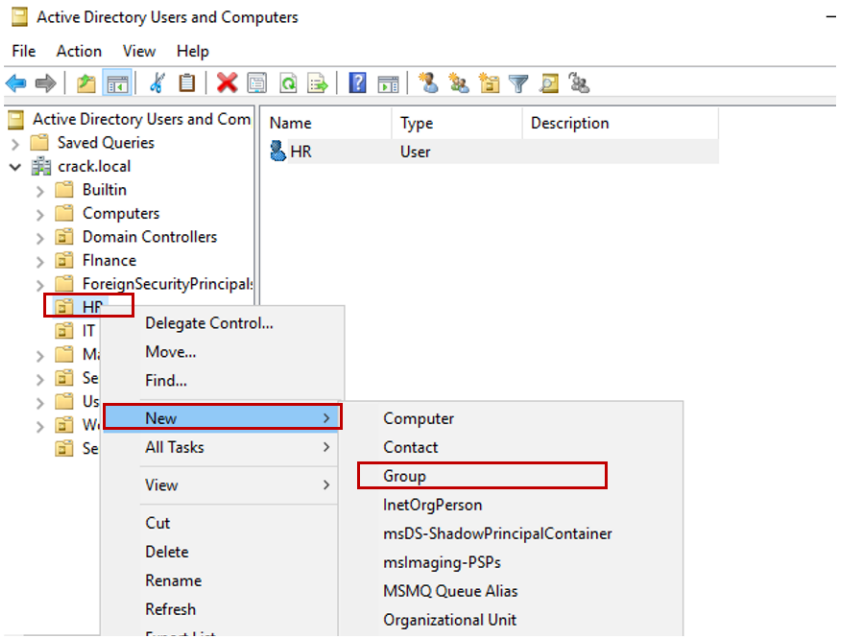


3. Password: Svcback123 → Uncheck "must change at next logon" → Finish

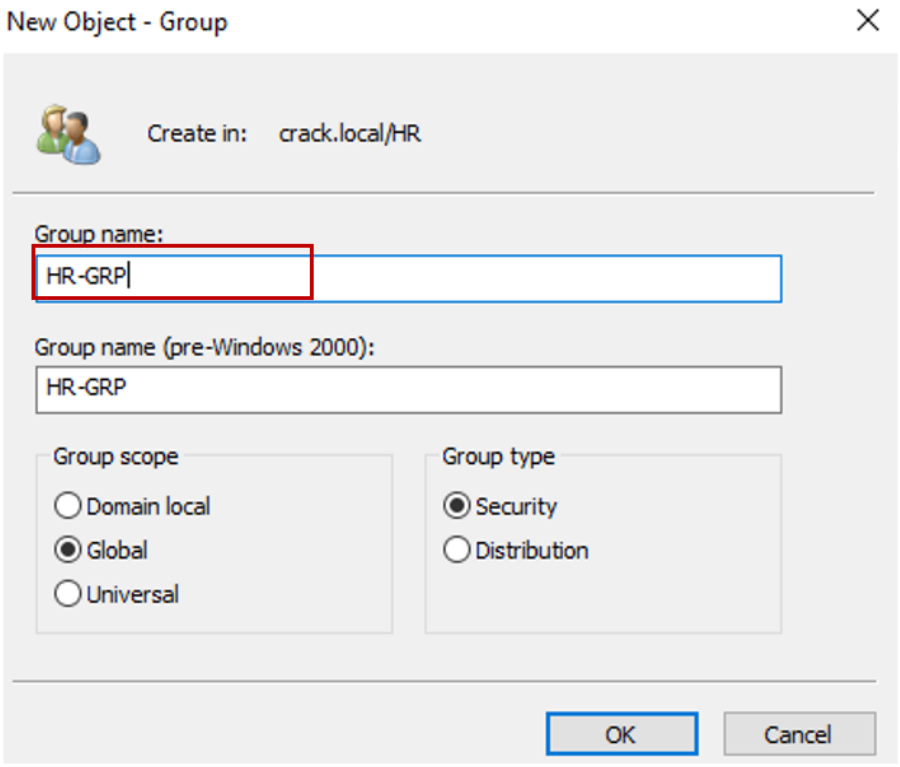


Create Security Groups:

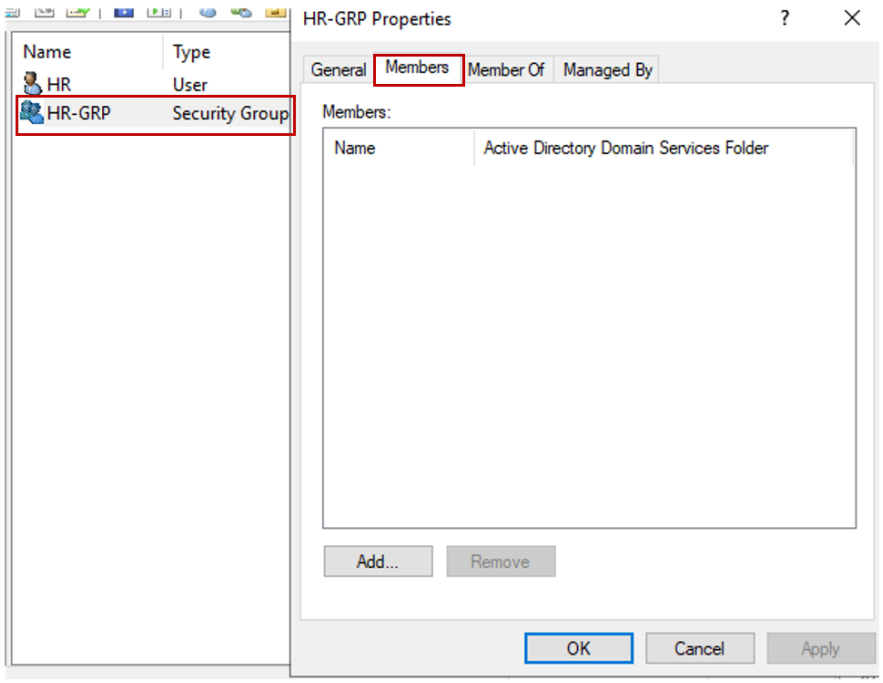
- Right-click **HR OU** → New → Group

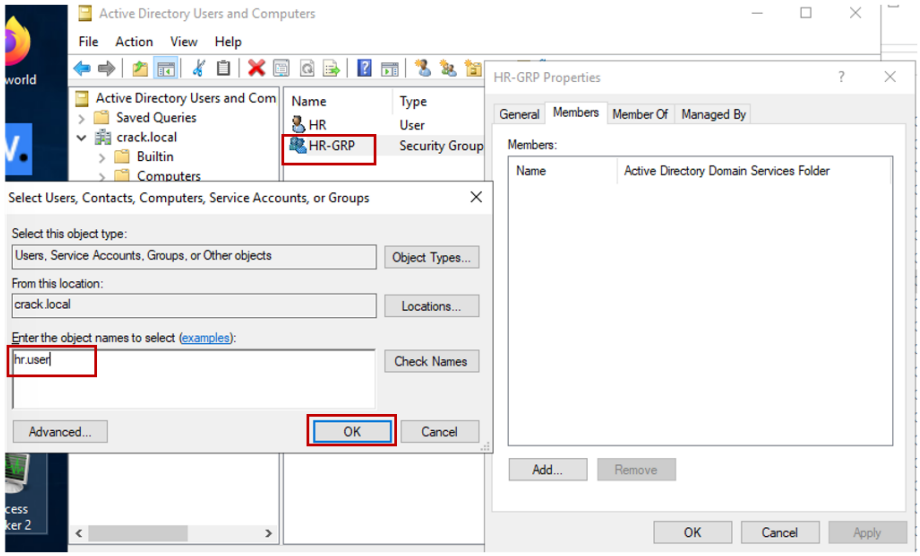


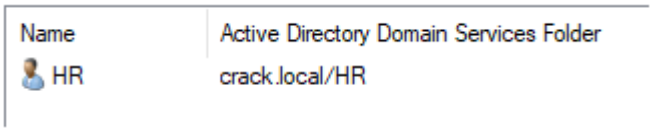
- Name: HR_GRP → Global → Security → OK



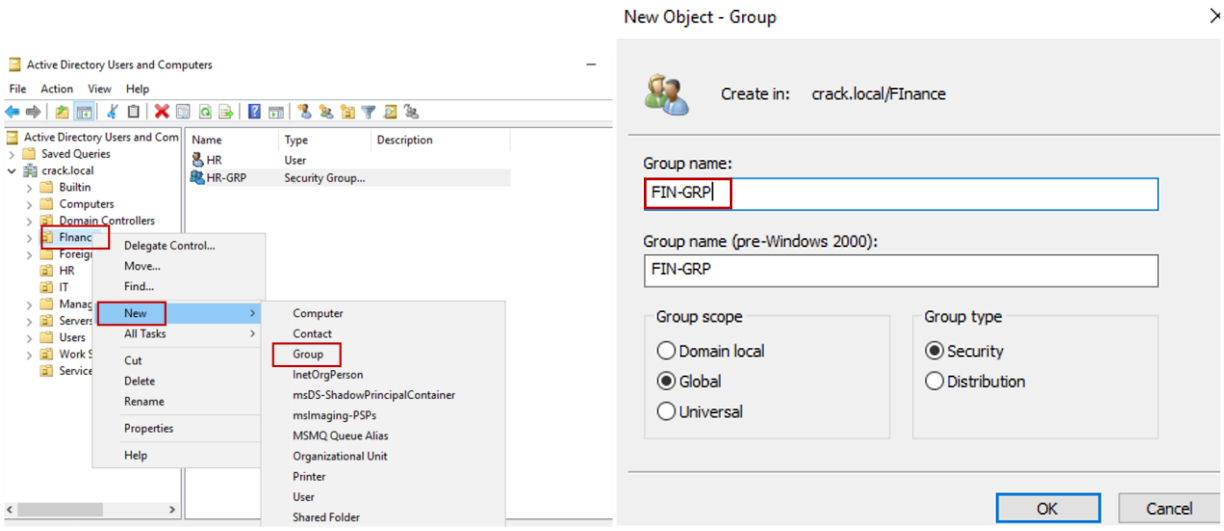
- Double-click GRP_HR → Members tab → Add → type hr.user → OK

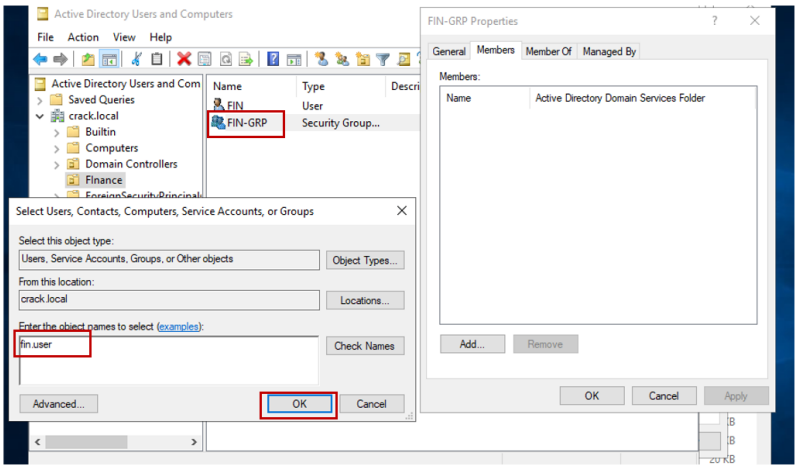


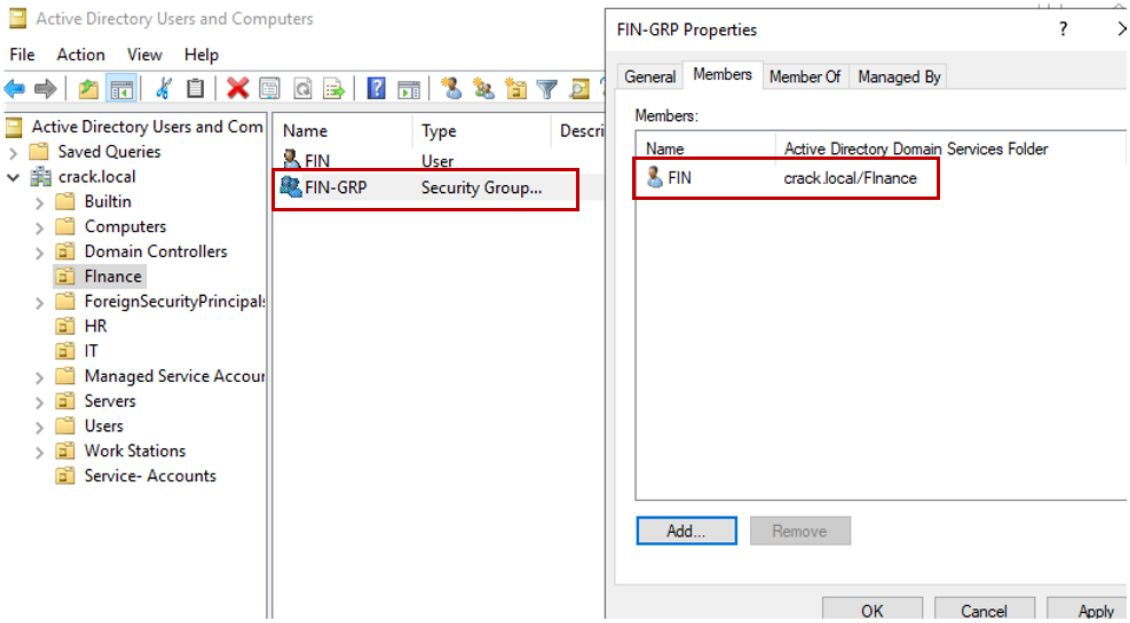




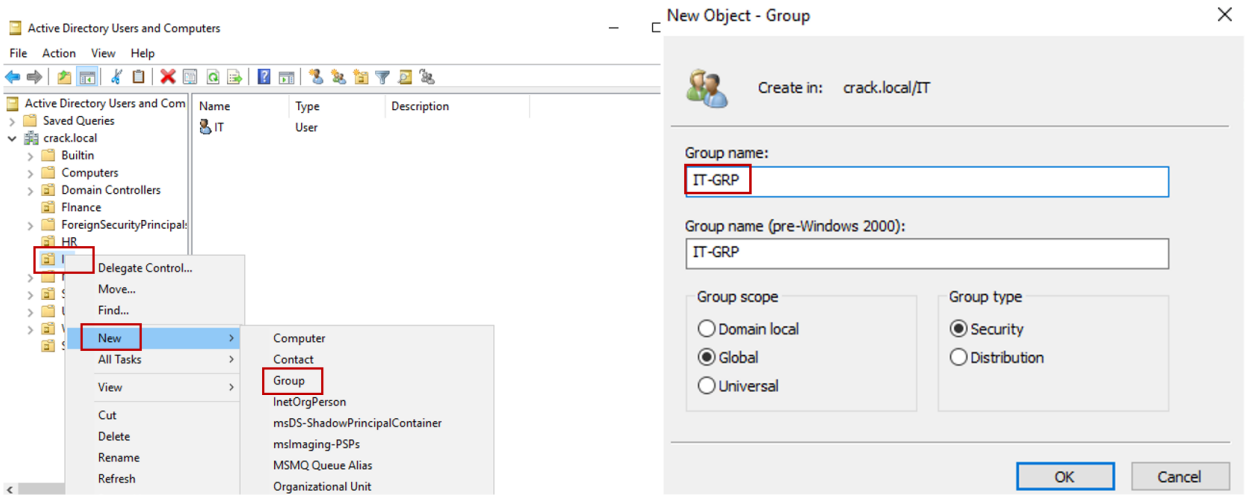
- Right-click **Finance OU** → New → Group → FIN_GRP→ add finance.user

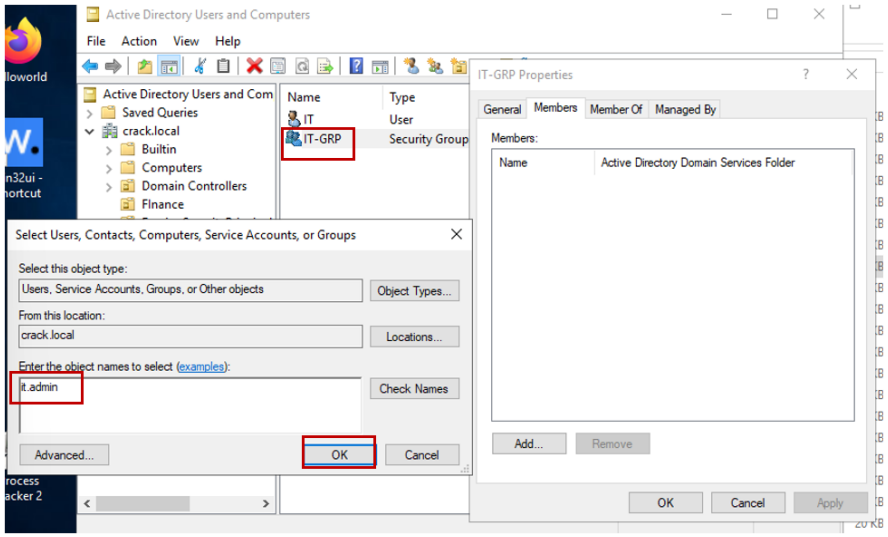


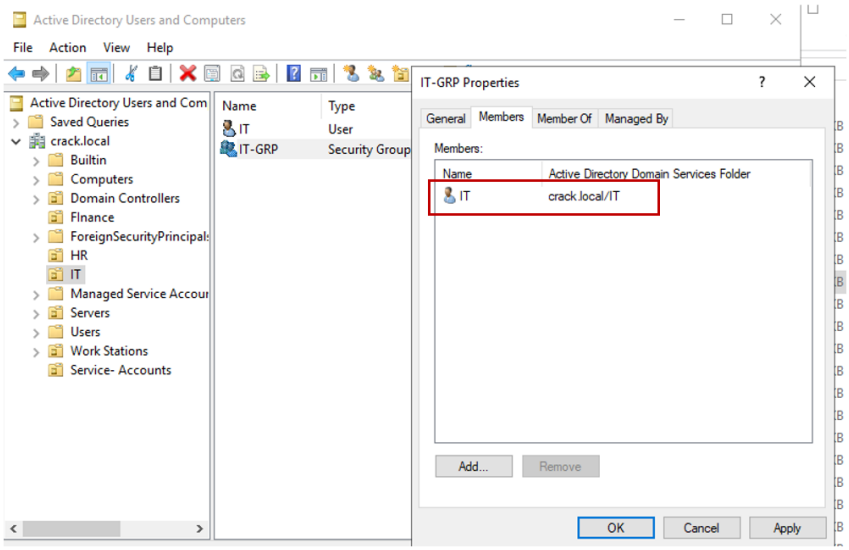




- Right-click **IT OU** → New → Group → IT_GRP→ add your IT admin user





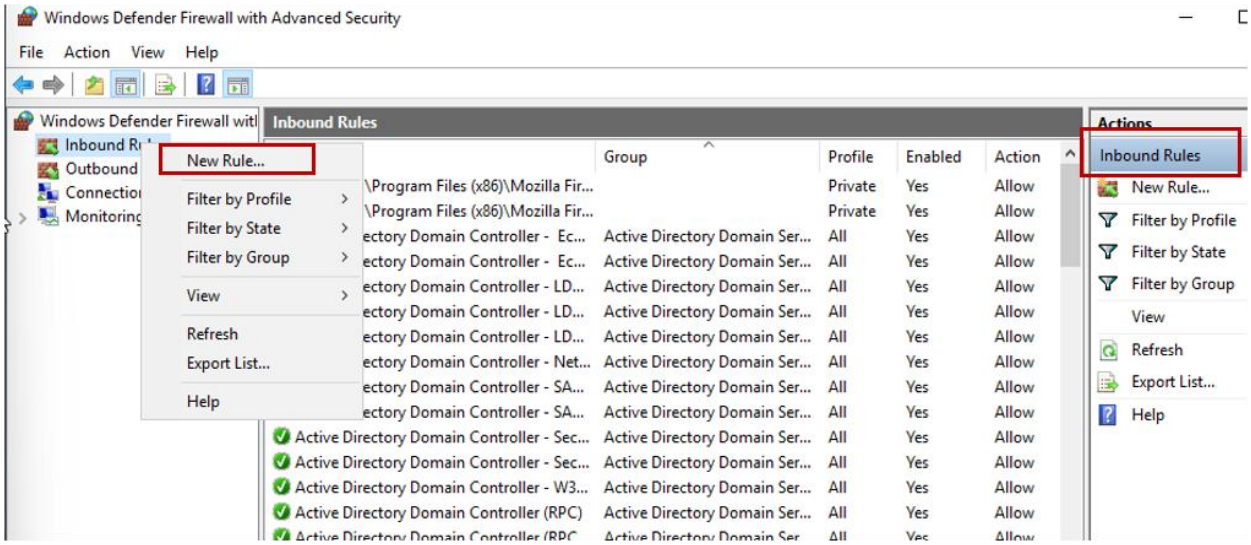


7. NETWORK SEGMENTATION (FIREWALL RULES ON DC01)

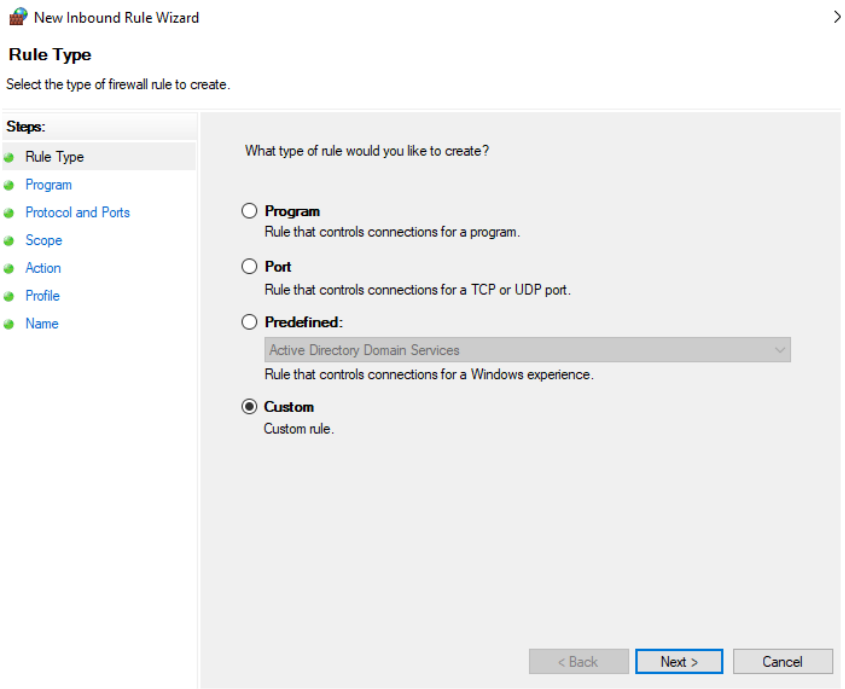
Press Win+R → type wf.msc → Enter on **DC01**

7.1 Block ICMP from CLIENT (CL01):

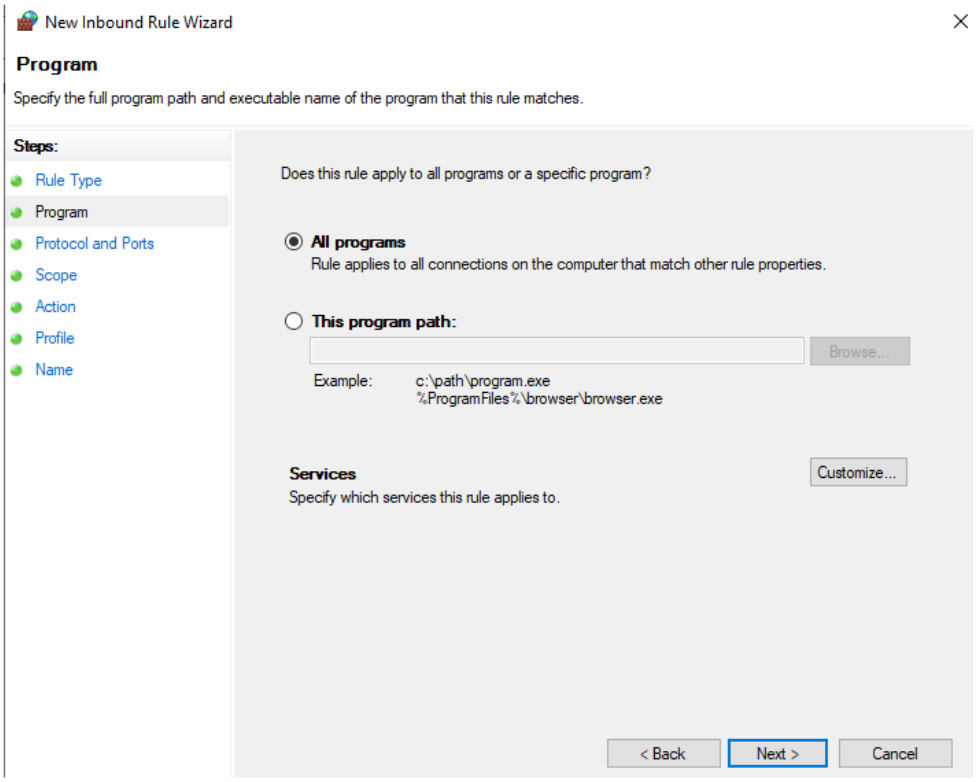
1. Right-click **Inbound Rules** → New Rule



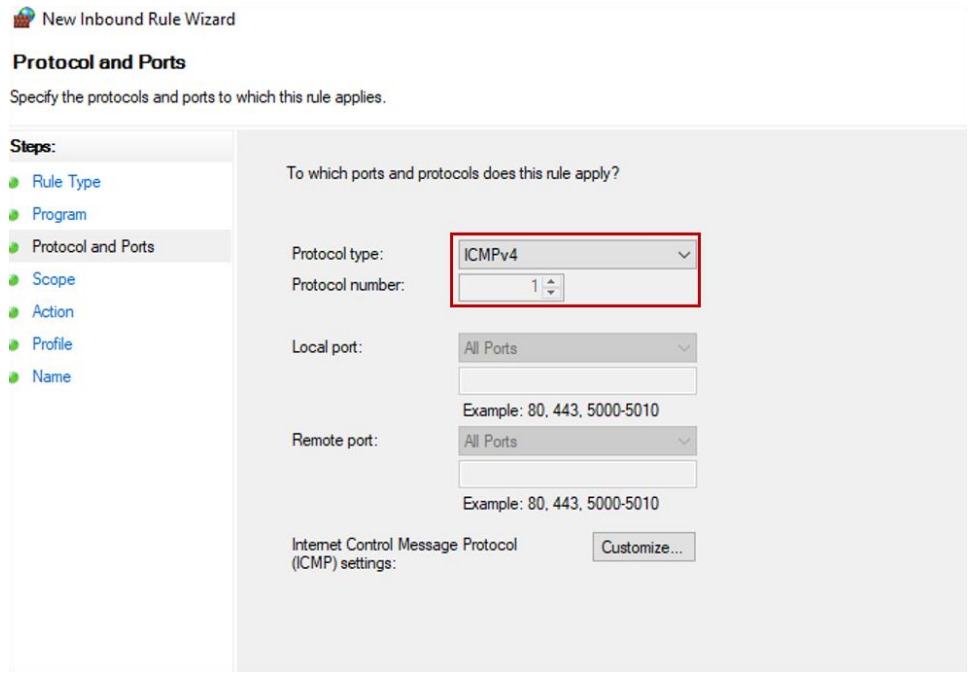
2. Rule Type: **Custom** → Next



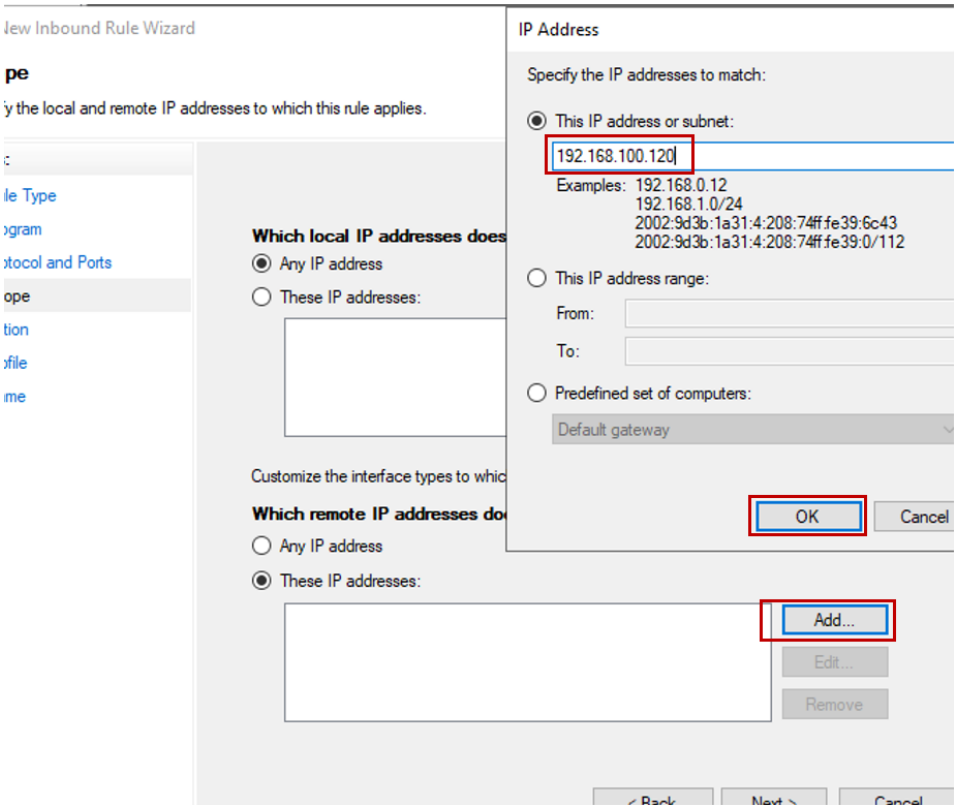
3. All Programs → Next



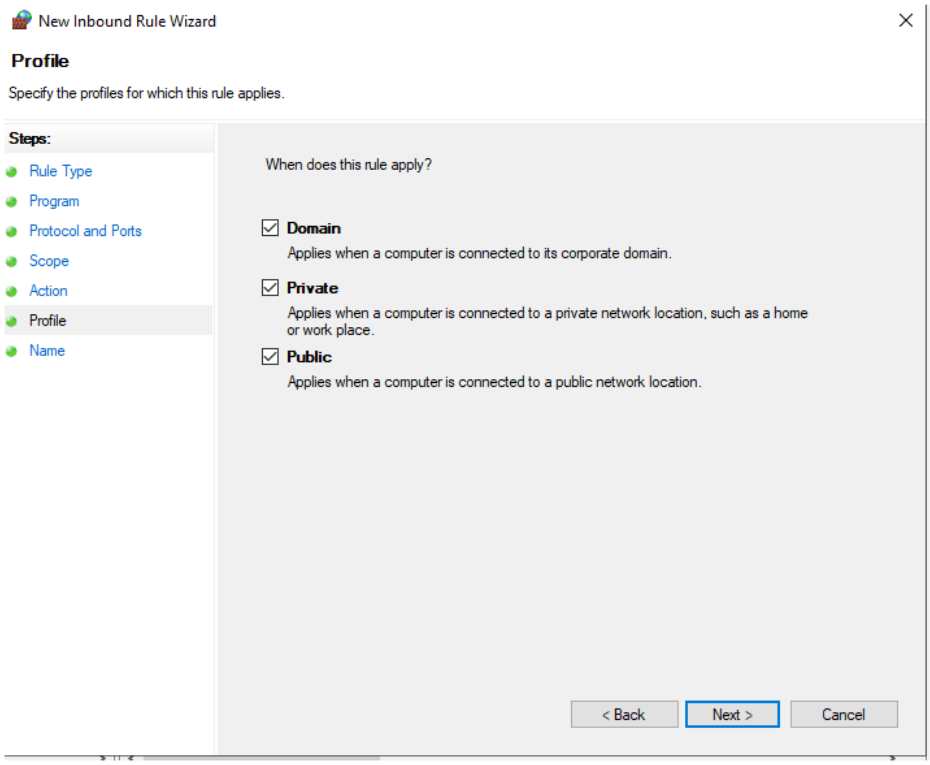
4. Protocol: **ICMPv4** → Next



5. Remote IP address: **These IP addresses** → Add → type 192.168.100.120 → Next

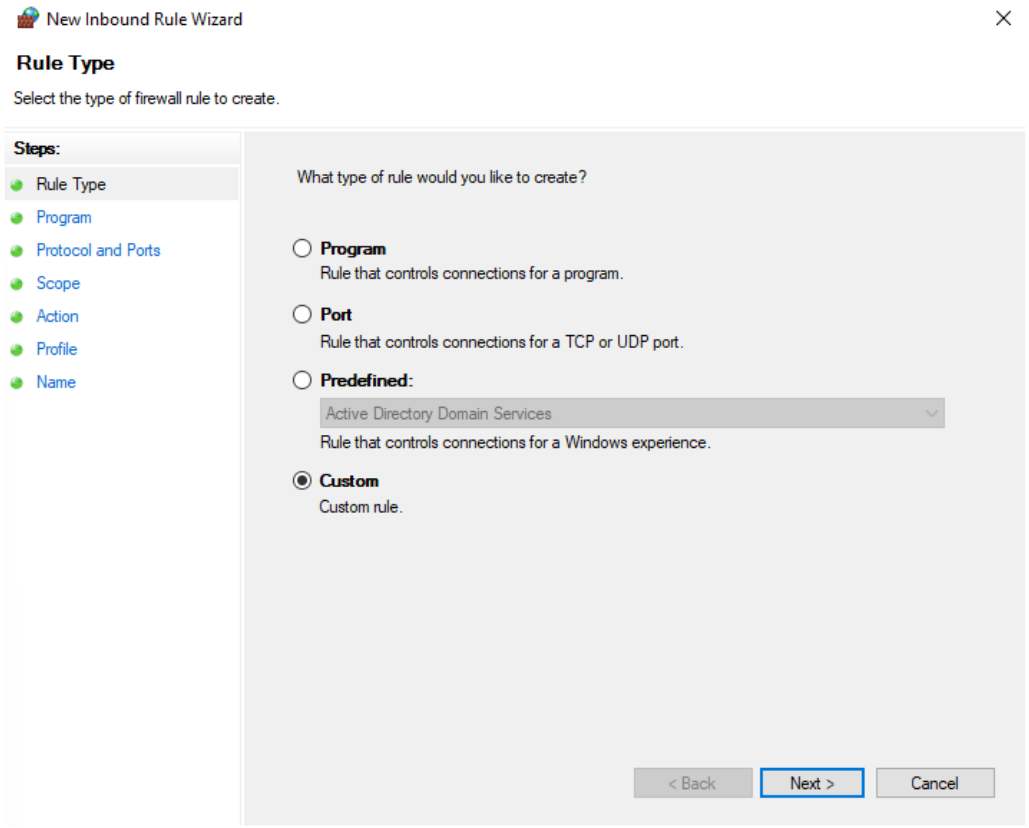


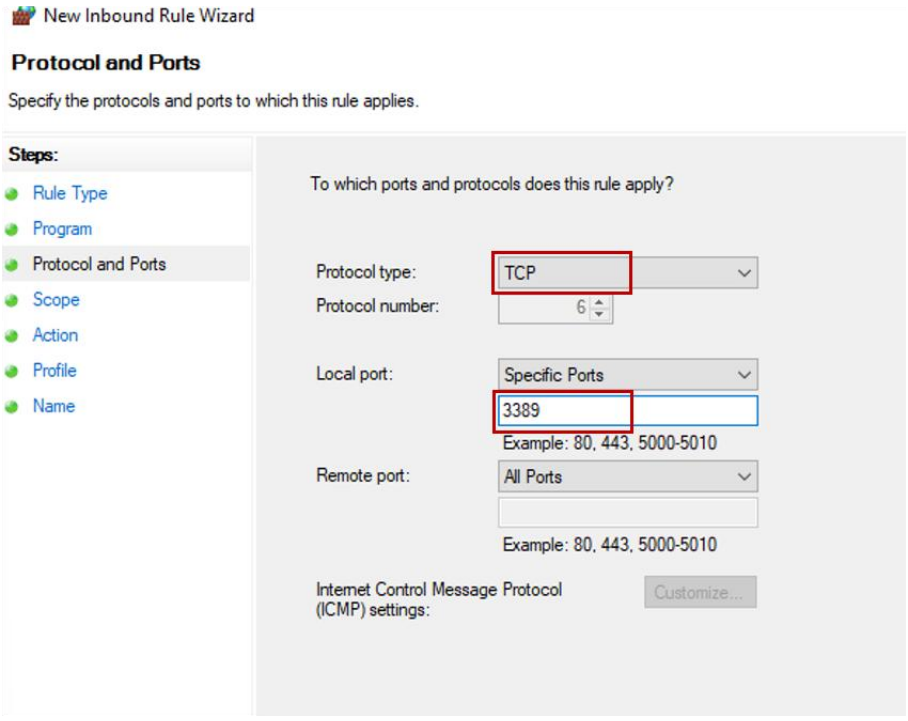
6. All profiles checked → Name: Block-ICMP-From-CL01 → Finish



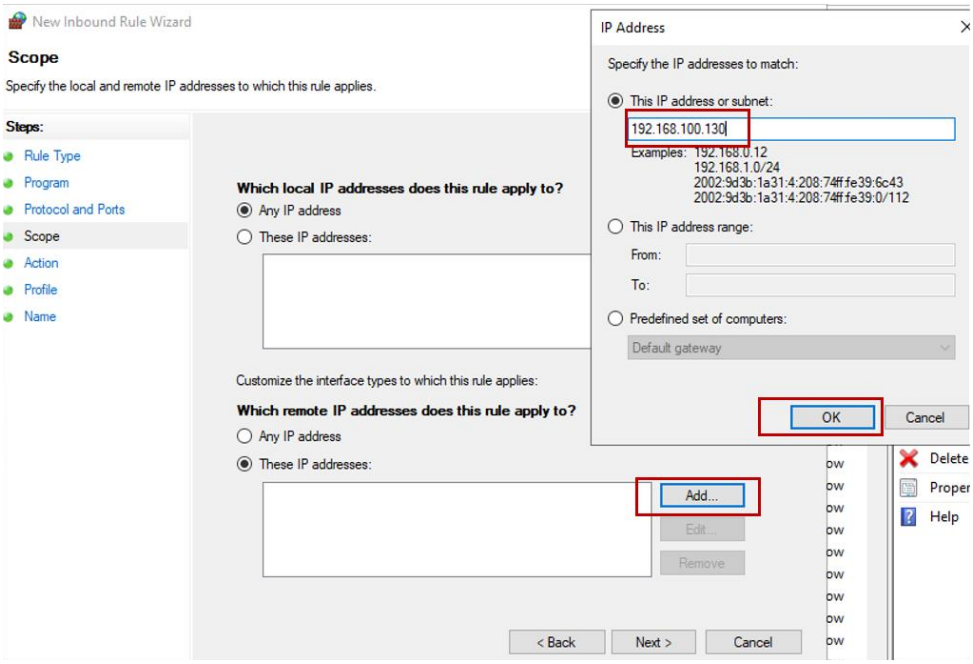
7.2 Block RDP from CL01 (users cannot remote into DC)

1. New Inbound Rule → Port → TCP → 3389

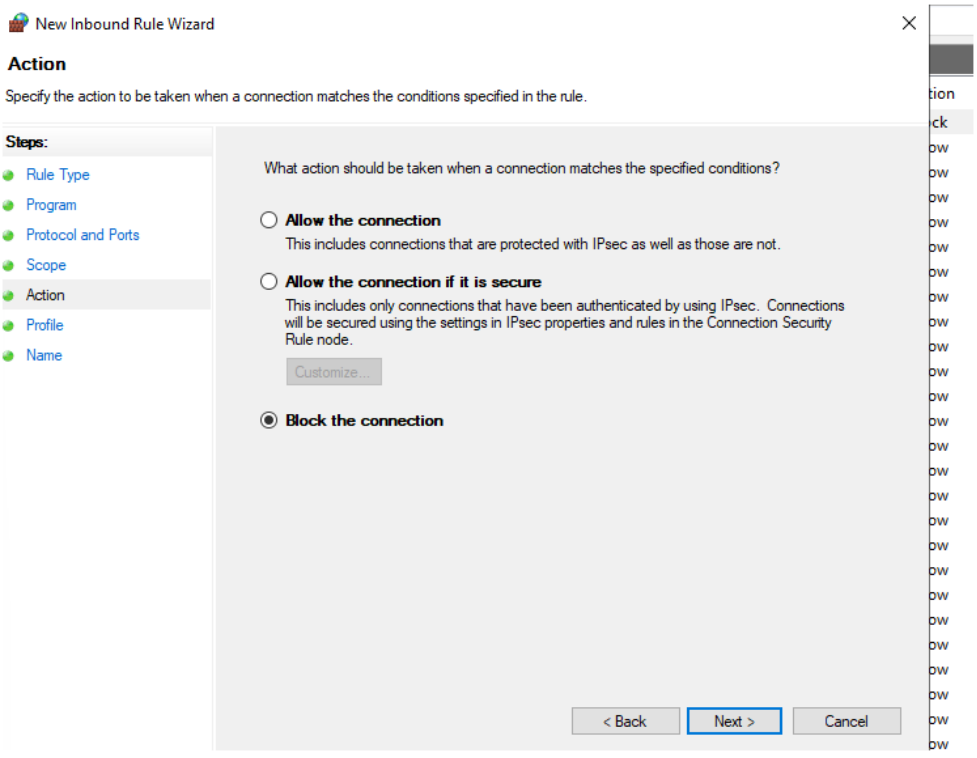


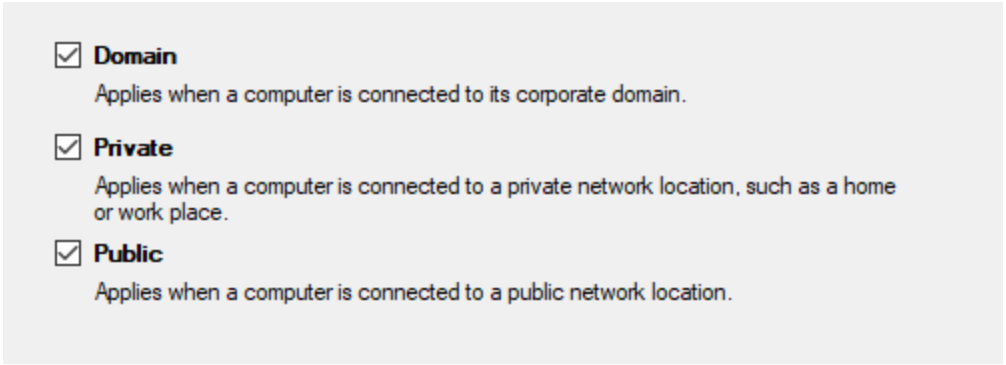


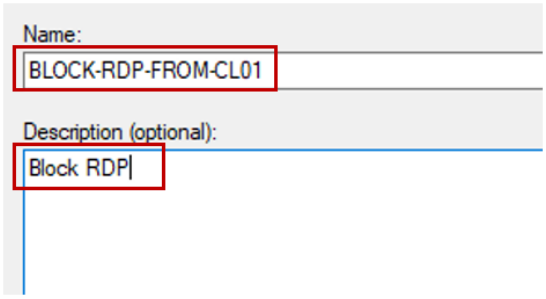
2. Remote IP: 192.168.100.130



3. Block → Name: Block-RDP-From-CL01 → Finish

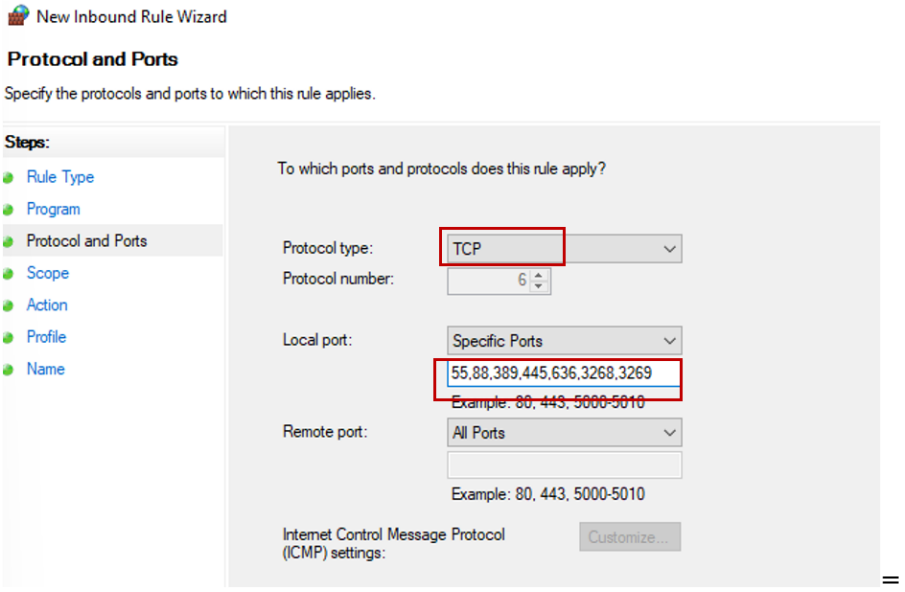



Domain
Applies when a computer is connected to its corporate domain.
Private
Applies when a computer is connected to a private network location, such as a home or work place.
Public
Applies when a computer is connected to a public network location.


Name:
BLOCK-RDP-FROM-CL01
Description (optional):
Block RDP

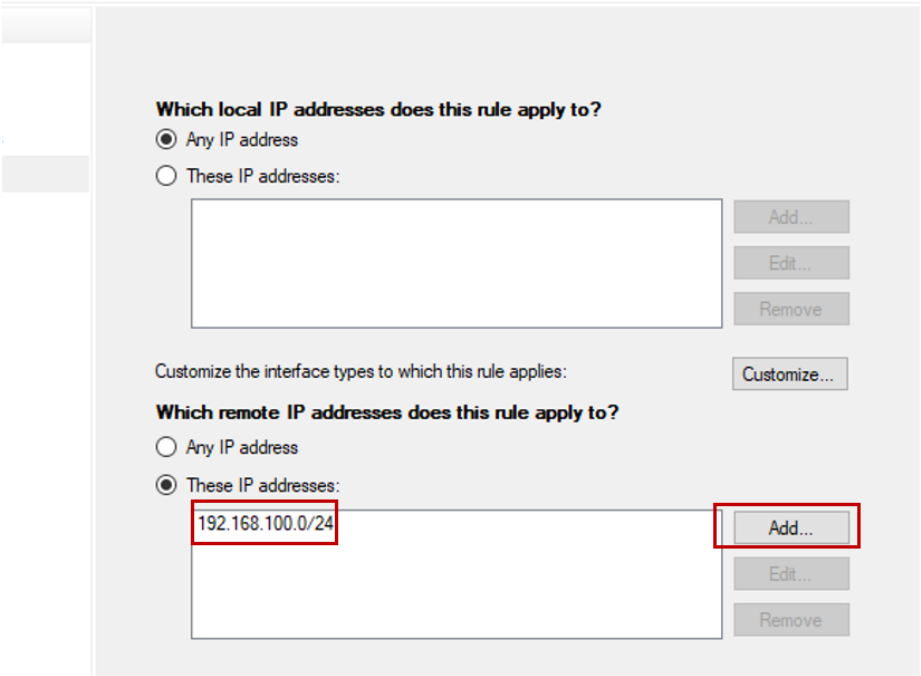
7.3 Allow AD services from all internal

1. New Inbound Rule → Port → TCP → 53,88,389,445,636,3268,3269

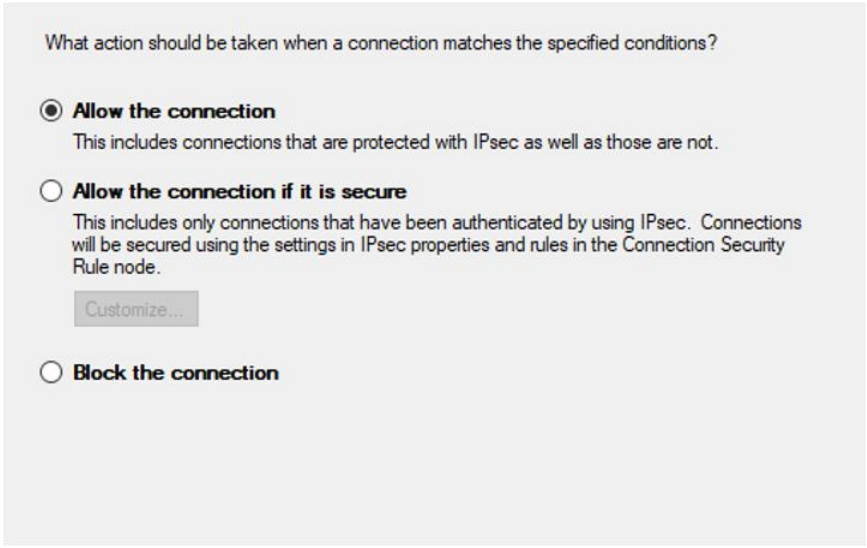


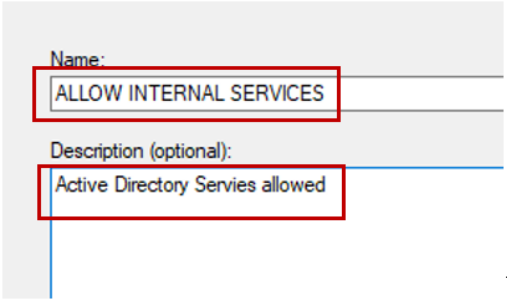
=

2. Remote IP: 192.168.100.0/24



3. Allow → Name: Allow-AD-Services-Internal → Finish



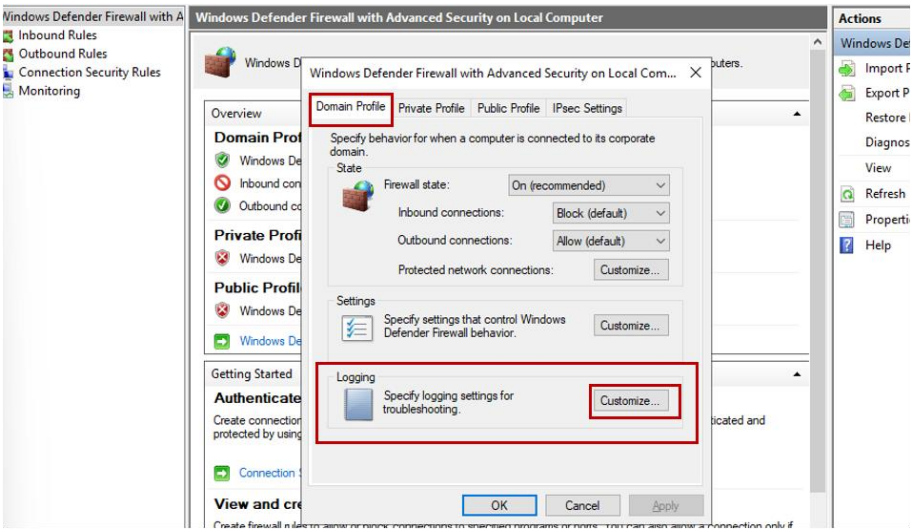


]

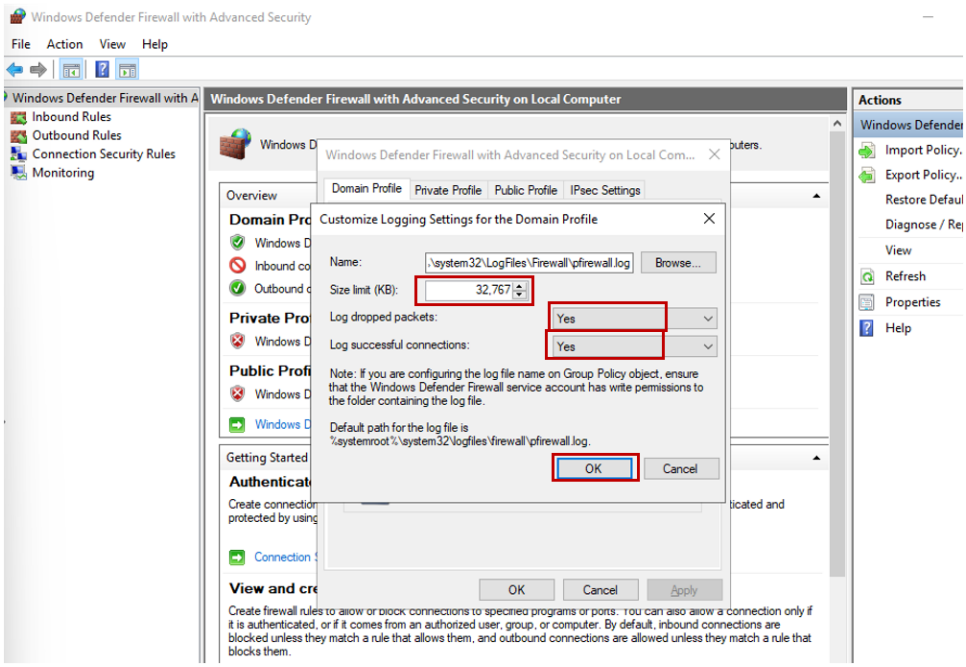


7.4 Enable Firewall Logging on DC01

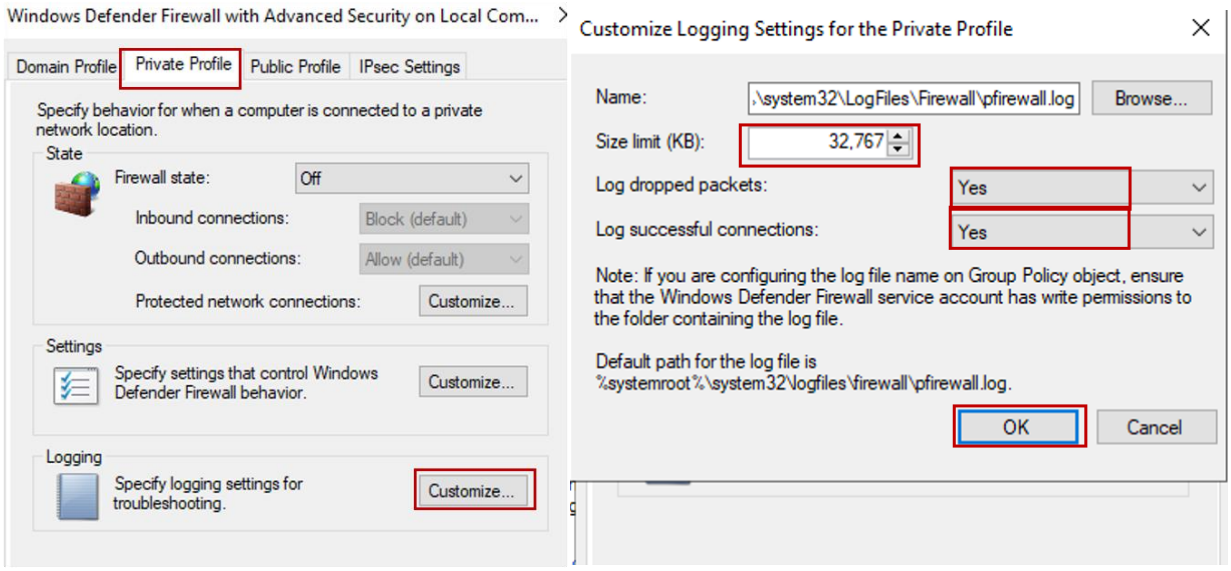
1. In wf.msc → Click **Windows Defender Firewall Properties** (right panel)
2. **Domain Profile** tab → Logging → **Customize**

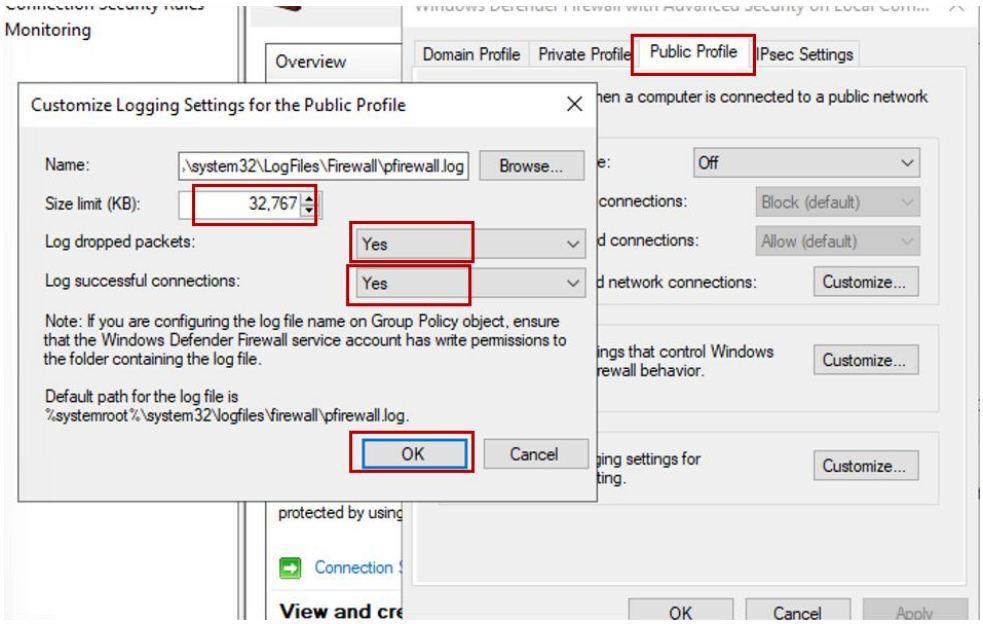


3. Log dropped packets: **Yes**
4. Log successful connections: **Yes**
5. Max file size: 32768 KB → OK



6. Repeat same for **Private Profile** and **Public Profile**





7.5 Verify segmentation works

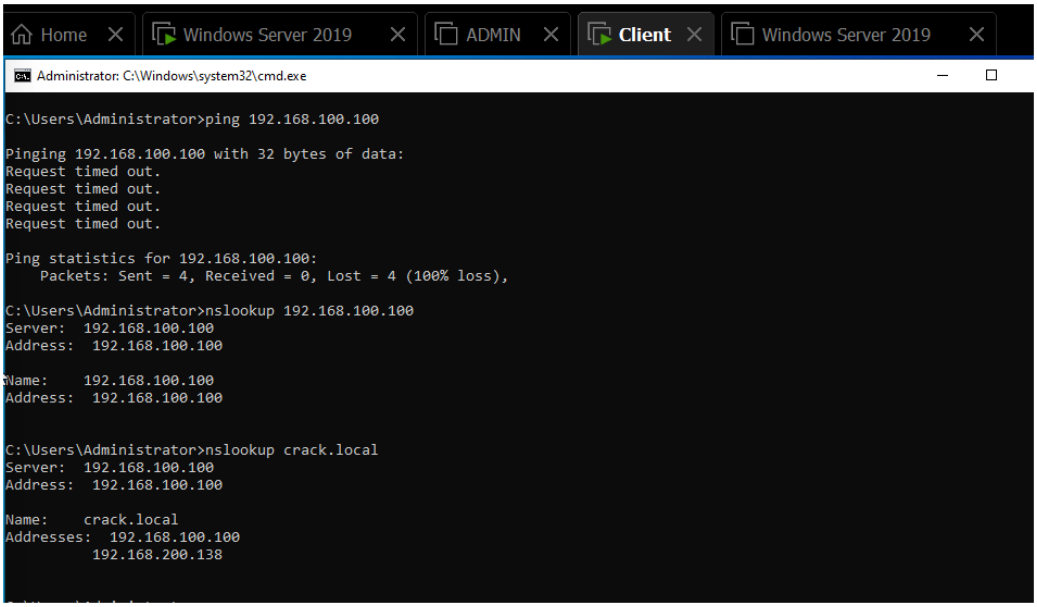
Go to **CL01** → open Command Prompt:

ping 192.168.100.100

- Result should be **Request timed out** (firewall rule is working)

nslookup crack.local

- Result should be DNS resolves correctly (port 53 still allowed)



8. DISABLE INSECURE PROTOCOLS

Run all on **DC01** in PowerShell as Administrator

8.1 Disable SMBv1:

What is SMB?

SMB = Server Message Block

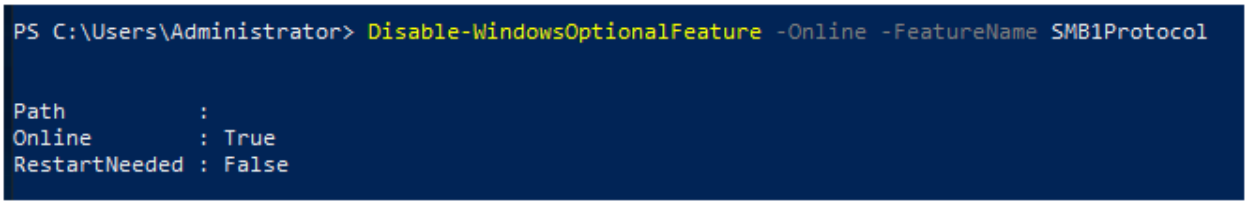
It is a network protocol used for:

- Sharing files between computers
- Sharing printers
- Group Policy delivery
- Authentication between Windows machines

SMBv1 Problems

Problem	**Explanation**
No encryption	Data travels in plain text
No integrity checking	Data can be tampered
EternalBlue exploit	NSA hacking tool that targets SMBv1
WannaCry/NotPetya	Ransomware that spreads via SMBv1

```
Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol
```




```
Set-ItemProperty -Path
"HKLM:\SYSTEM\CurrentControlSet\Services\LanmanServer\Parameters" -Name
"SMB1" -Value 0
```



8.2 Verify SMBv2 is still active

```
Get-SmbServerConfiguration | Select EnableSMB2Protocol
```

```
PS C:\Users\Administrator> Get-SmbServerConfiguration | Select EnableSMB2Protocol

EnableSMB2Protocol
------------------
              True
```

8.3 Disable LLMNR (via GPO)

LLMNR is a protocol that helps computers **find each other by name** on a network when DNS fails.

```
Computer asks: "Hey where is FILESERVER ?"

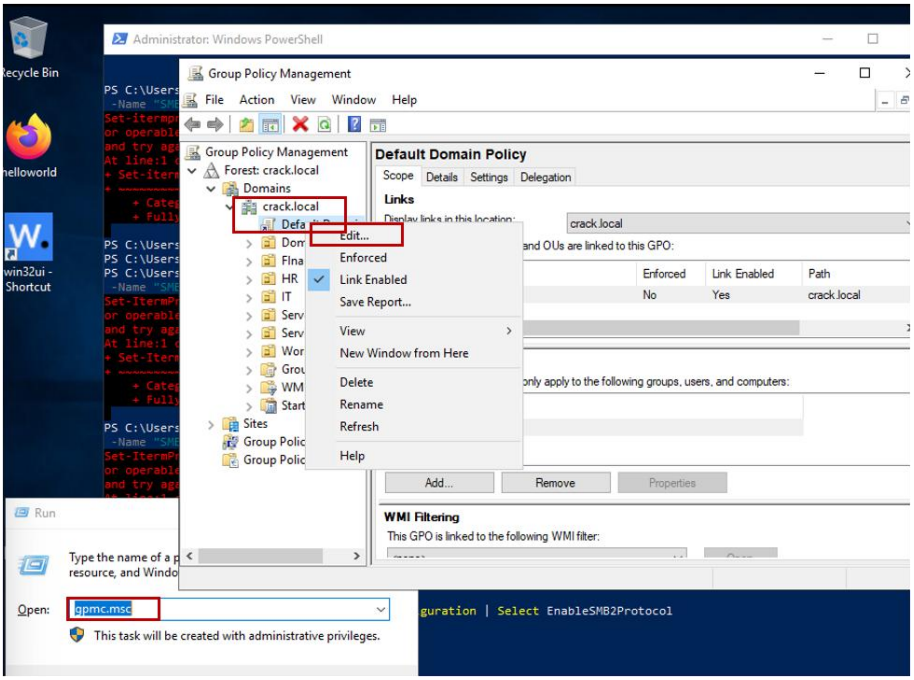
DNS says:      "I don't know ❌"

LLMNR says:    "Let me ask EVERYONE on the network 📢"

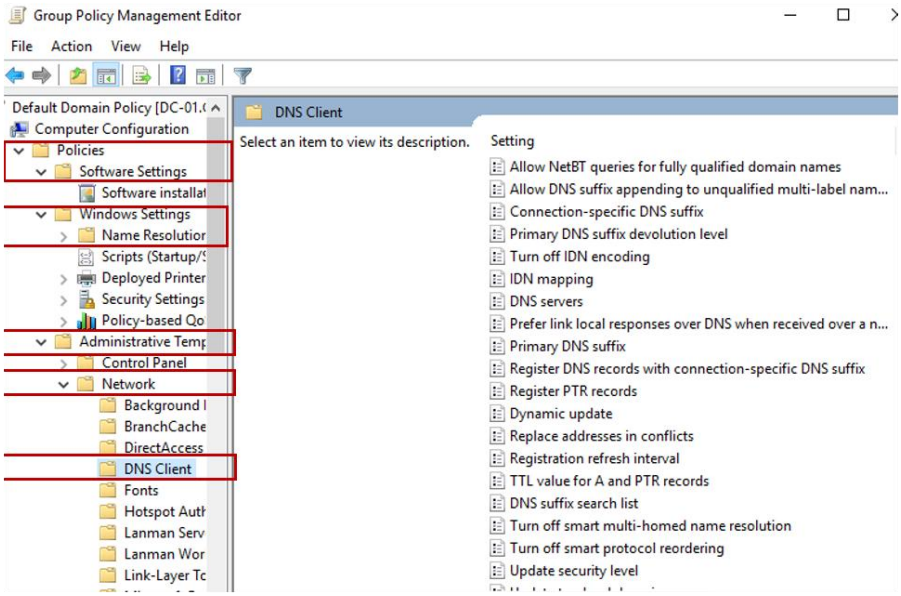
               Broadcasts to all devices →→→→

               "Does anyone know where FILESERVER is?"
```

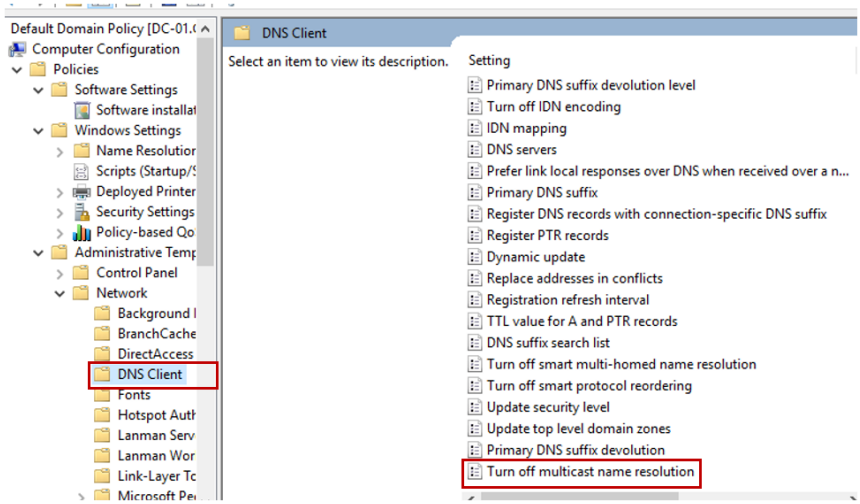
1. On DC01: Win+R → gpmc.msc
2. Right-click **Default Domain Policy** → Edit

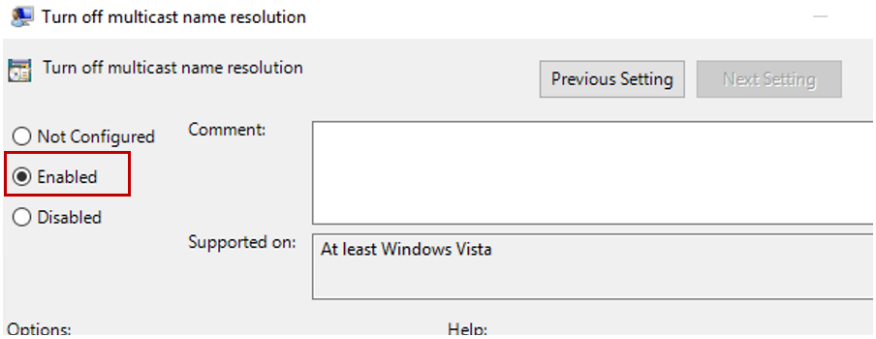


3. Navigate: Computer Configuration → Policies → Administrative Templates → Network → **DNS Client**



4. Find: **"Turn off multicast name resolution"** → double-click → Set to **Enabled** → OK





5. Run on DC01: gpupdate /force

```
C:\Users\Administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
```

8.4 Disable NetBIOS via Registry on DC01

```
$adapters = Get-WmiObject Win32_NetworkAdapterConfiguration

foreach ($adapter in $adapters) {

    $adapter.SetTcpipNetbios(2) }
```

```
PS C:\Users\Administrator> $adapters = Get-WmiObject Win32_NetworkAdapterConfiguration ;
foreach ($adapter in $adapters) { $adapter.SetTcpipNetbios(2) }


__GENUS          : 2
__CLASS          : __PARAMETERS
__SUPERCLASS     :
__DYNASTY        : __PARAMETERS
__RELPATH        :
__PROPERTY_COUNT : 1
__DERIVATION     : {}
__SERVER         :
__NAMESPACE      :
__PATH           :
ReturnValue      : 84
PSComputerName   :

__GENUS          : 2
__CLASS          : __PARAMETERS
__SUPERCLASS     :
__DYNASTY        : __PARAMETERS
__RELPATH        :
__PROPERTY_COUNT : 1
__DERIVATION     : {}
__SERVER         :
__NAMESPACE      :
__PATH           :
ReturnValue      : 0
PSComputerName   :
```

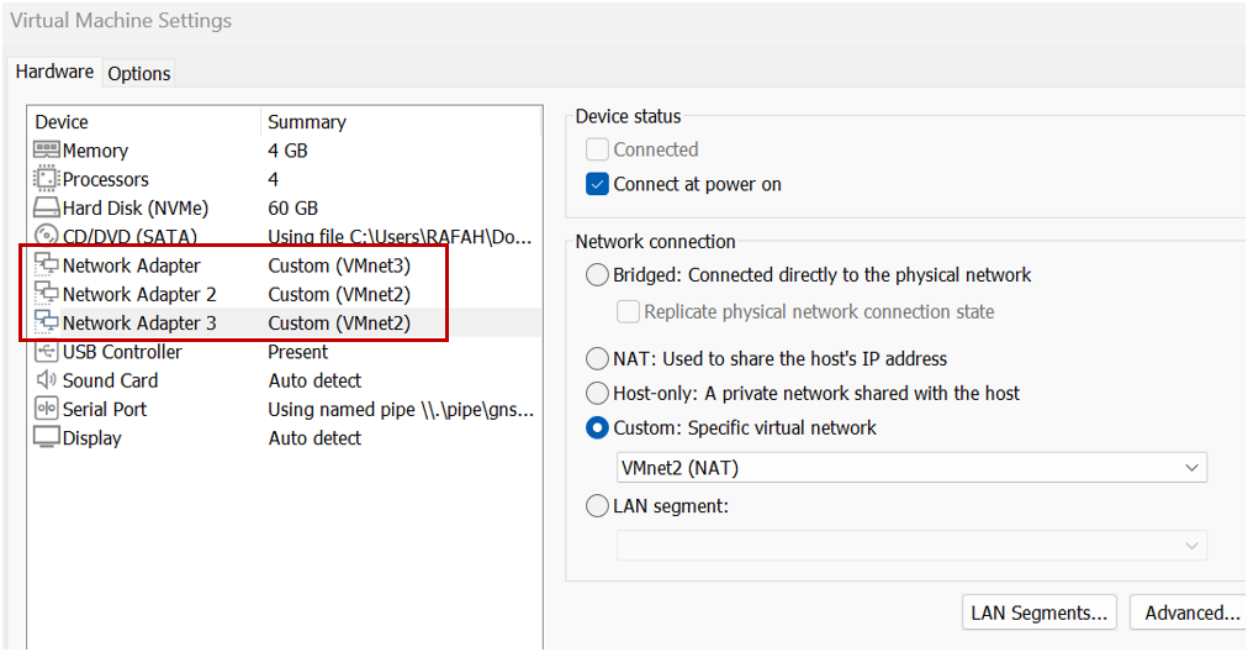
8.5 Disable mDNS in Registry

```
reg add "HKLM\SYSTEM\CurrentControlSet\Services\Dnscache\Parameters" /v
EnableMDNS /t REG_DWORD /d 0 /f
```

```
PS C:\Users\Administrator> reg add "HKLM\SYSTEM\CurrentControlSet\Services\Dnscache\Parameter" /v EnableMDNS /t REG_DWORD /d 0 /f
The operation completed successfully.
PS C:\Users\Administrator>
```

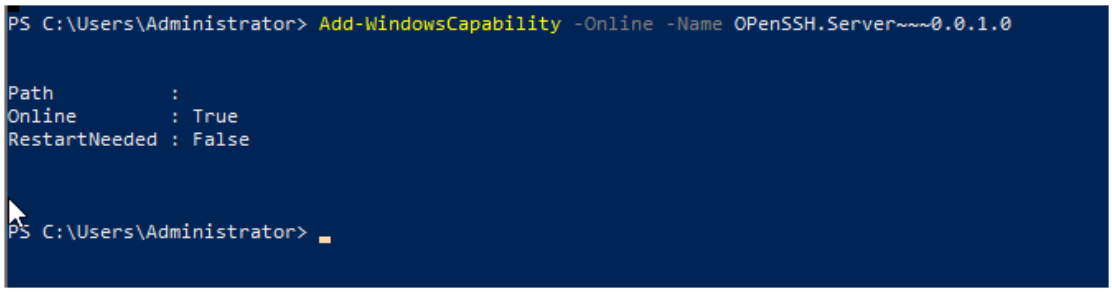
8.6 Install and Enable SSH (replace Telnet)

For installing OpenSSH.Server, we need internet access. We will add another adapter and set it to NAT to get internet access for our windows server Virtual Machine.



After adding the adapter, run these commands below in the power shell.

```
1. Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0
```

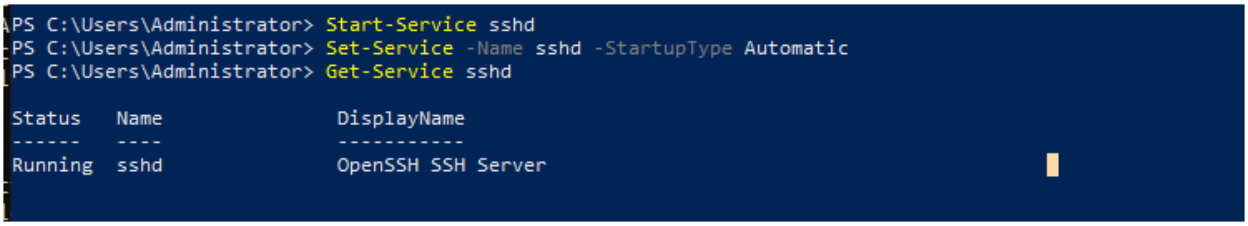


Verify SSH is running

```
Start-Service sshd

Get-Service sshd

Set-Service -Name sshd -StartupType Automatic
```



9. TLS HARDENING

Still on **DC01** in PowerShell as Administrator.

```
$base =
"HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols"
```

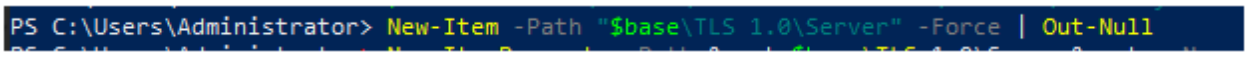


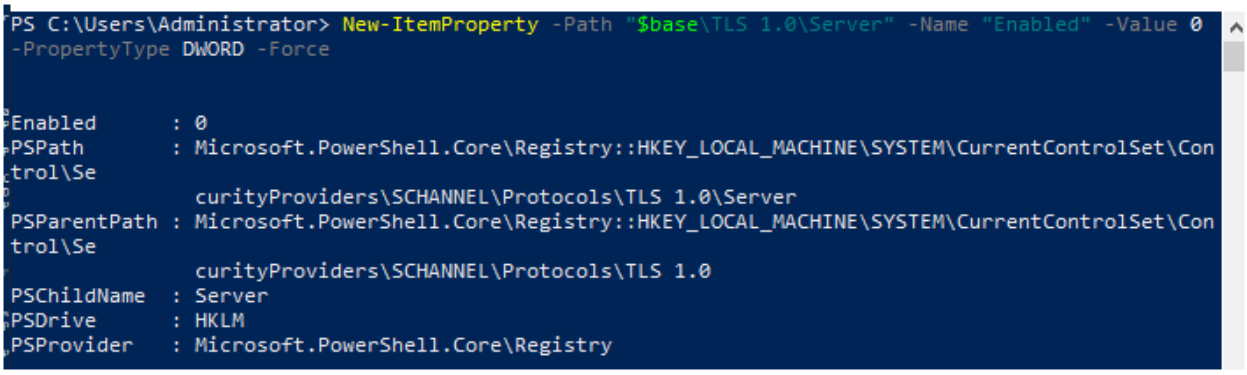
Disable TLS 1.0 Server

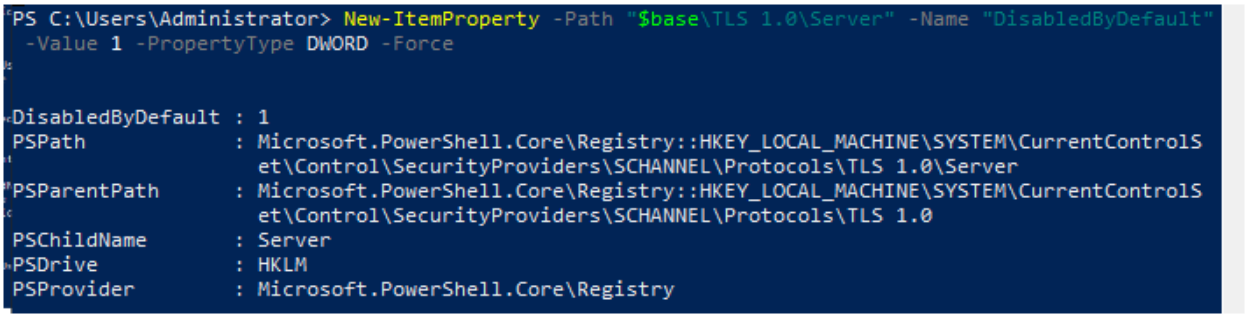
```
New-Item -Path "$base\TLS 1.0\Server" -Force | Out-Null

New-ItemProperty -Path "$base\TLS 1.0\Server" -Name "Enabled" -Value 0 -
PropertyType DWORD -Force

New-ItemProperty -Path "$base\TLS 1.0\Server" -Name "DisabledByDefault" -
Value 1 -PropertyType DWORD -Force
```







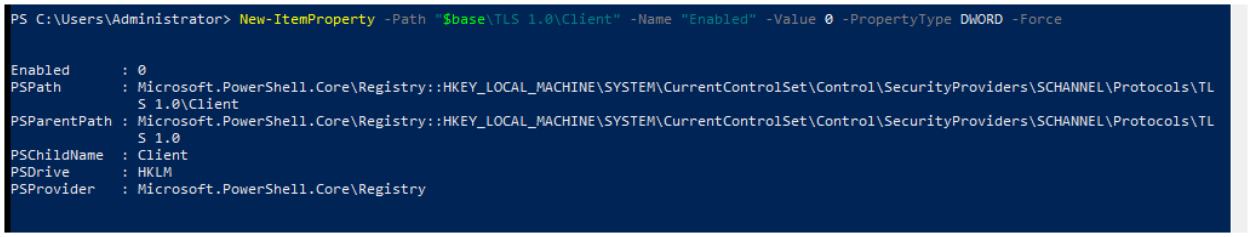
Disable TLS 1.0 Client

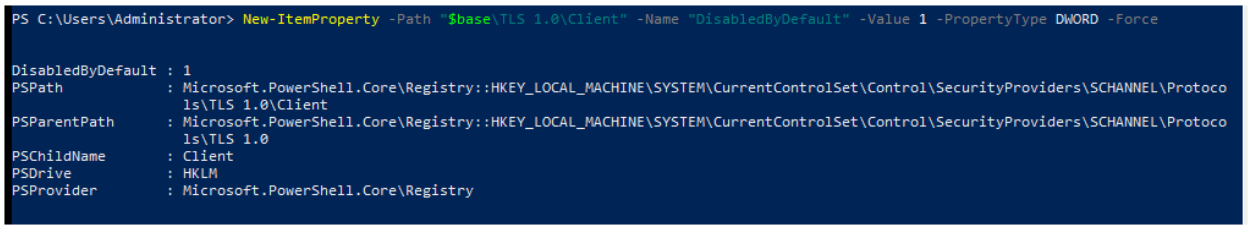
```
New-Item -Path "$base\TLS 1.0\Client" -Force | Out-Null

New-ItemProperty -Path "$base\TLS 1.0\Client" -Name "Enabled" -Value 0 -
PropertyType DWORD -Force

New-ItemProperty -Path "$base\TLS 1.0\Client" -Name "DisabledByDefault" -
Value 1 -PropertyType DWORD -Force
```





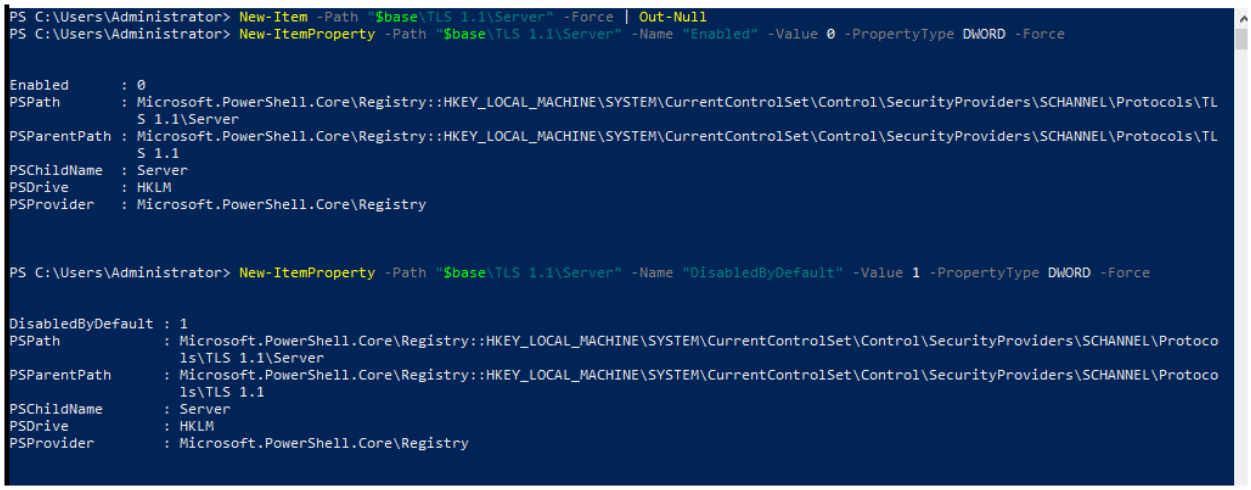


Disable TLS 1.1 Server

```
New-Item -Path "$base\TLS 1.1\Server" -Force | Out-Null

New-ItemProperty -Path "$base\TLS 1.1\Server" -Name "Enabled" -Value 0 -
PropertyType DWORD -Force

New-ItemProperty -Path "$base\TLS 1.1\Server" -Name "DisabledByDefault" -
Value 1 -PropertyType DWORD -Force
```

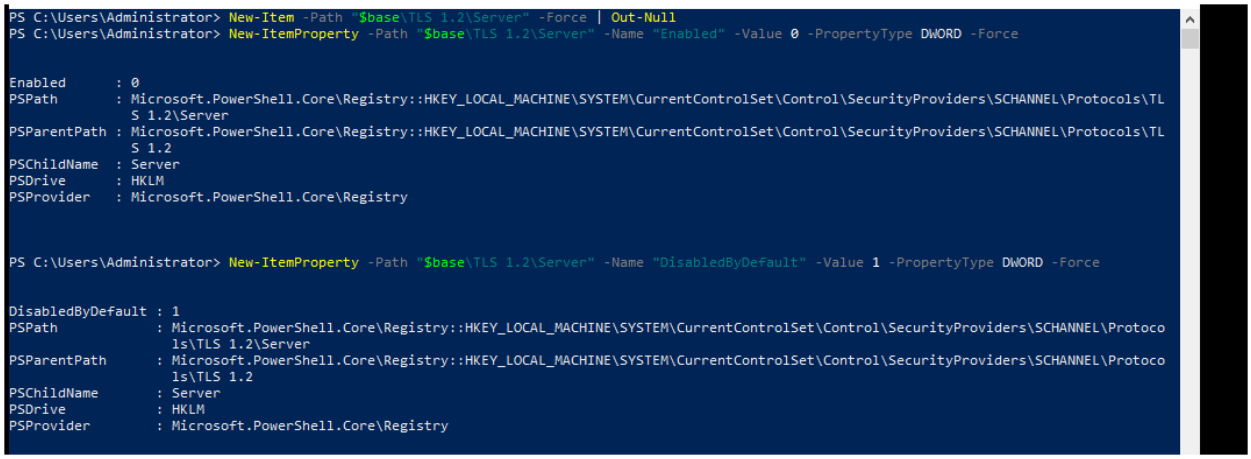


Enable TLS 1.2 Server

```
New-Item -Path "$base\TLS 1.2\Server" -Force | Out-Null

New-ItemProperty -Path "$base\TLS 1.2\Server" -Name "Enabled" -Value 1 -
PropertyType DWORD -Force

New-ItemProperty -Path "$base\TLS 1.2\Server" -Name "DisabledByDefault" -
Value 0 -PropertyType DWORD -Force
```

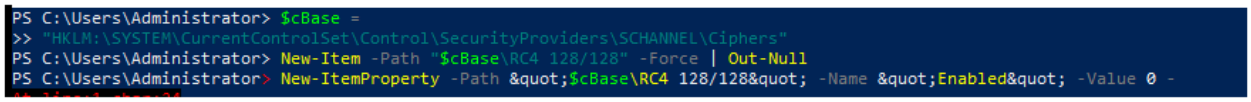


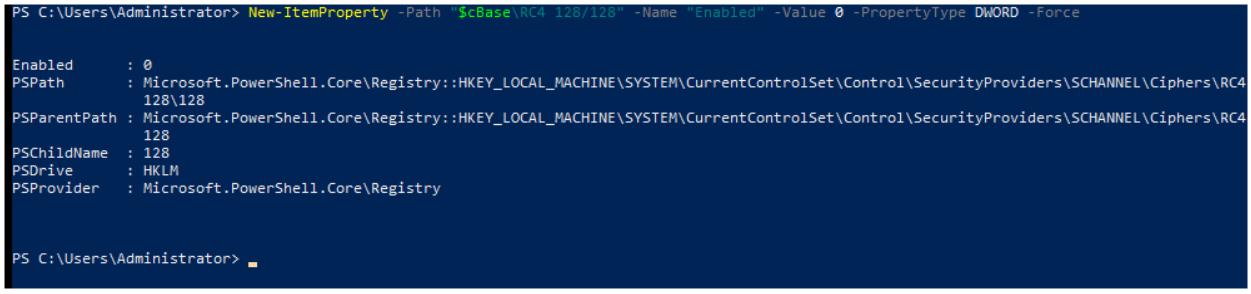
Disable RC4

```
$cBase =
"HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Ciphers"

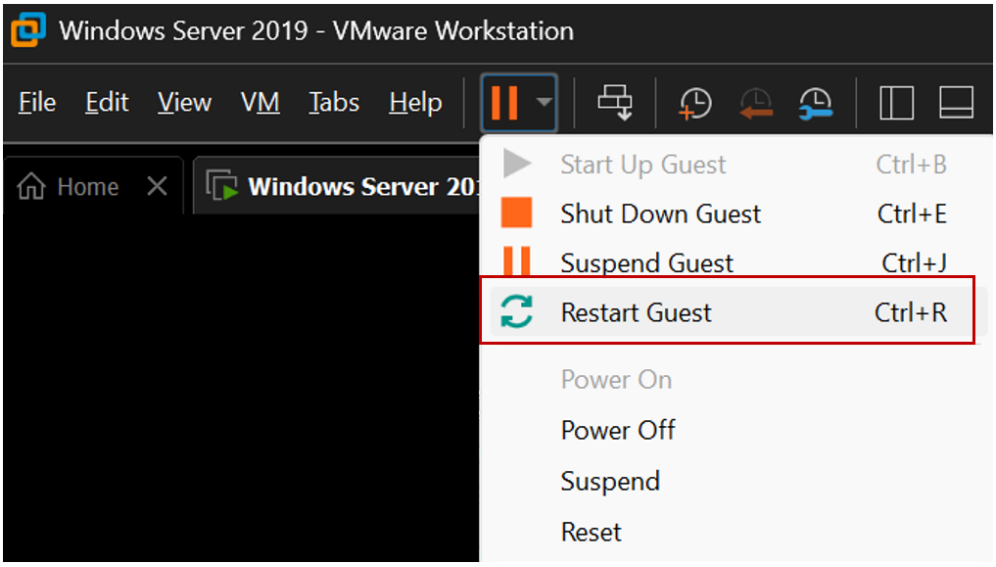
New-Item -Path "$cBase\RC4 128/128" -Force | Out-Null

New-ItemProperty -Path "$cBase\RC4 128/128" -Name "Enabled" -Value 0 -
PropertyType DWORD -Force
```





Now restart DC01:

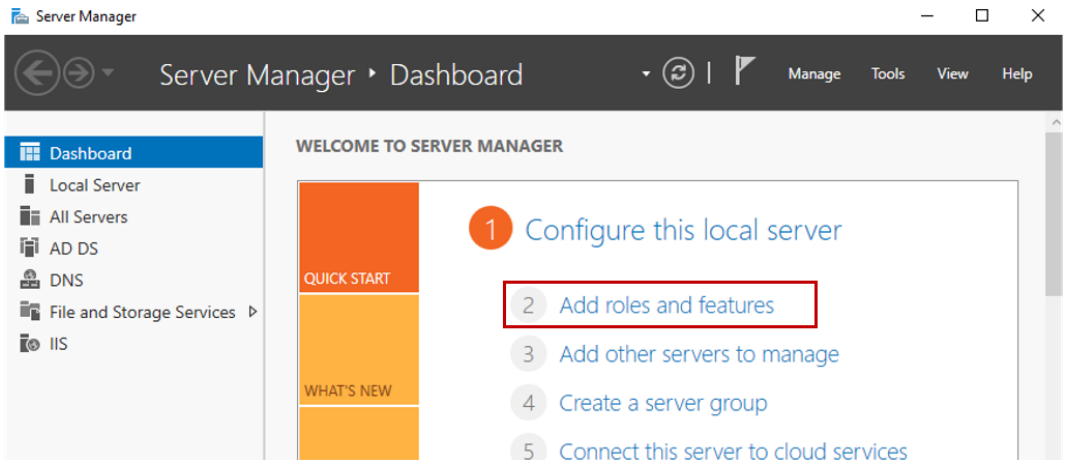


10. CERTIFICATE AUTHORITY (PKI)

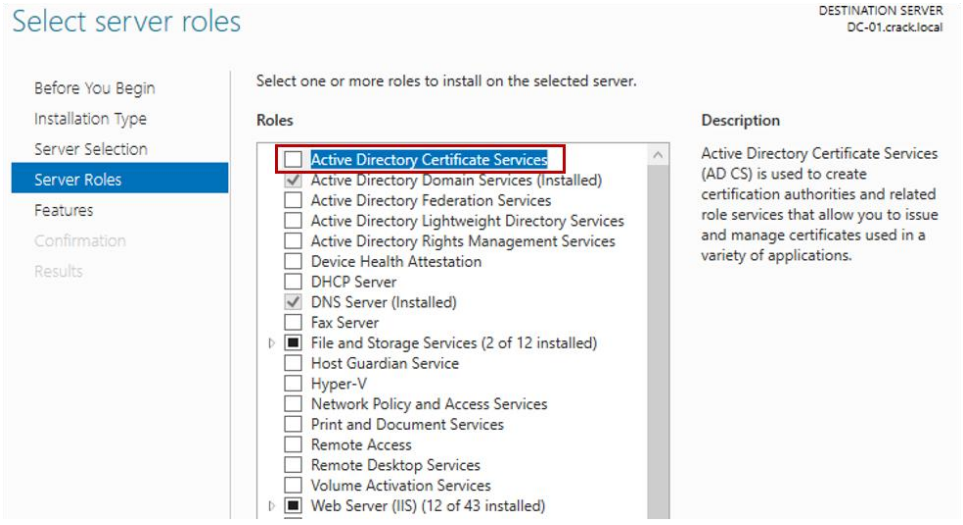
On **DC01**, logged in as CRACK\Administrator

10.1 INSTALL AD CERTIFICATE SERVICES

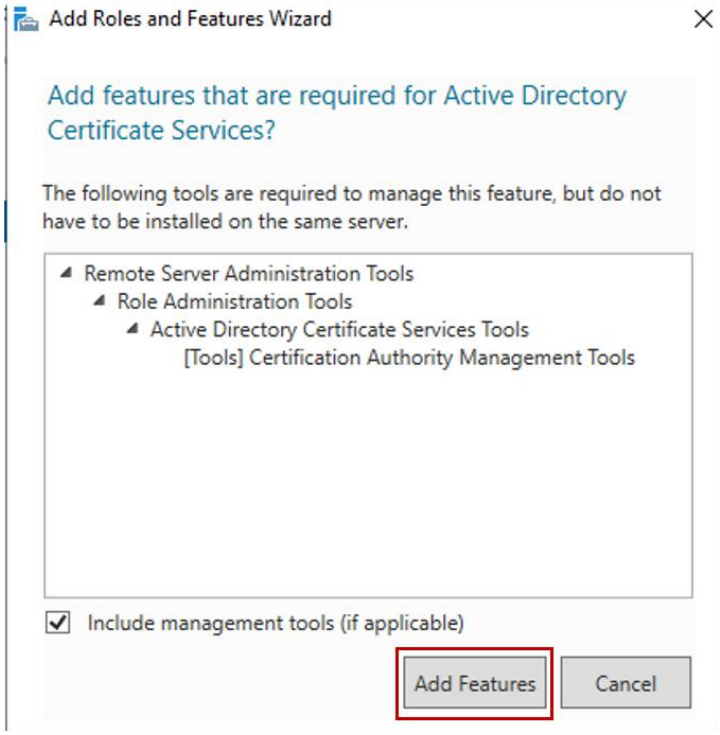
1. Open **Server Manager**
2. Click **"Add Roles and Features"**



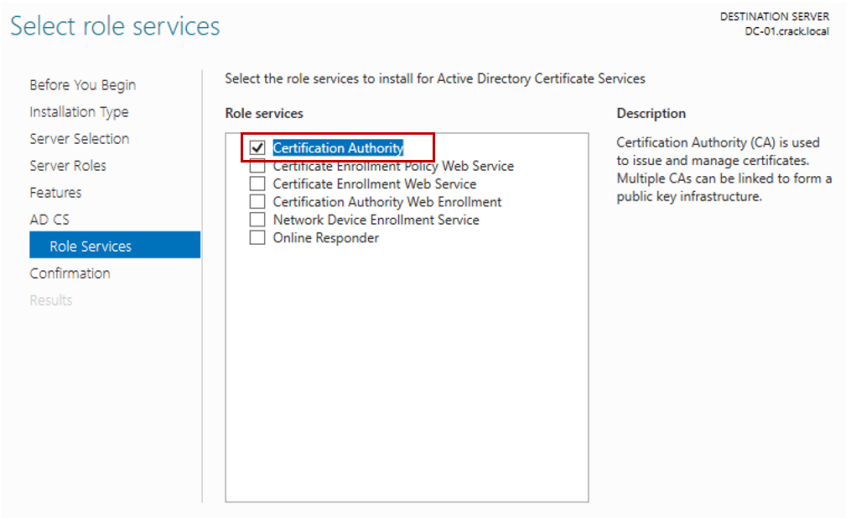
3. Installation type: **Role-based or feature-based installation** → Next
4. Server selection: your DC01 should be selected → Next
5. Server Roles: scroll down and check **Active Directory Certificate Services**



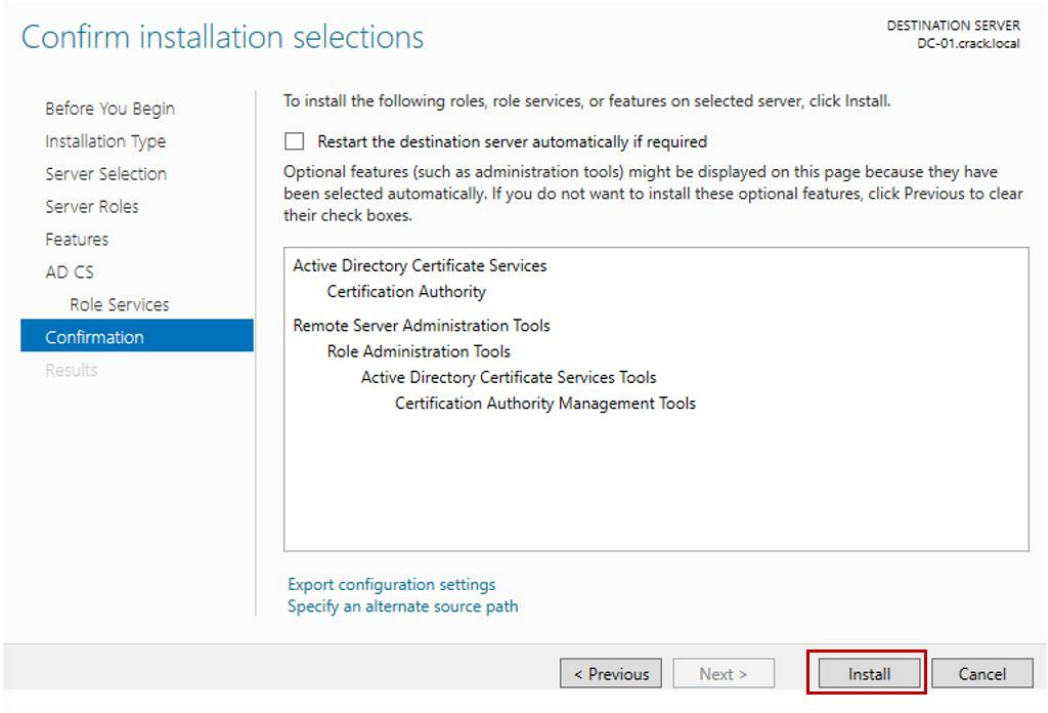
6. A popup appears → click **"Add Features"** → Next



7. Features: don't change anything → Next
8. Features Page: click Next
9. AD CS page: click Next
10. Role Services: make sure **Certification Authority** is checked → Next



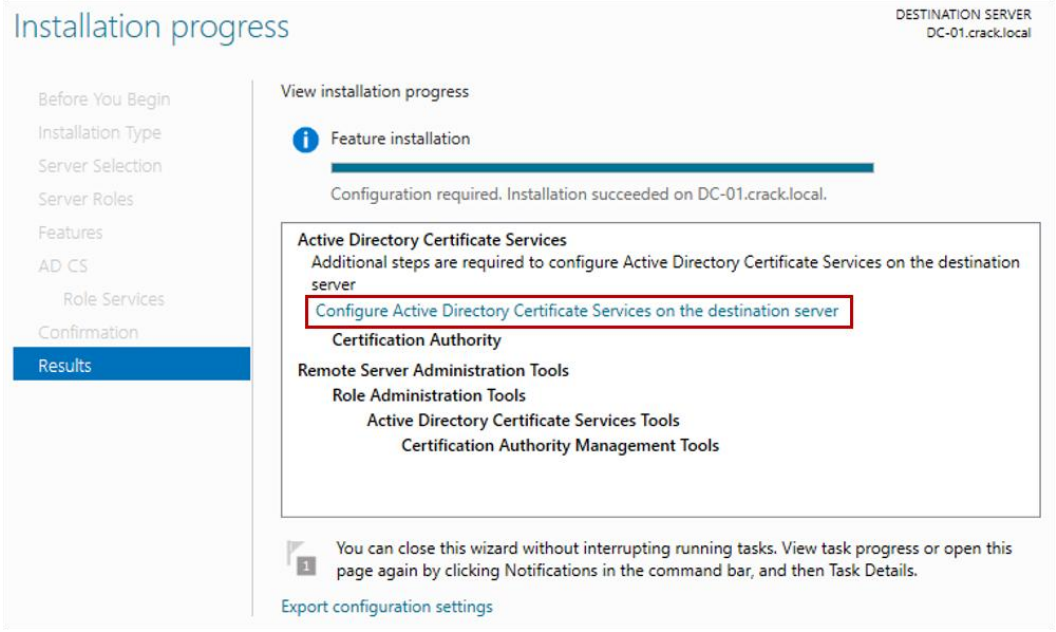
11. Confirmation: → click **Install**



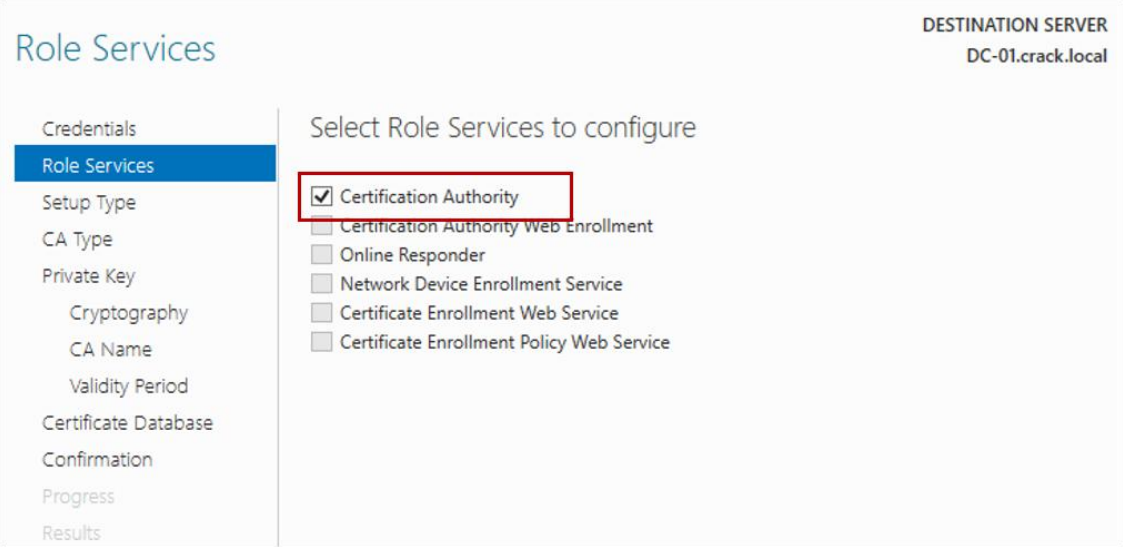
12. Wait for installation to finish (about 2 minutes)

10.2 CONFIGURE THE CERTIFICATE AUTHORITY

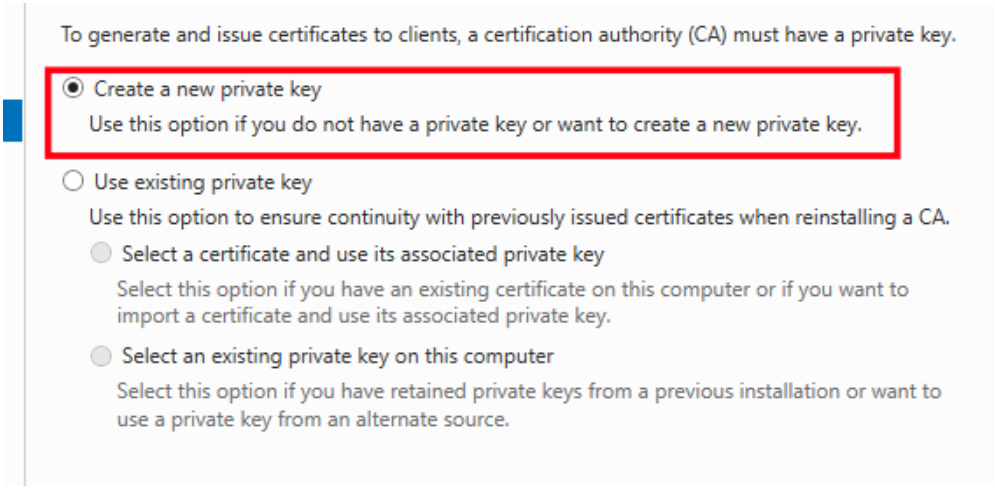
1. click **"Configure Active Directory Certificate Services on the destination server"**



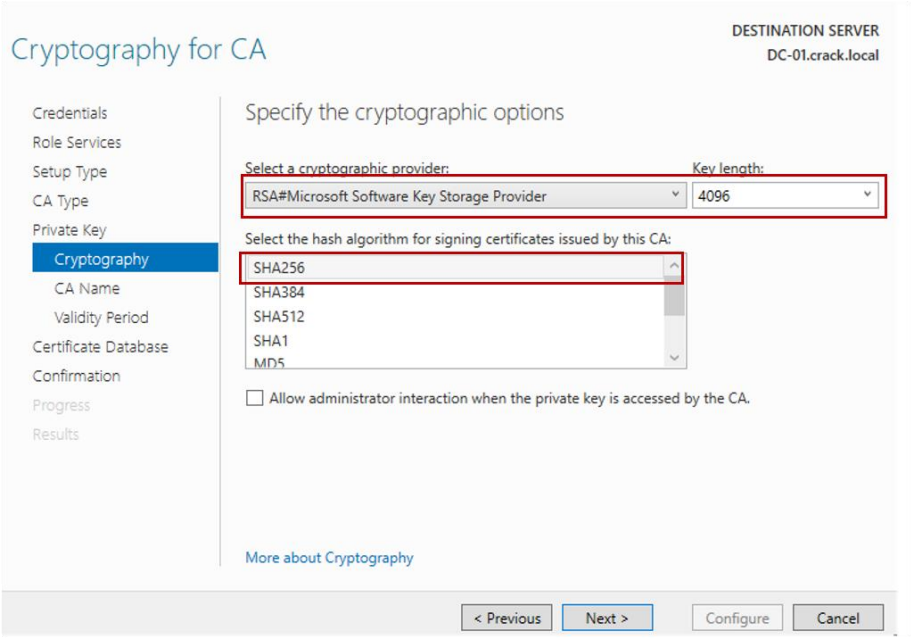
2. Credentials: CRACK\Administrator is already filled in → click Next
3. Role Services: check **Certification Authority** → Next



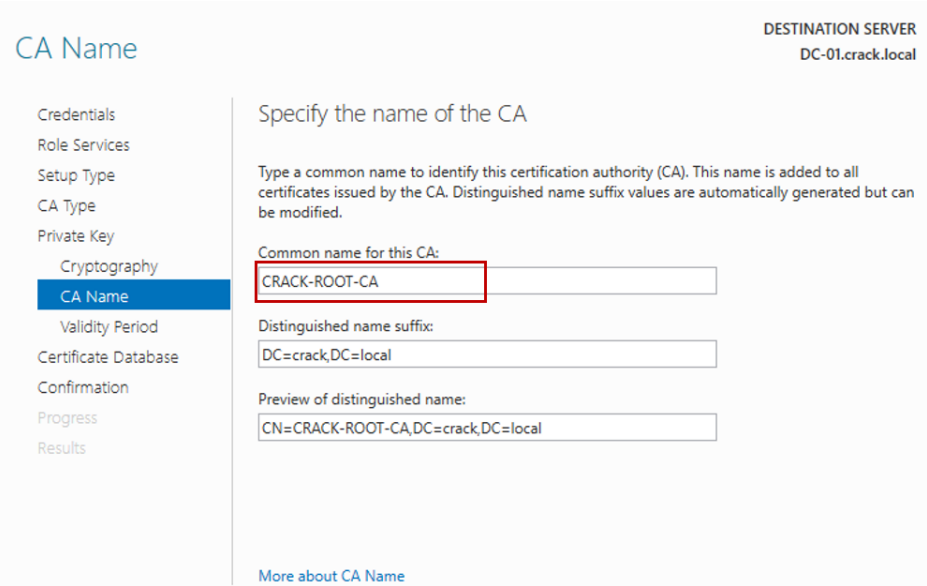
4. Setup type: **Enterprise CA** → Next
5. CA type: **Root CA** → Next
6. Private key: **Create a new private key** → Next



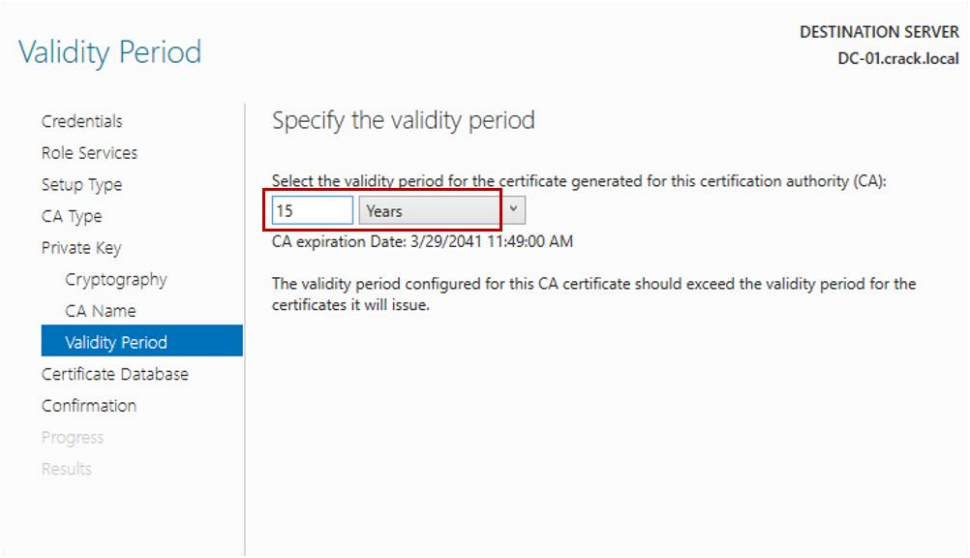
7. Cryptography:
    - Provider: RSA#Microsoft Software Key Storage Provider
    - Key length: **4096**
    - Hash algorithm: **SHA256** → Next



8. CA name: CRACK-Root-CA → Next

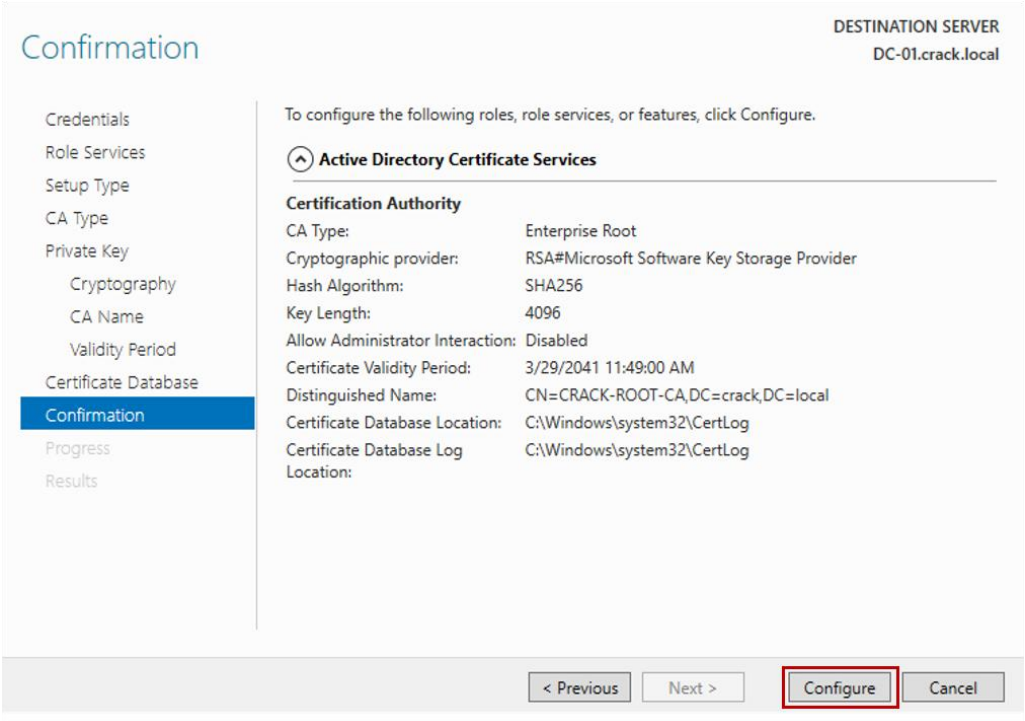


9. Validity period: type 10 → Years → Next

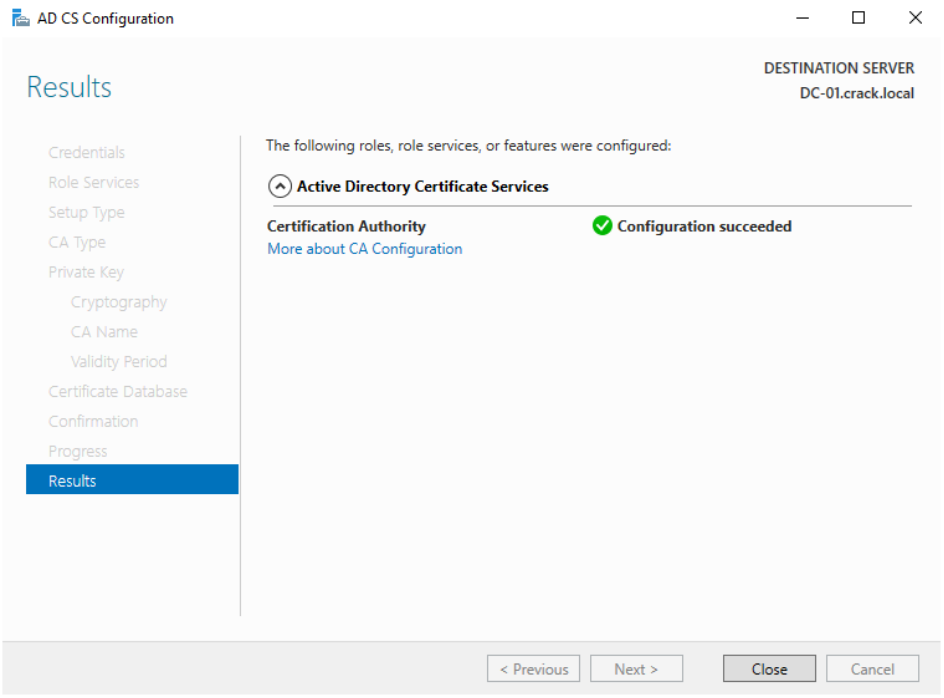


10. Certificate database: leave paths as default → Next

11. Click **Configure**

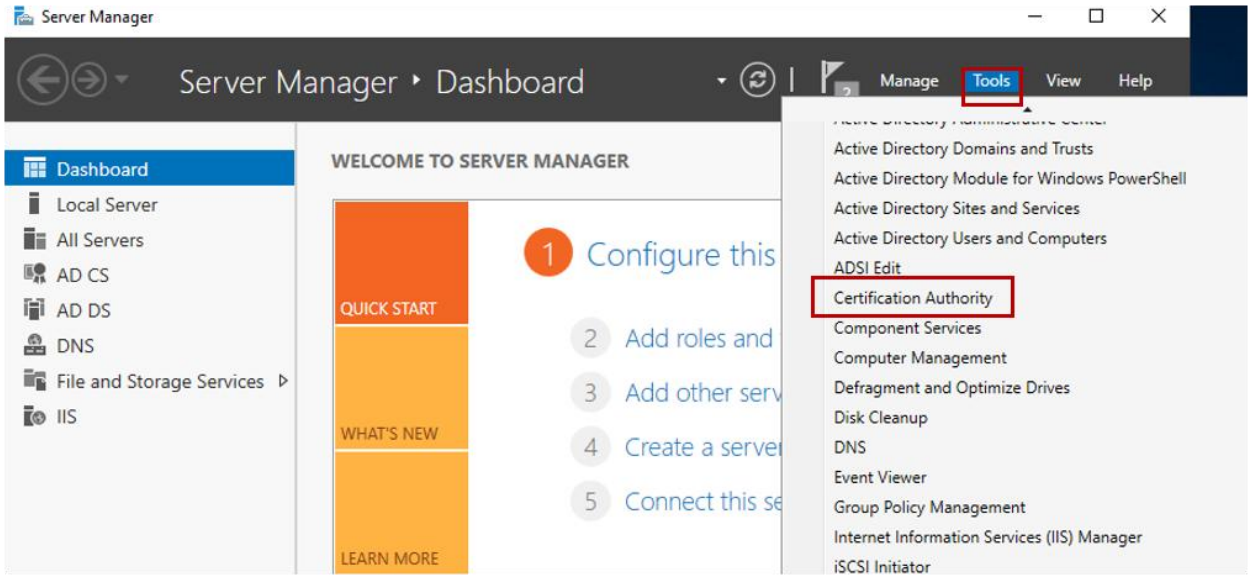


12. When done: click Close

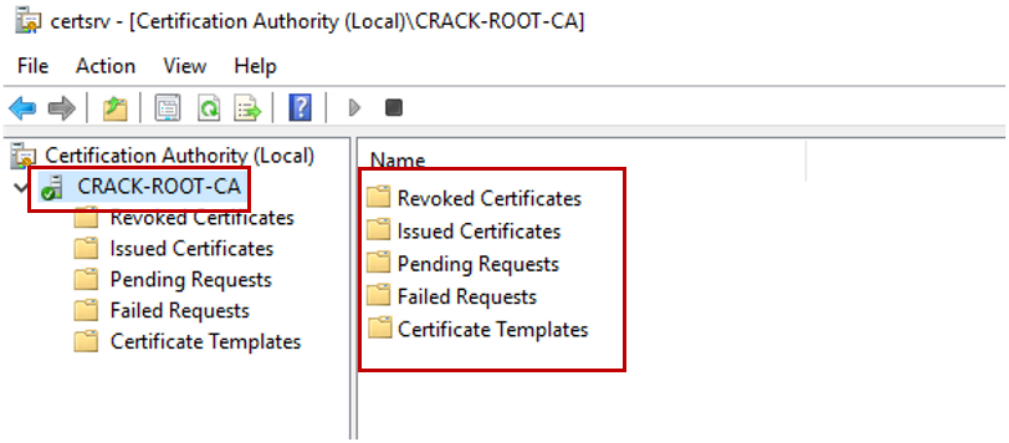


10.3 VERIFY THE CA IS RUNNING

1. Server Manager → Tools → **Certification Authority**

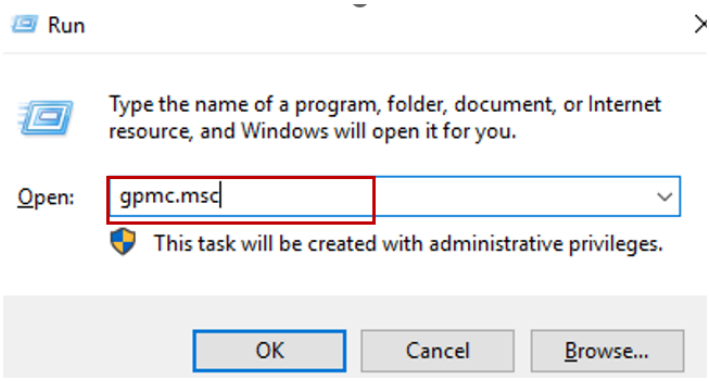


2. You see: CRACK-Root-CA in the tree
3. Expand it — you should see folders: Revoked Certificates, Issued Certificates, etc
4. There should be a green checkmark or no red X next to the CA name

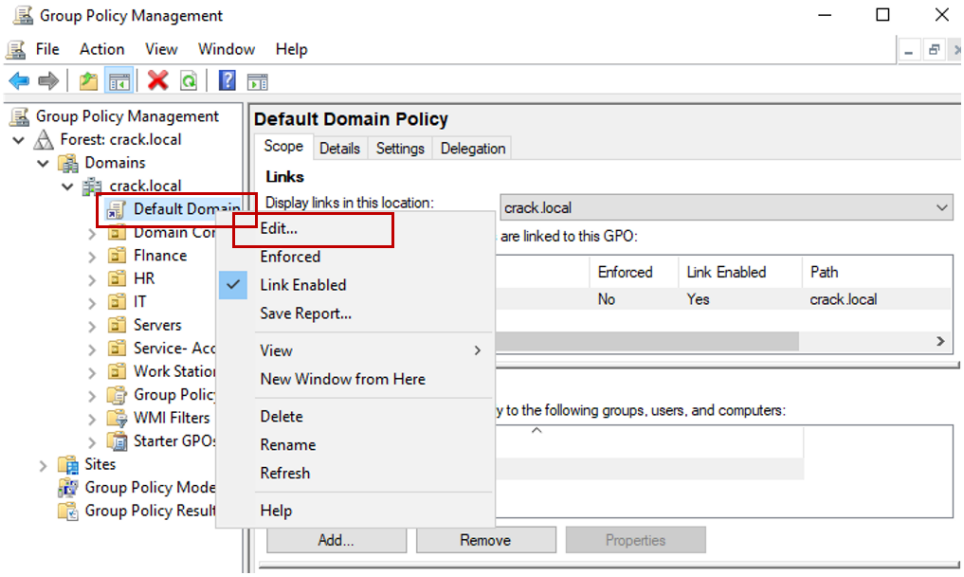


11. ENABLE CERTIFICATE AUTO-ENROLLMENT VIA GPO

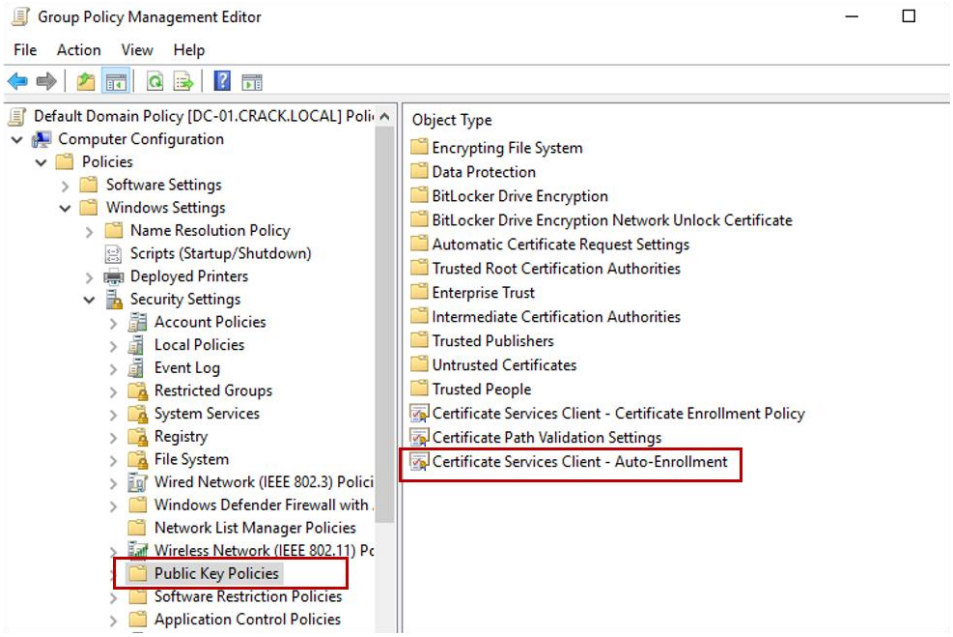
1. Press Win+R → gpmc.msc → Enter



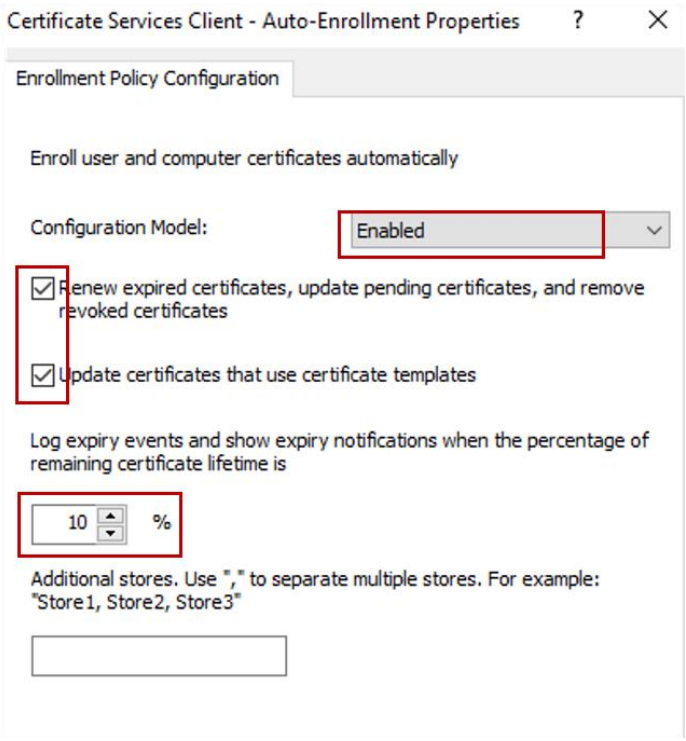
2. Expand Forest → Domains → crack.local
3. Right-click **Default Domain Policy → Edit**



4. Navigate:
    - Computer Configuration
    - → Policies
    - → Windows Settings
    - → Security Settings
    - → **Public Key Policies**
5. In the right panel: double-click **"Certificate Services Client – Auto-Enrollment"**



6. Configuration Model: change to **Enabled**
7. Check these two boxes:
    - Renew expired certificates, update pending certificates
    - Update certificates that use certificate templates
8. Click Apply and then OK → close GPO editor



Apply:

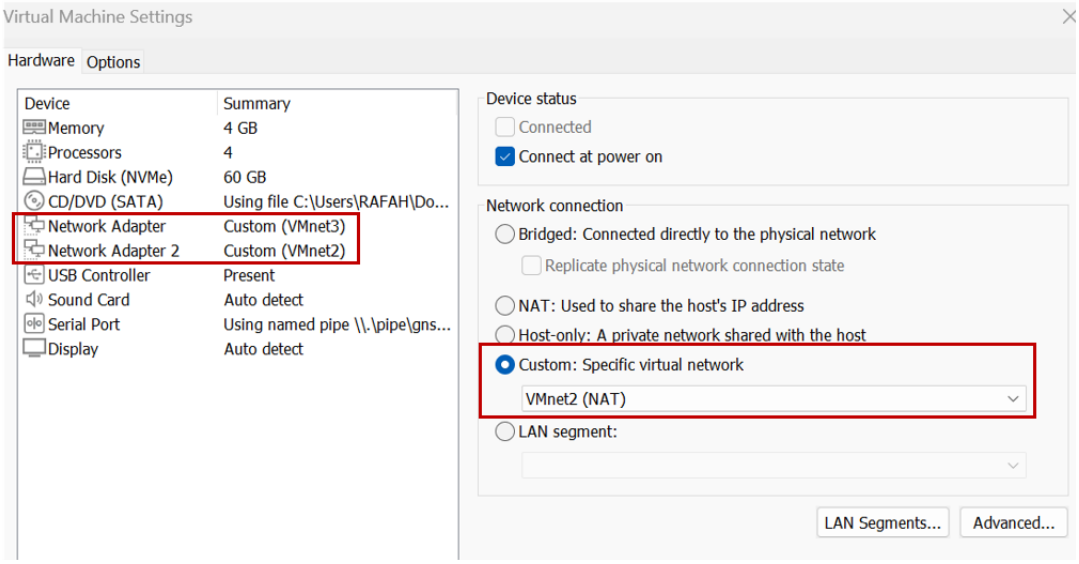
```
gpupdate /force
```

```
C:\Users\Administrator>gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
```

12. VPN SERVER (Soft Ether on VPN01)

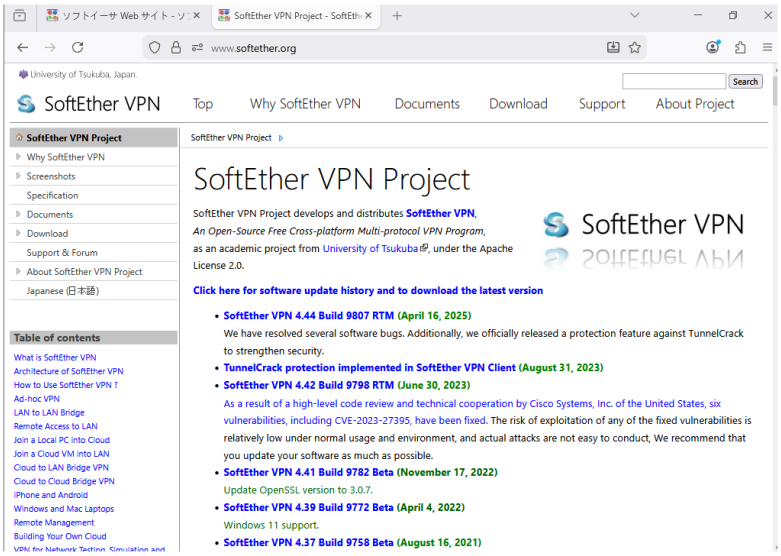
For installing VPN, we need internet access. We will add another adapter and set it to NAT to get internet access for our windows server Virtual Machine.

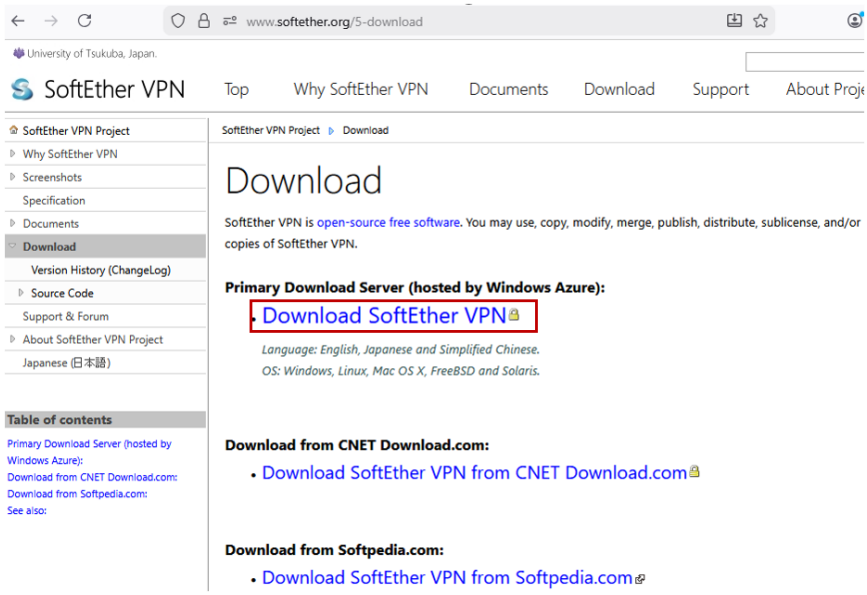


12.1 INSTALL SOFTETHER ON VPN01

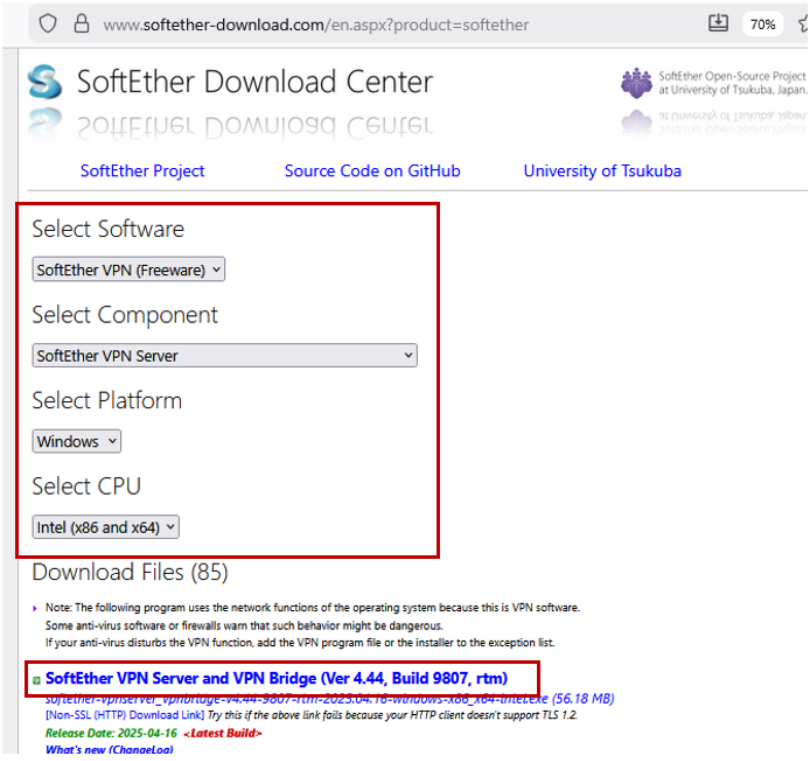
Go to **VPN01** VM.

1. On VPN01, open **Internet Explorer or Edge**
2. Go to: softether.org

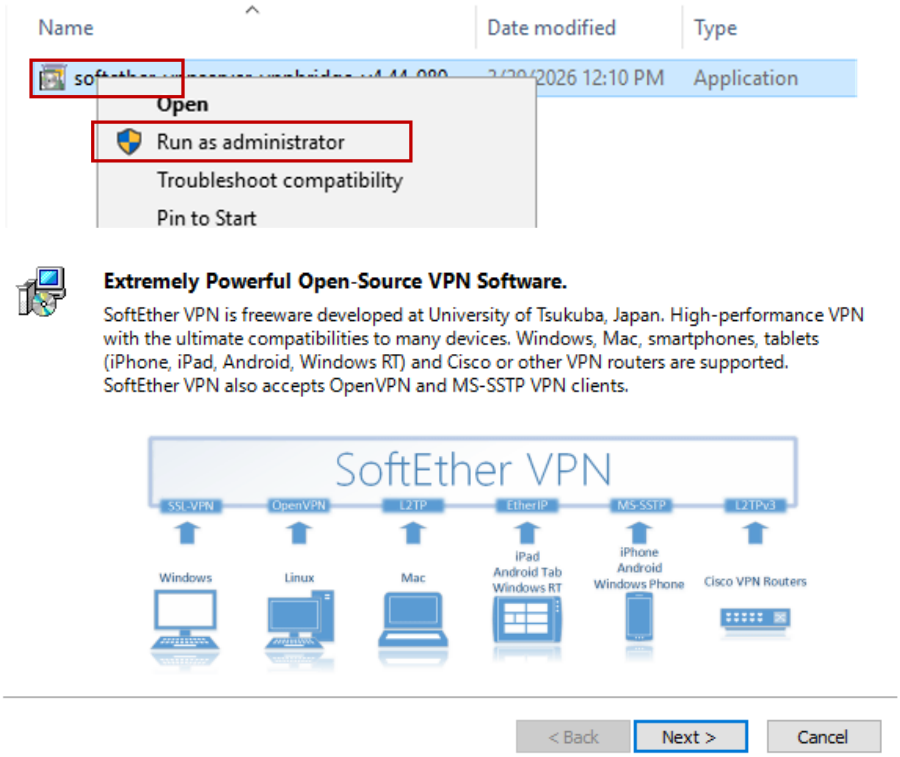




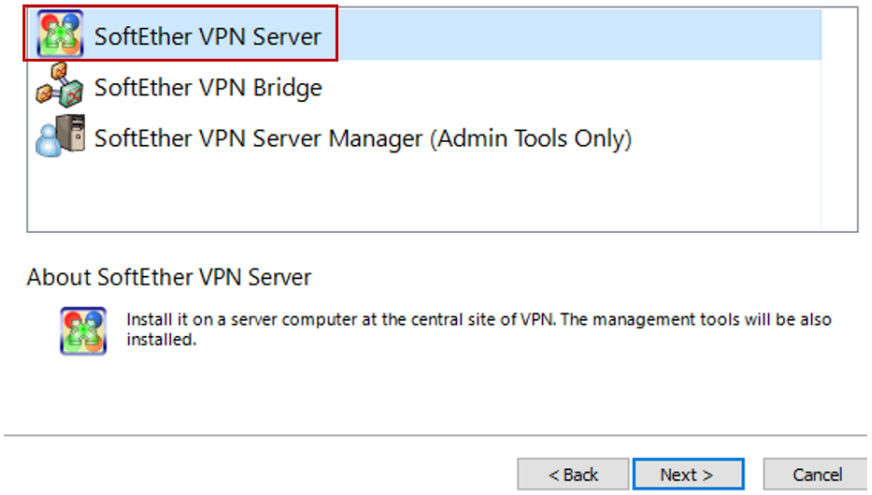
3. Platform: **Windows** → CPU: **Intel (x86 and x64)**
4. Download the latest stable version (.exe file)



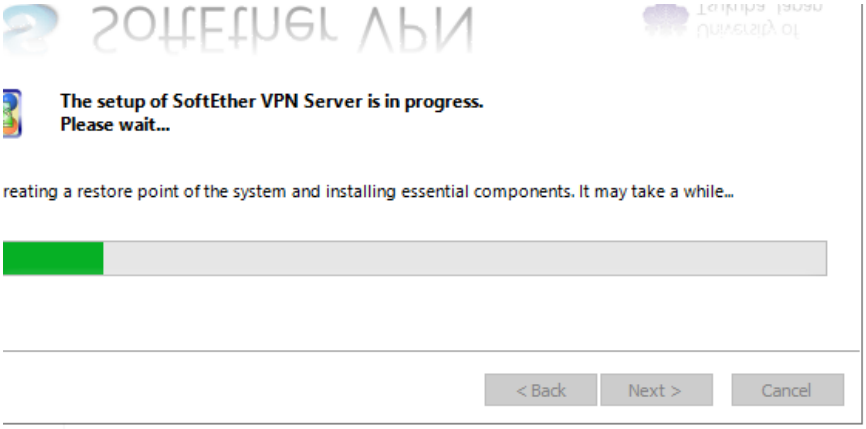
5. Run the downloaded installer as Administrator



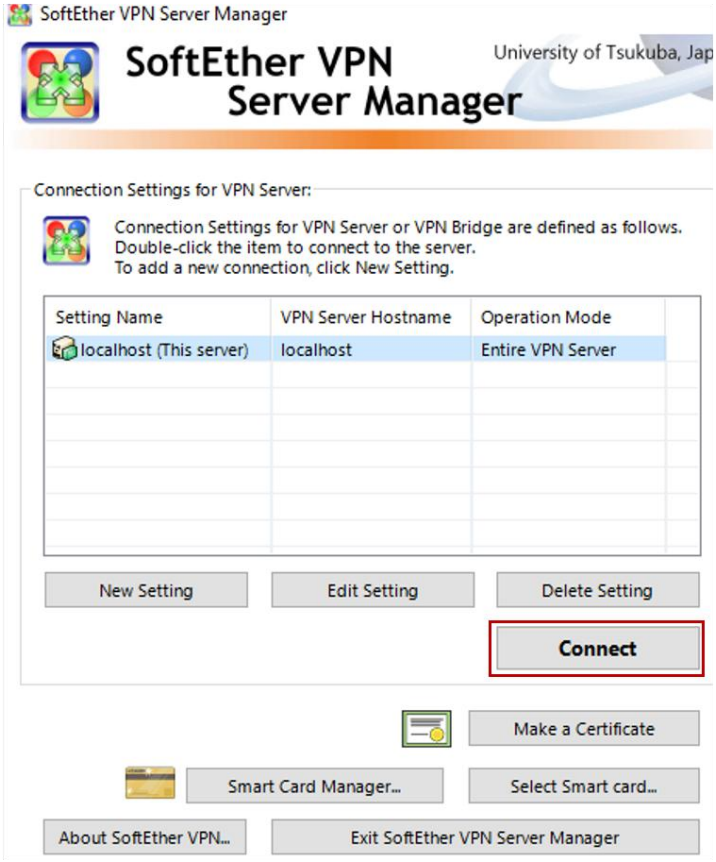
6. Component to install: **SoftEther VPN Server** → Next



7. Accept the license → Next → Next
8. Click **Install** → Finish



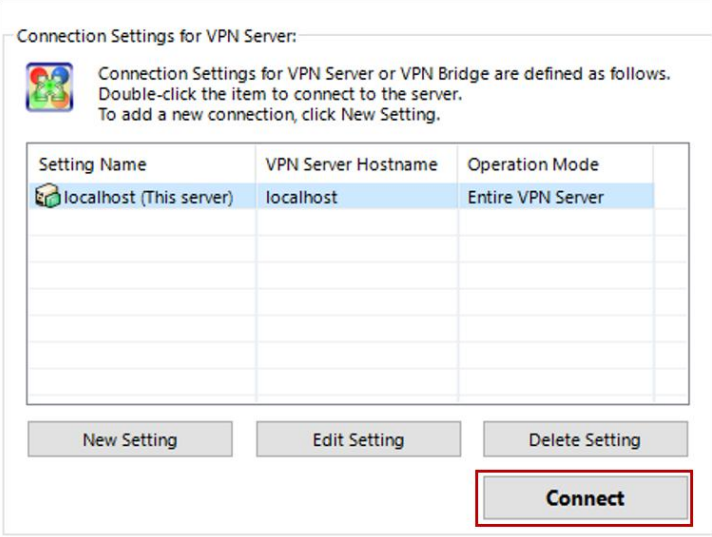
9. The SoftEther VPN Server Manager opens automatically



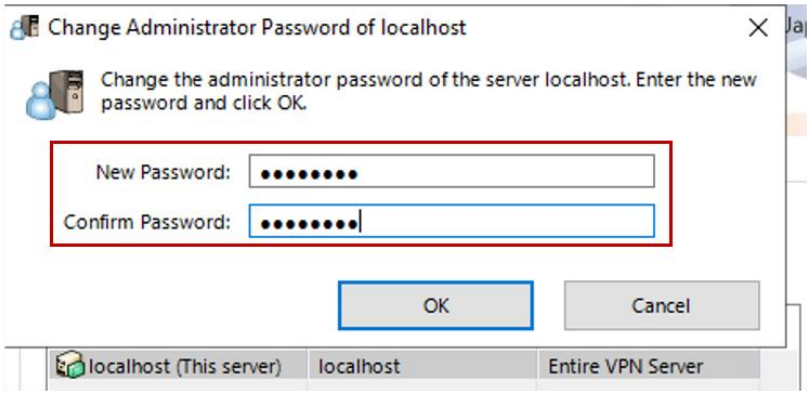
12.2 CONFIGURE VPN SERVER

When SoftEther VPN Server Manager opens on VPN01:

1. The server localhost is already listed — click it → click **Connect**



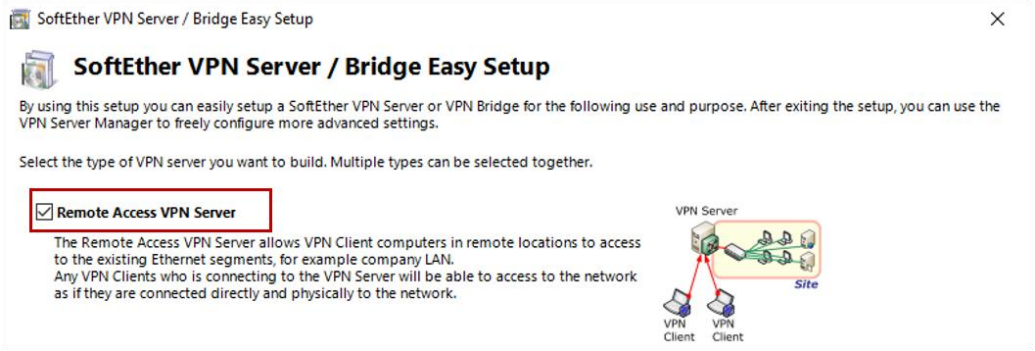
2. First time connecting: it asks you to set an admin password



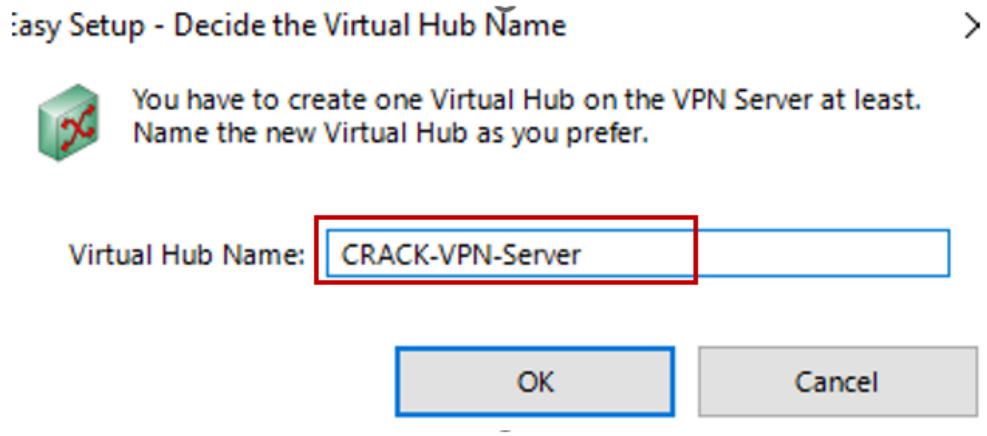
3. Set password: Admin123 → OK

12.3 CREATE A VIRTUAL HUB

1. Soft Ether VPN server page will appear. Select Remote Access VPN server



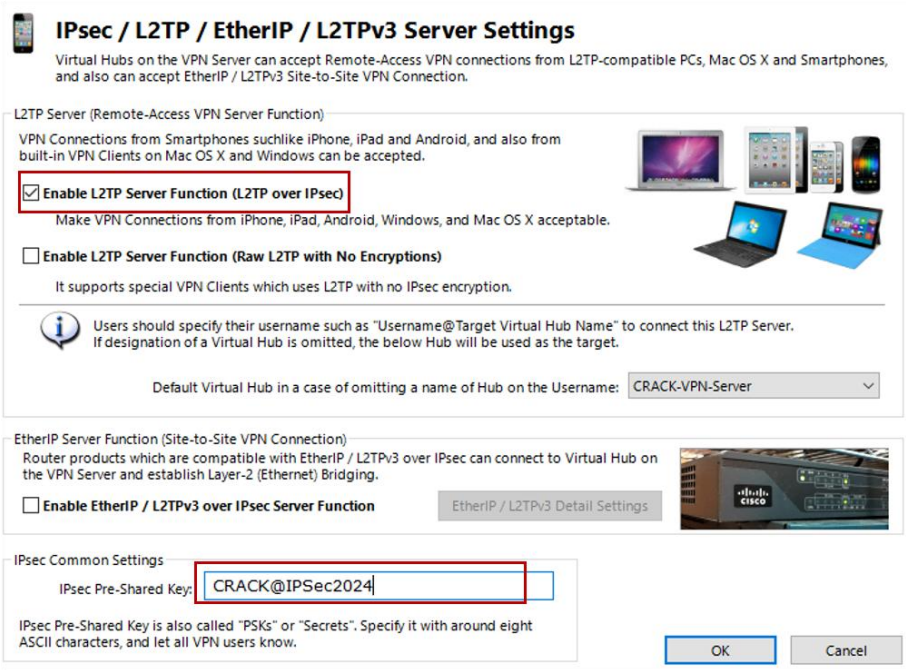
2. Virtual Hub Name: type CORP-VPN



3. For Dynamic DNS Simply click **Exit**

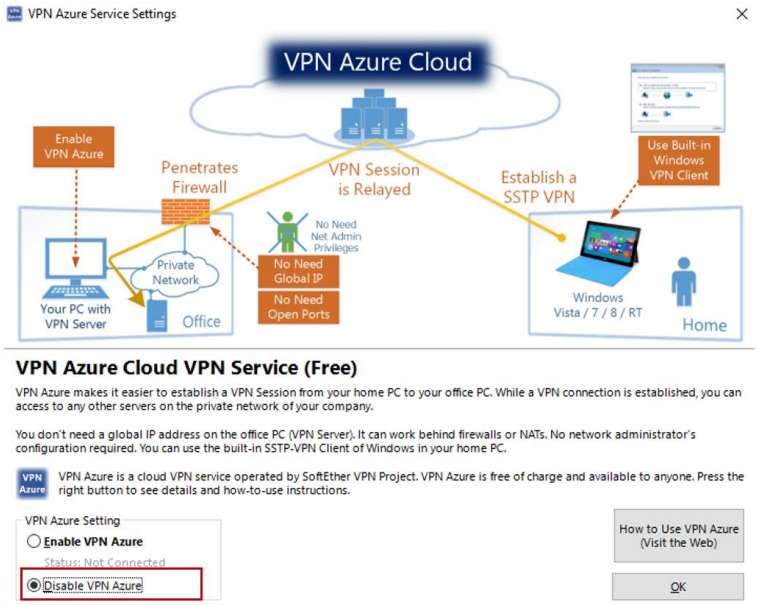
12.4 ENABLE L2TP/IPSEC

1. Check: **"Enable L2TP Server Function (L2TP over IPsec)"**
2. IPsec Pre-Shared Key: type Corp@IPSec2024
3. Click OK



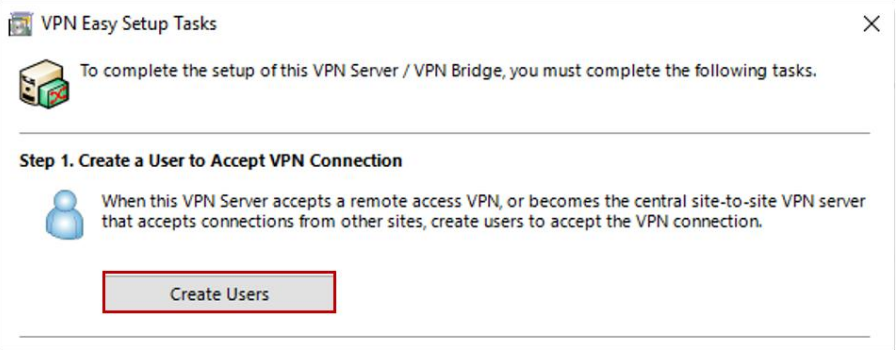
12.5 VPN Azure Cloud

Select **"Disable VPN Azure"** → Click **OK**

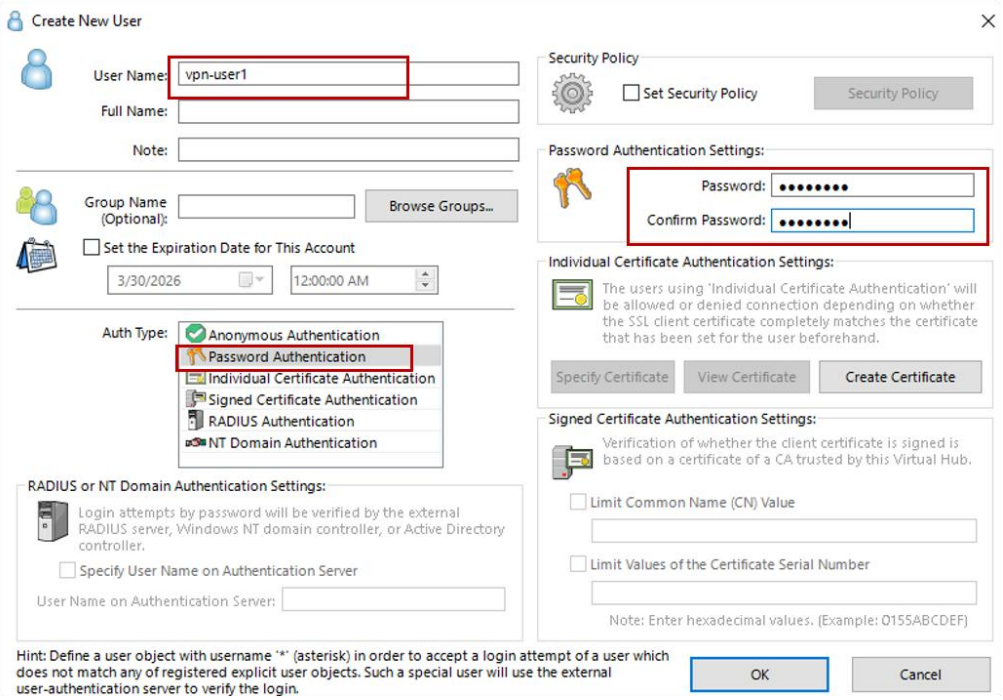


12.6 CREATE VPN USERS

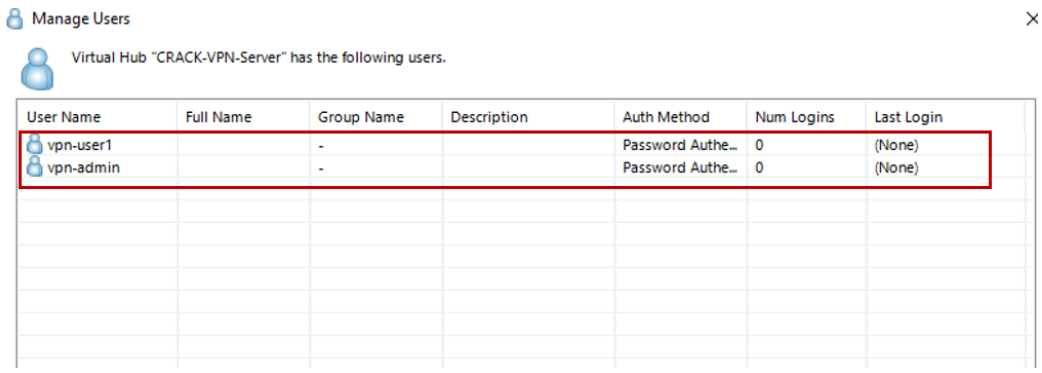
1. Click **"Create Users"** button



2. Username: vpn-user1
3. Auth Type: **Password Authentication**
4. Password: Admin123 → Confirm: Admin123 → Click OK
5. Click **OK**

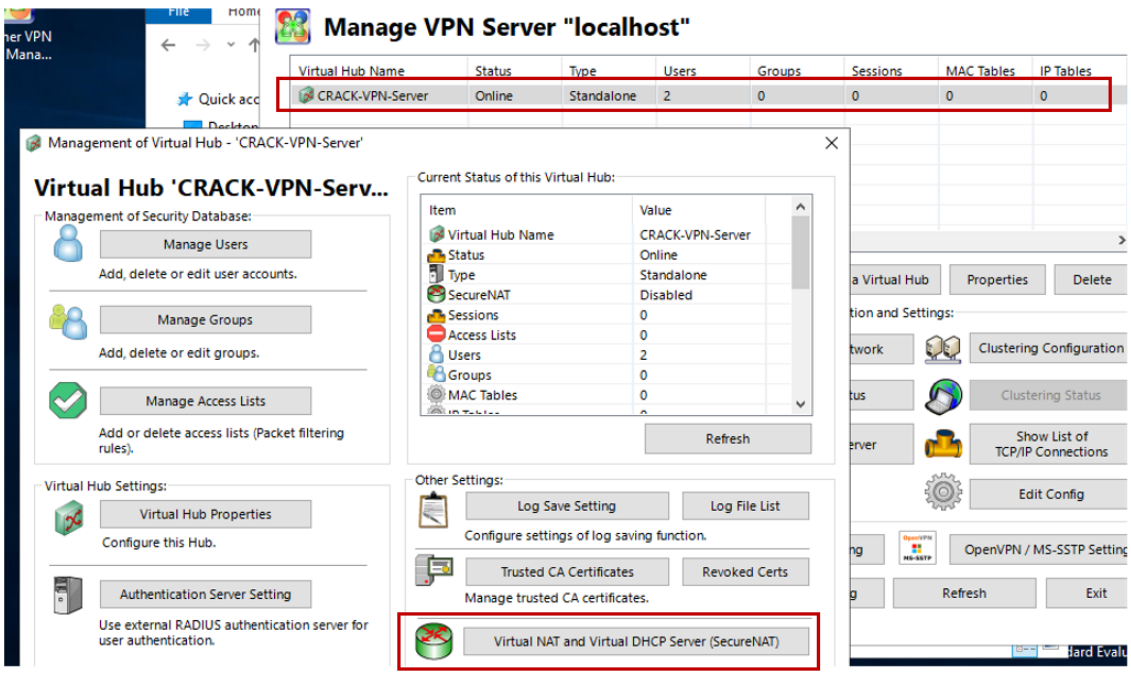


6. Click **New** again & repeat the same process for vpn-admin
7. Username: vpnadmin → Password: Admin123 → OK
8. Close the user management window

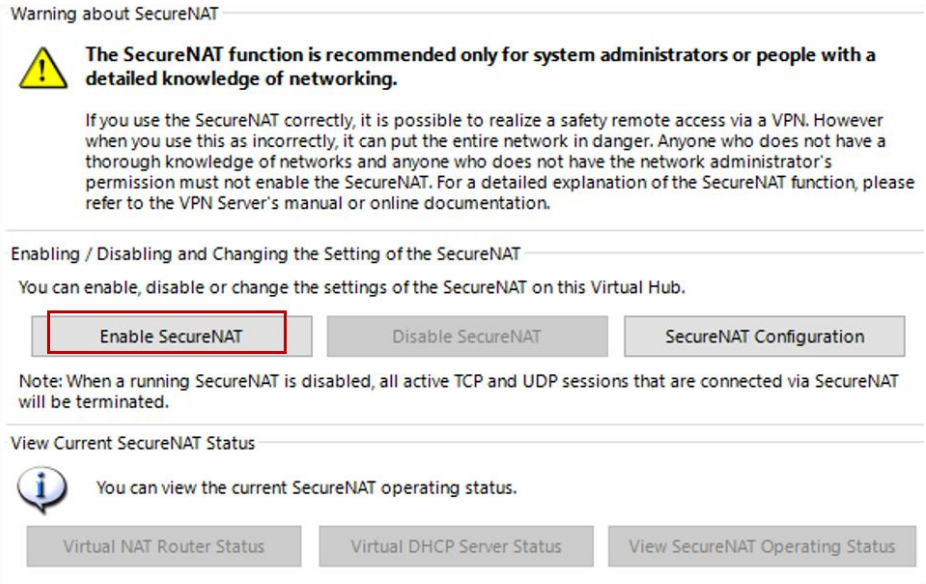


12.7 ENABLE SECURENT (DHCP FOR VPN CLIENTS)

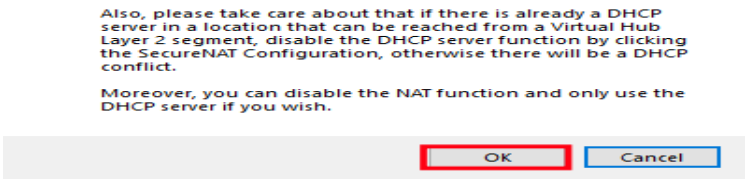
1. Double click CORP VPN Hub
2. Click **"Virtual NAT and Virtual DHCP Server (SecureNAT)"**



3. Click **"Enable SecureNAT"**



4. A confirmation appears → click OK

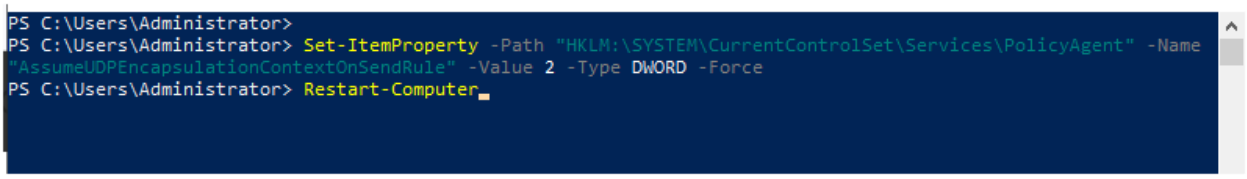


5. Click Exit

On VPN01 PowerShell as Administrator:

```
Set-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Services\PolicyAgent"
-Name "AssumeUDPEncapsulationContextOnSendRule" -Value 2 -Type DWORD -Force
```

Now Restart VPN01



After restarting Turn on the VPN

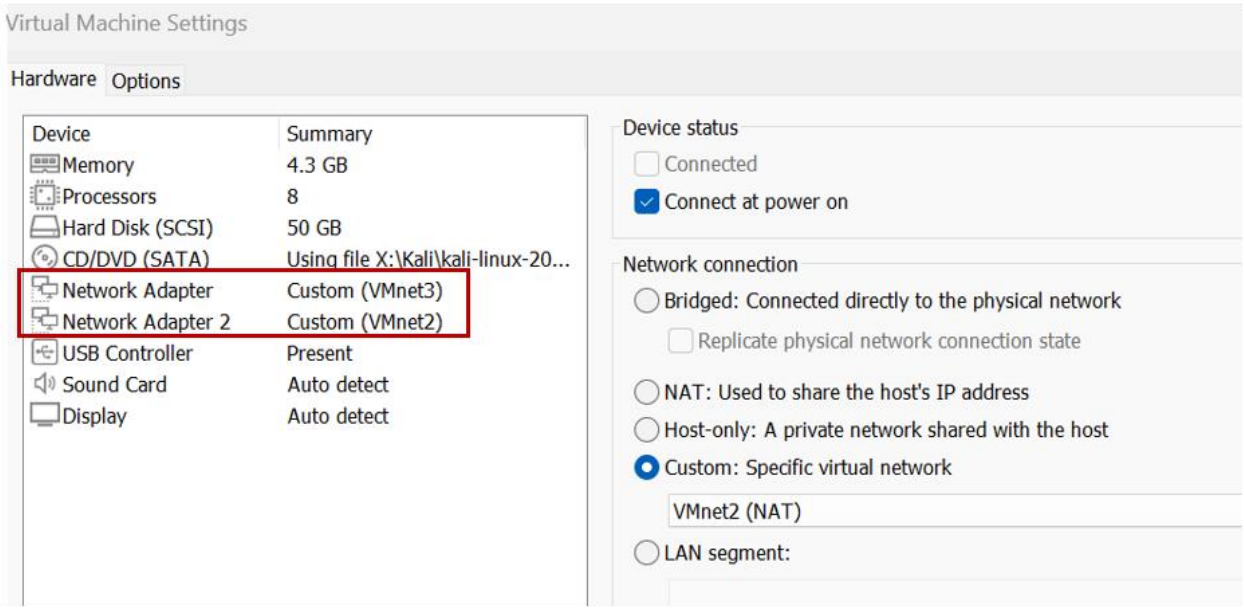


VPN server is now ready.

13. INTRUSION DETECTION SYSTEM (Suricata on IDS01)

Open Kali Linux Virtual Machine

For installing Suricata, we need internet access. We will add another adapter and set it to NAT to get internet access for our Kali Linux Virtual Machine.



13.1 INSTALL SURICATA

Type these commands one at a time, pressing Enter after each:

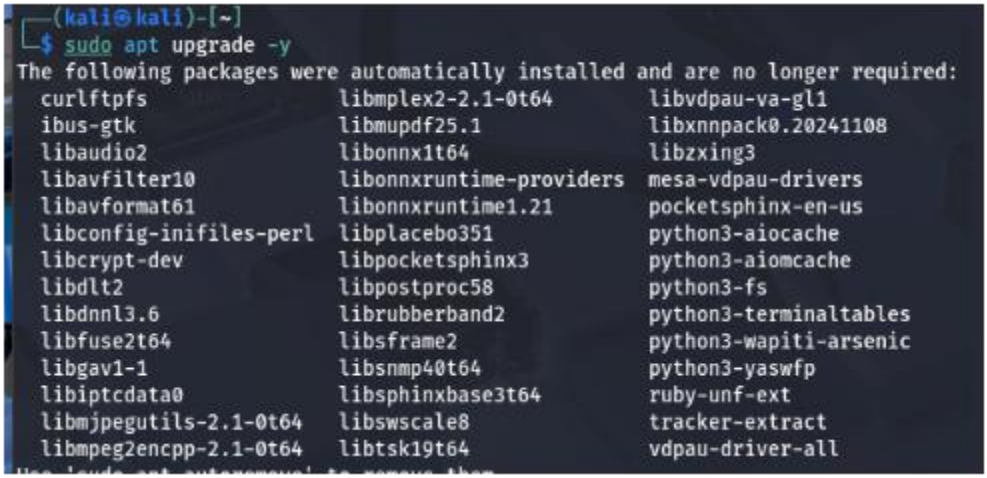
```
sudo apt update
```

Enter password when asked

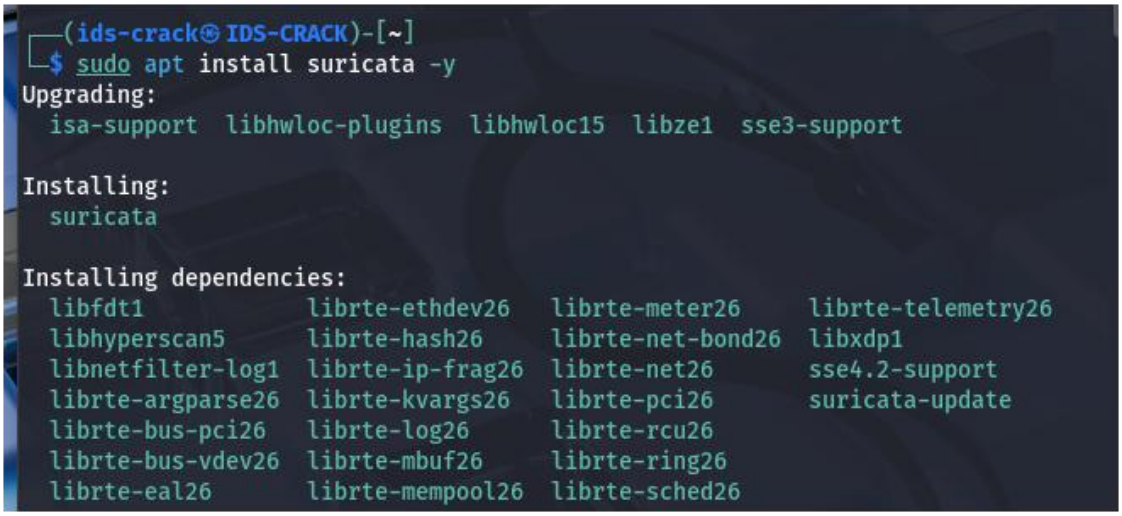


```
sudo apt upgrade -y
```

Wait — this may take a few minutes.



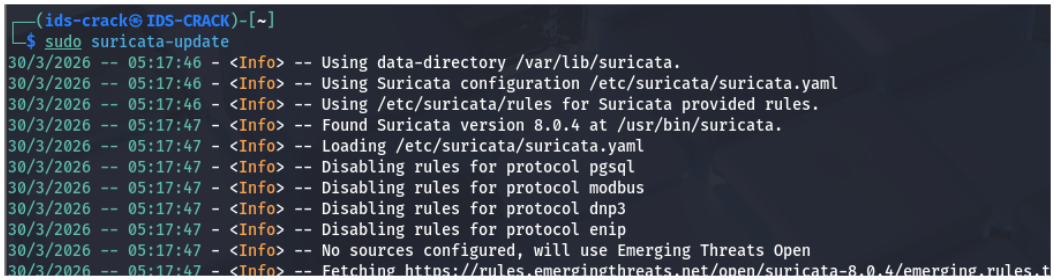

```
sudo apt install suricata -y
```



Wait for installation to complete.

13.2 UPDATE DETECTION RULES

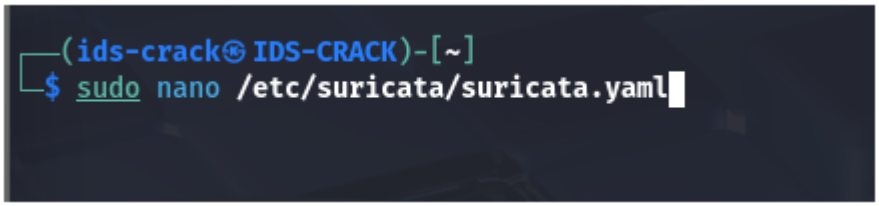
```
sudo suricata-update
```



13.3 CONFIGURE SURICATA

```
sudo nano /etc/suricata/suricata.yaml
```

This opens a very long file. You need to find and change specific lines.



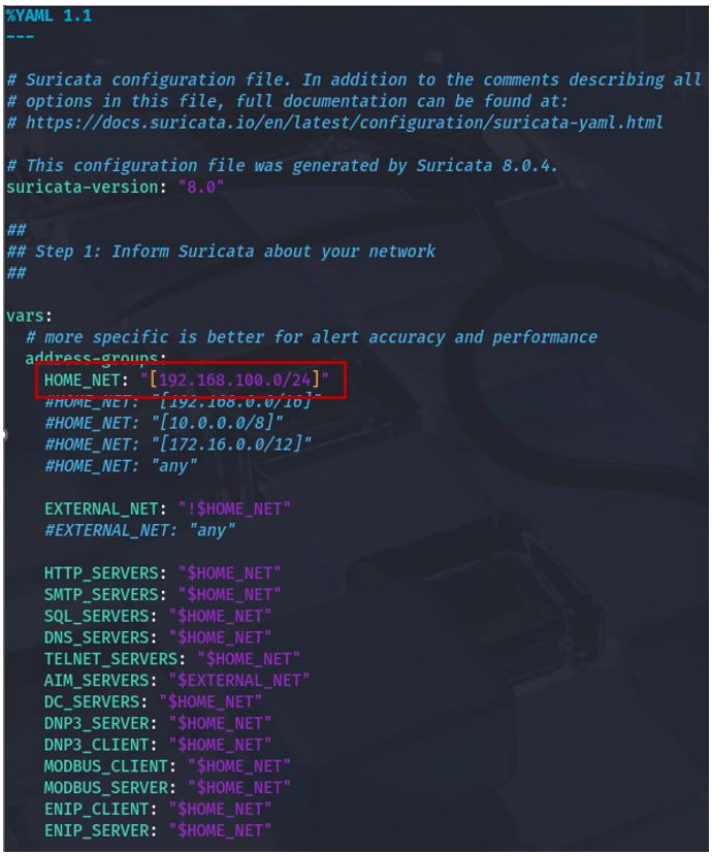
Set your network:

Press Ctrl+W (search) → type `HOME_NET` → press Enter

You will see a line like: HOME_NET: "[192.168.0.0/16,10.0.0.0/8,172.16.0.0/12]"

Change it to:

- HOME_NET: "[Your Host Only Network IP address]"

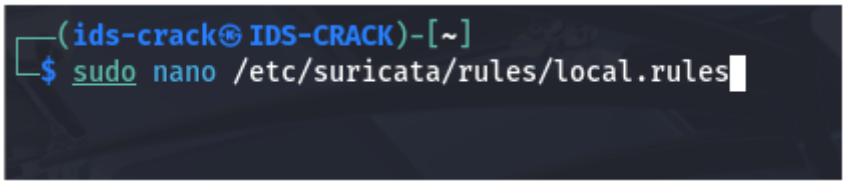


If you are using Kali Linux Virtual Machine then you do not need to change the interface. It is by default set to eth0. If your interface is not eth0 then change it.

Save and exit: **Ctrl+S** and then Ctrl+**X.**

13.4 CREATE YOUR CUSTOM DETECTION RULES

```
sudo nano /etc/suricata/rules/local.rules
```

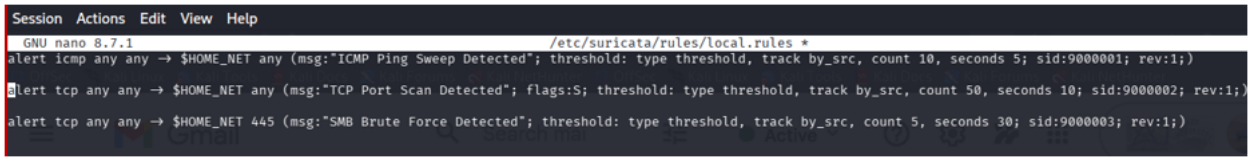


This creates a new empty file. Type or paste these rules:

```
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Sweep Detected";
threshold: type threshold, track by_src, count 10, seconds 5; sid:9000001;
rev:1;)

alert tcp any any -> $HOME_NET any (msg:"TCP Port Scan Detected"; flags:S;
threshold: type threshold, track by_src, count 50, seconds 10; sid:9000002;
rev:1;)

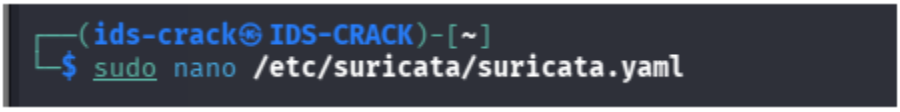
alert tcp any any -> $HOME_NET 445 (msg:"SMB Brute Force Detected";
threshold: type threshold, track by_src, count 5, seconds 30; sid:9000003;
rev:1;)
```



Save and exit: **Ctrl+S** and then Ctrl+**X.**

13.5 ADD LOCAL RULES TO SURICATA CONFIG

```
sudo nano /etc/suricata/suricata.yaml
```



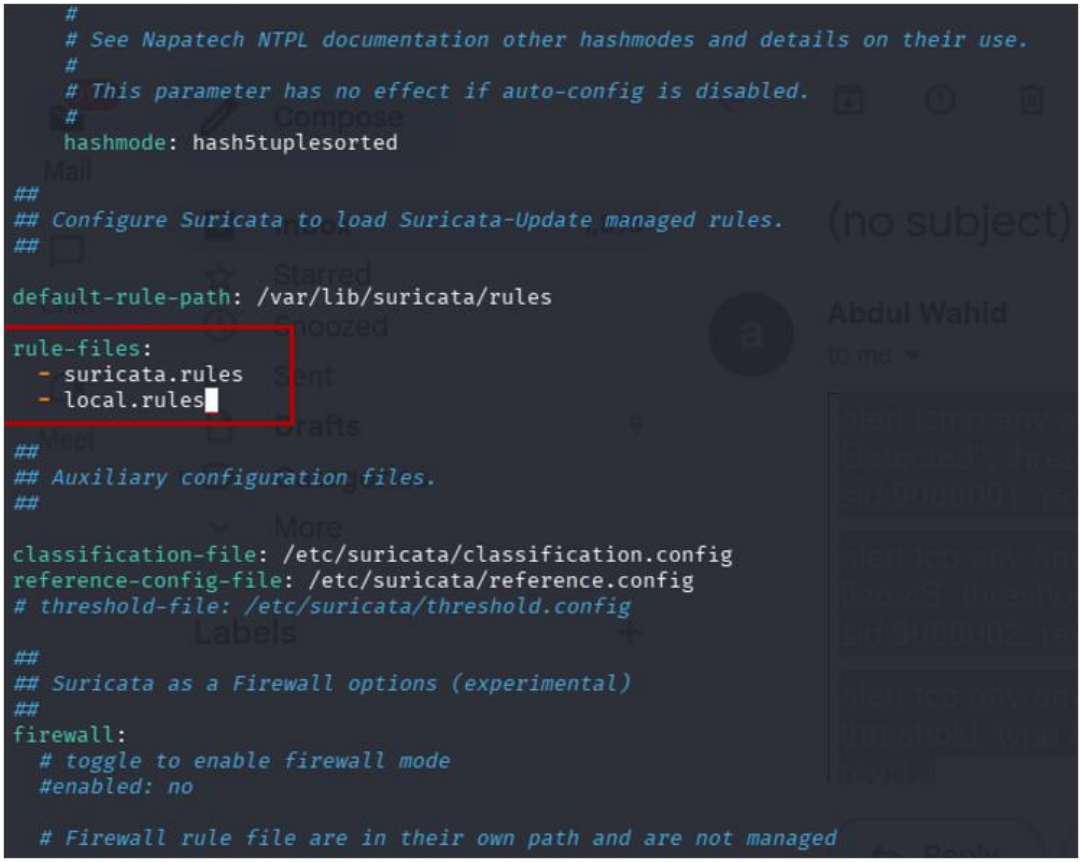
Press Ctrl+W → type rule-files → Enter.

You will see a section like:

rule-files:

- suricata.rules

Add your local rules file under it:

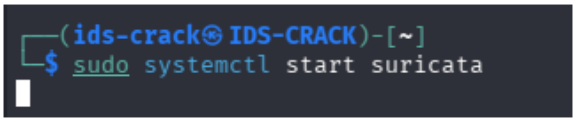


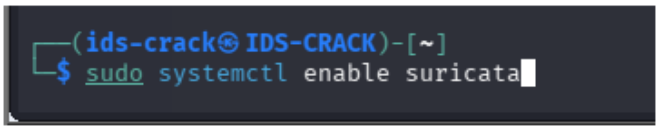
Save and exit: **Ctrl+S** and then Ctrl+**X.**

13.6 START SURICATA

```
sudo systemctl start suricata

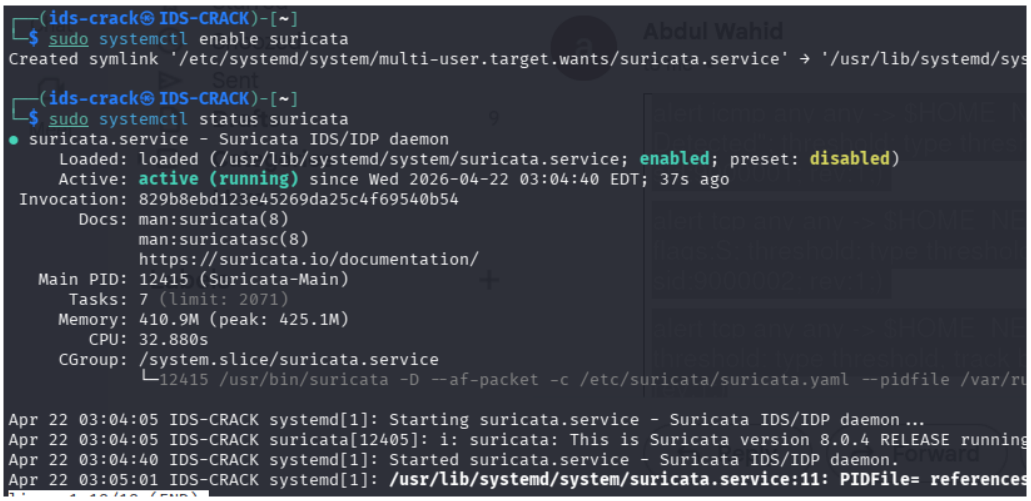
sudo systemctl enable suricata
```





Check it is running:

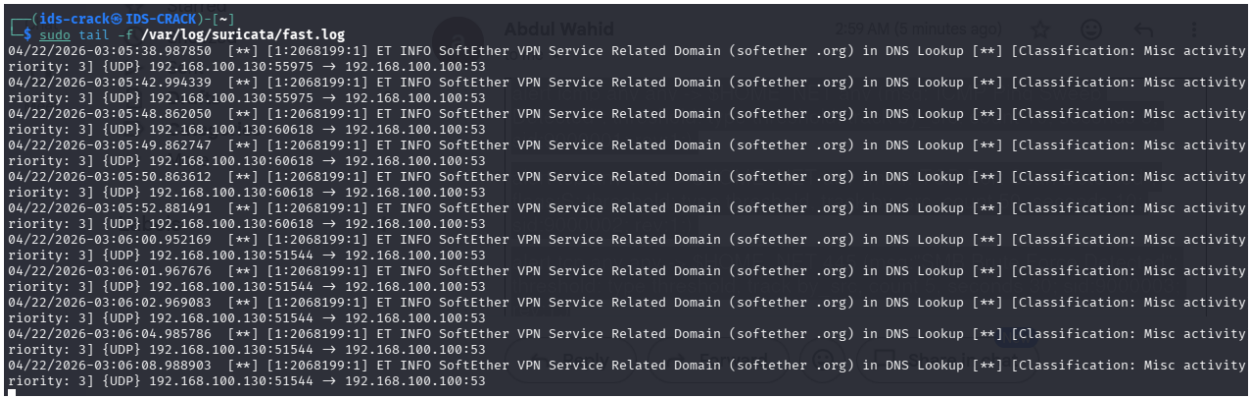
```
sudo systemctl status suricata
```



Press Q to exit the status view.

13.7 OPEN THE LIVE ALERT MONITOR

```
sudo tail -f /var/log/suricata/fast.log
```

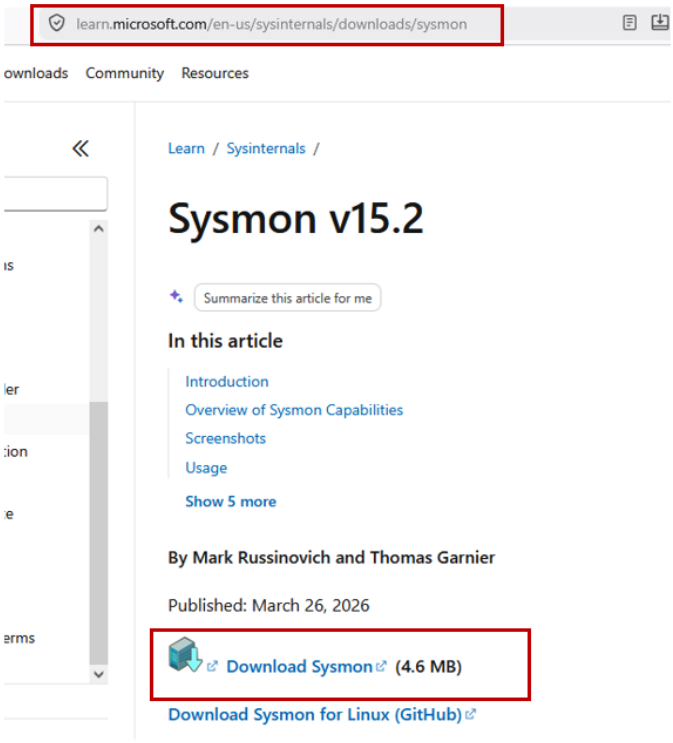


Leave this running. You will come back to this window during the attack phase to watch alerts appear live.

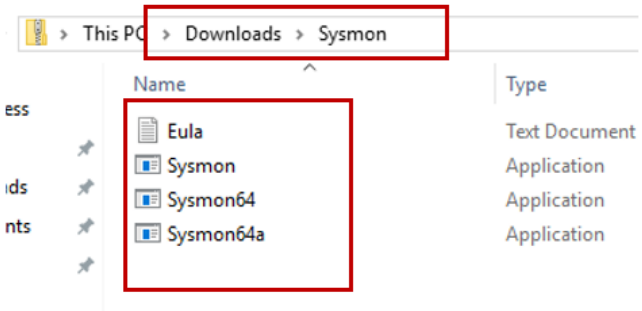
14. SYSMON (ENHANCED LOGGING ON DC01 AND ADMIN01)

14.1 Sysmon Installation on DC01

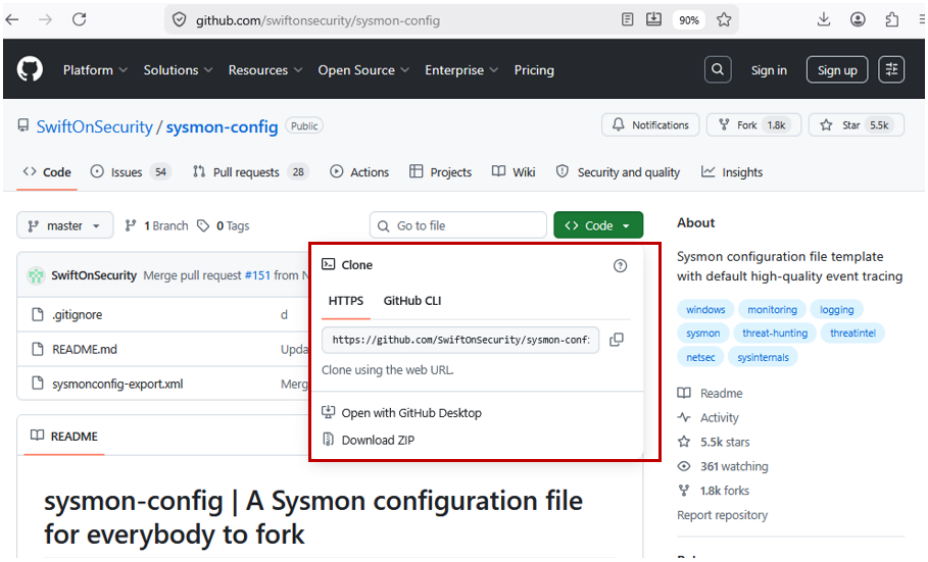
1. On DC01, open **Edge browser**
2. Search Microsoft Sysmon Download



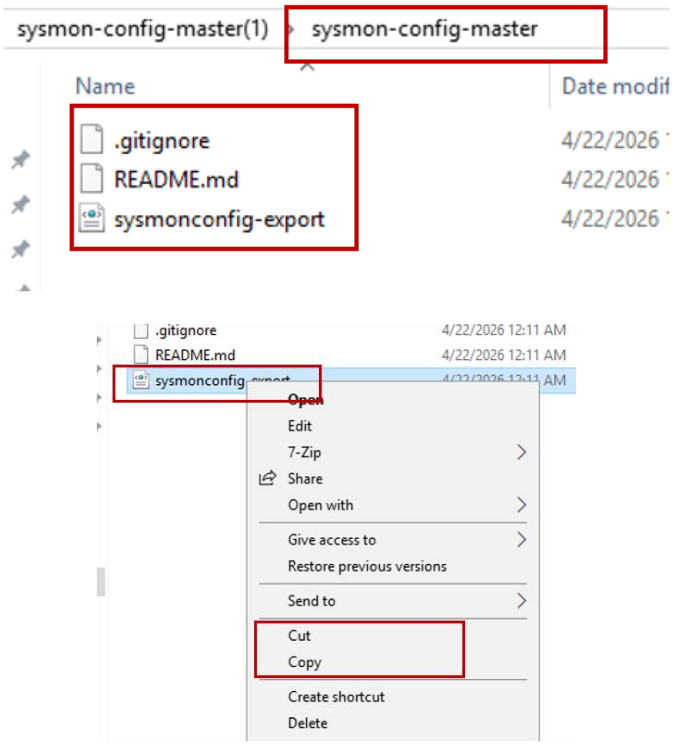
3. Click **Download Sysmon** → save the ZIP
4. Extract the ZIP to C:\Tools\Sysmon\



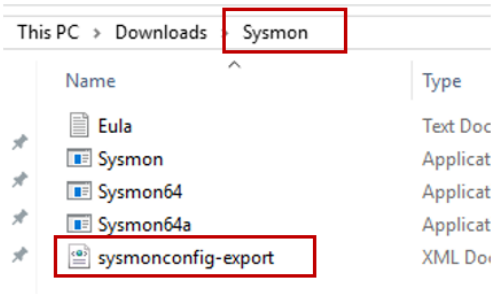
5. Download the configuration file:
    - In Edge, go to: raw.githubusercontent.com/SwiftOnSecurity/sysmon-config
    - Click on Download Zip



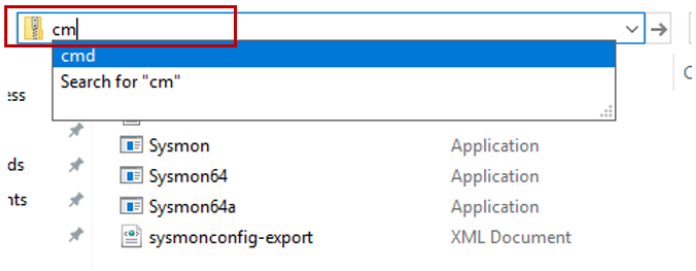
6. Extract the downloaded sysmon config zip file and copy the file sysmonconfig-export and paste it with the sysmon installation file



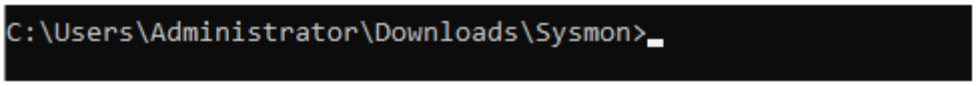
Save the file with .xml in the same folder where sysmon.exe is downloaded

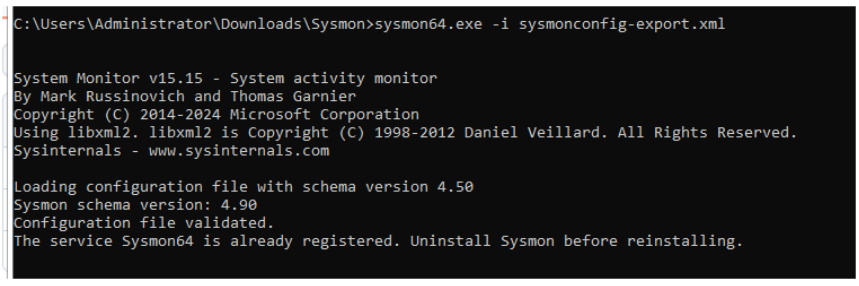


7. Stay in the directory where sysmon.exe file is present and enter cmd on the file location and press Enter



8. Run the following command in the command prompt
    - sysmon64.exe -i sysmonconfig-export.xml





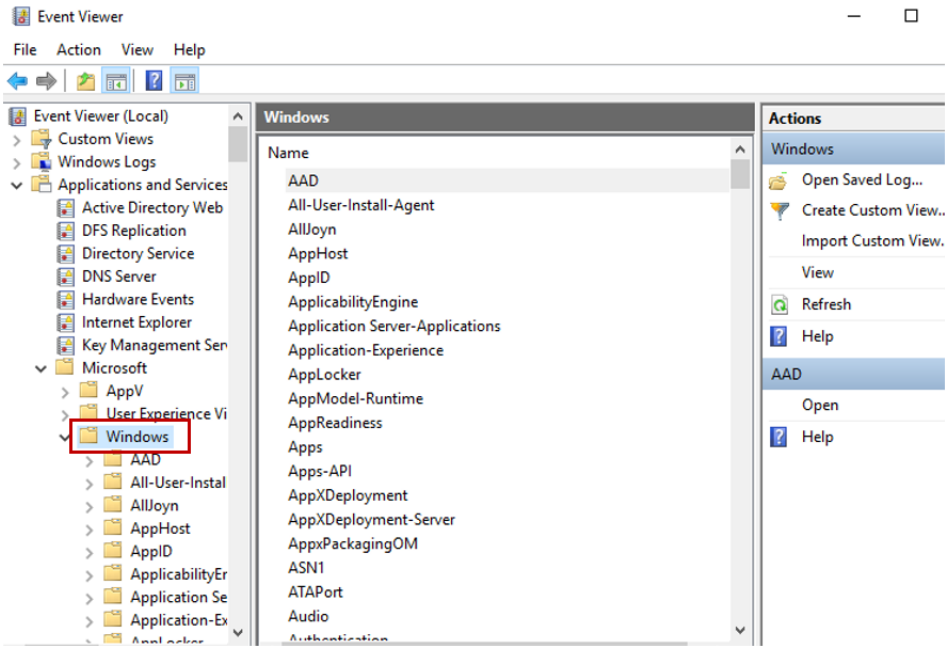
9. If the execution is successful, you will see this in your terminal.

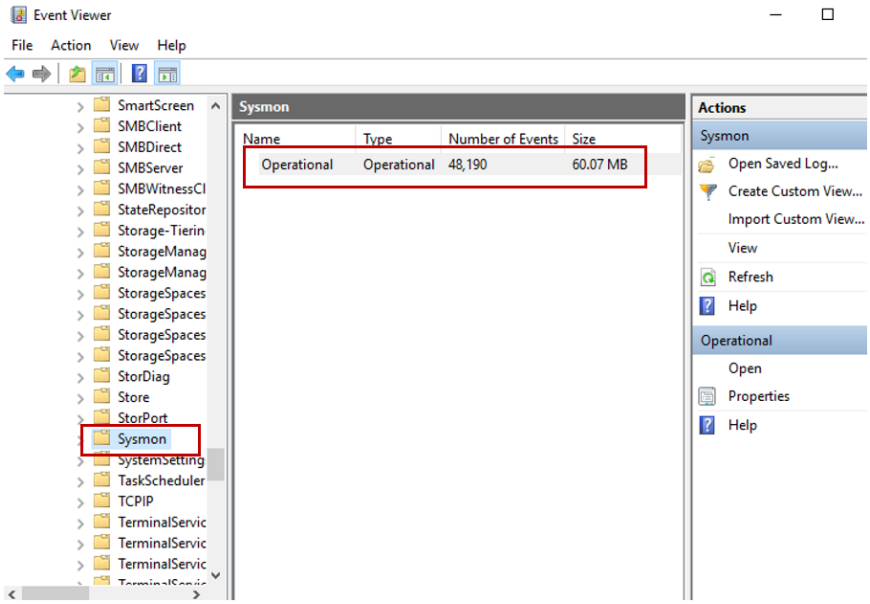
    Successful installation. Sysmon does not require a reboot.

14.2 Confirm Sysmon Logs in Event Viewer

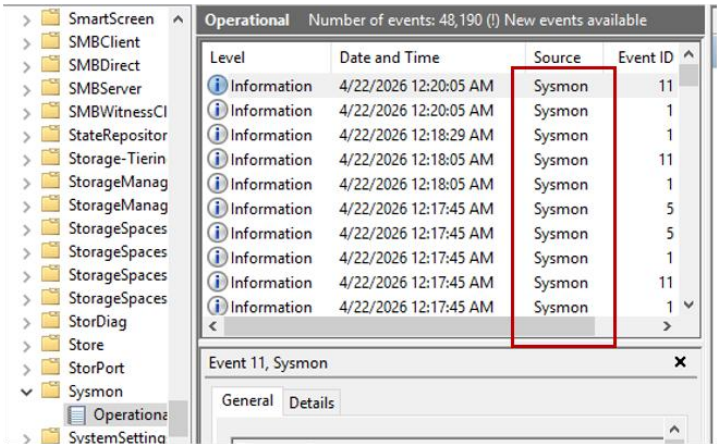
Open Event Viewer

- Go to Application and Services
- Expand Microsoft > Windows > System > Operational



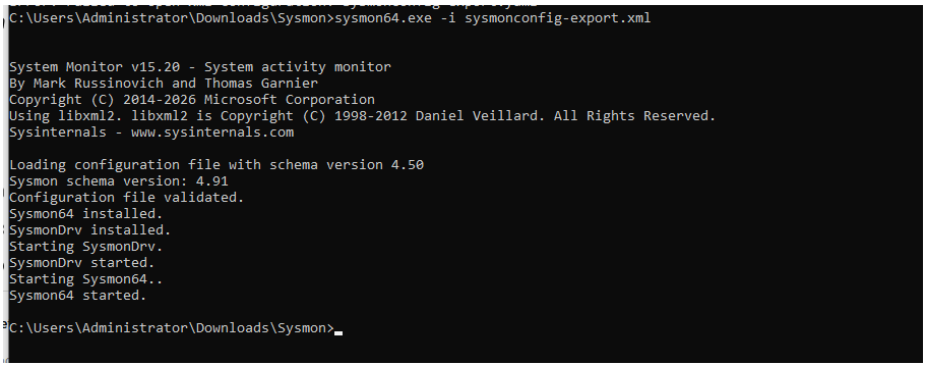


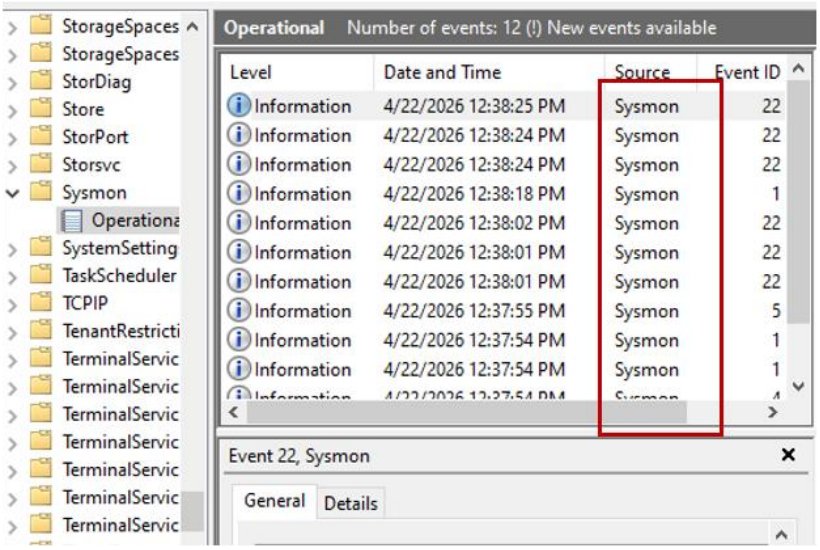
Sysmon in Event Viewer



ON ADMIN01

Repeat the exact same steps — download Sysmon and the config to C:\Downoads\Sysmon\ and run the same install command.



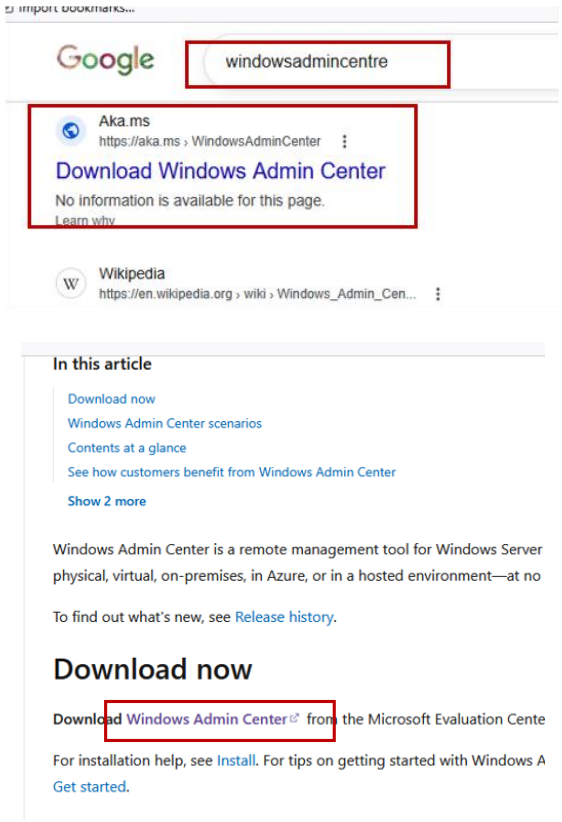


15. WINDOWS ADMIN CENTER

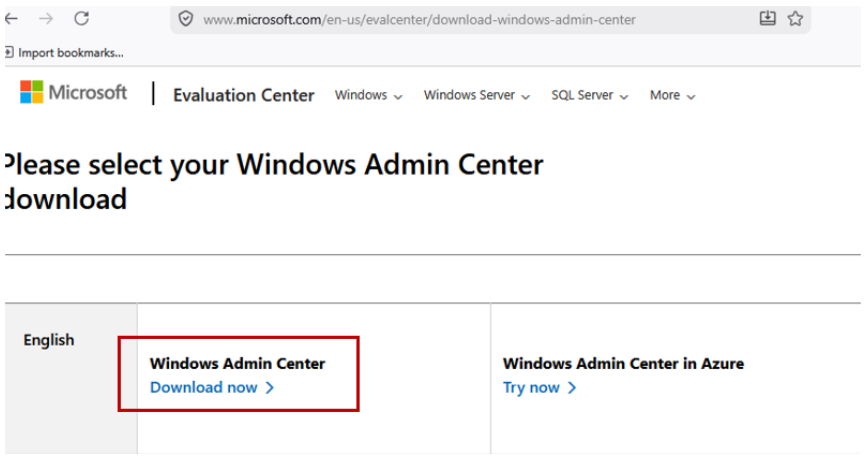
INSTALL ON ADMIN01

Go to **ADMIN01** VM.

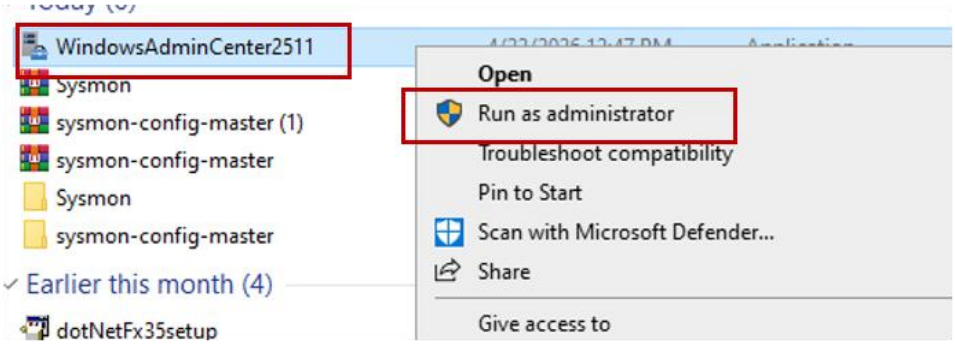
1. Open Edge browser
2. Go to: aka.ms/windowsadmincenter

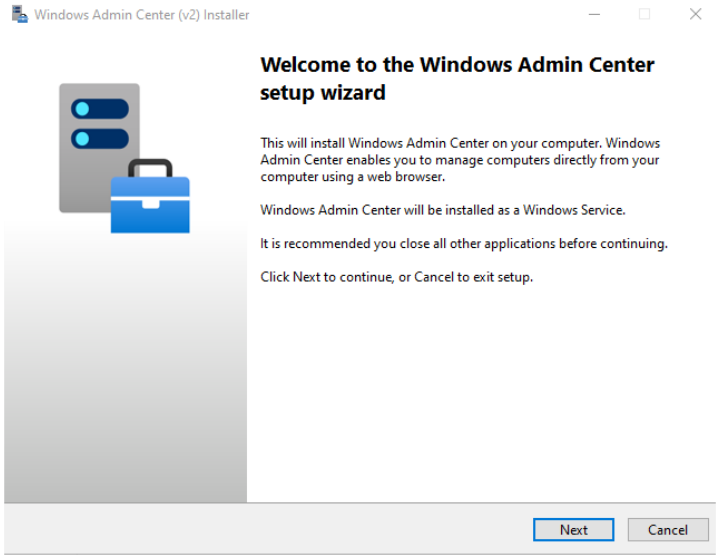


3. Download the .msi file

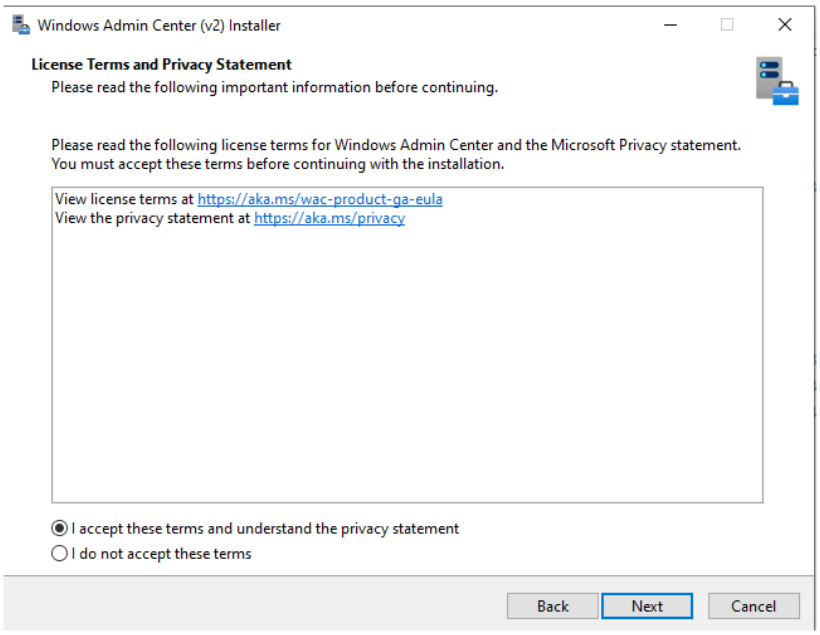


4. Right-click the downloaded file → **Run as administrator**

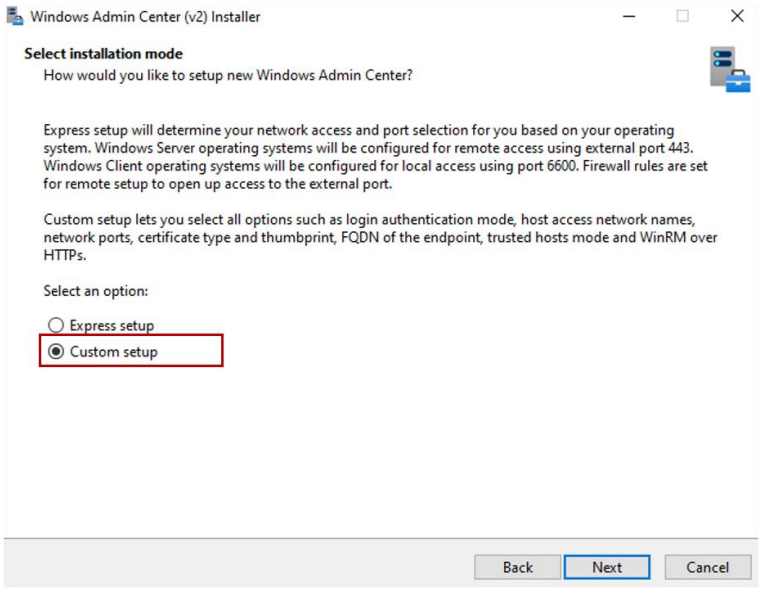




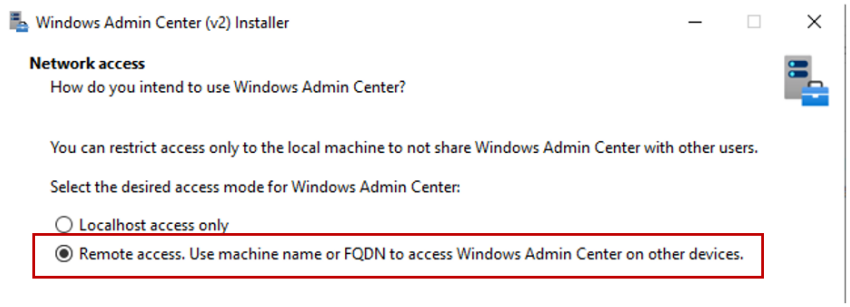
5. Accept the license → Next



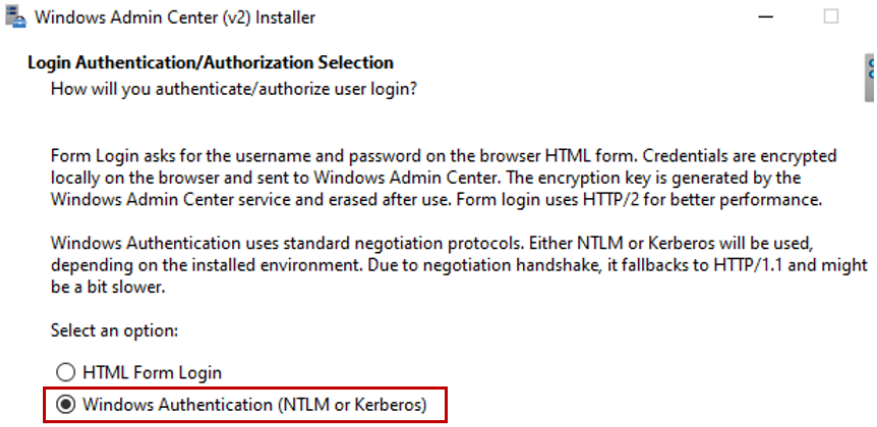
6. Select **Custom** Setup



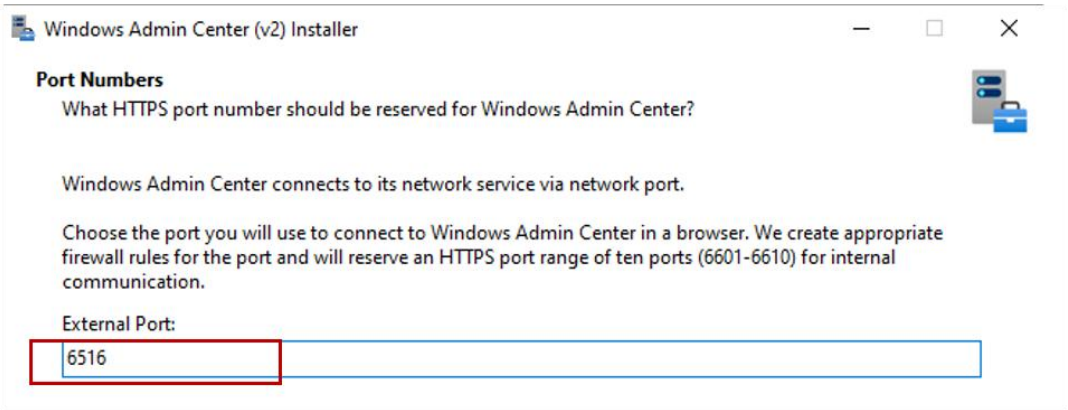
7. Set Network Access to Remote Access



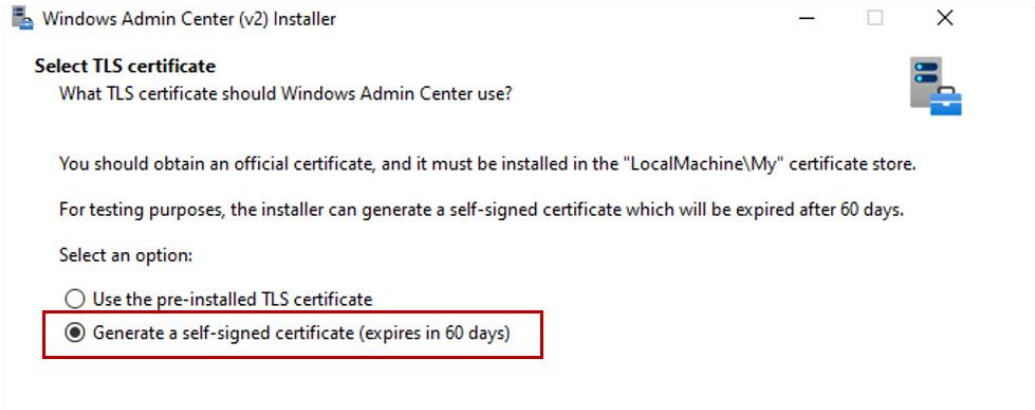
8. Select "Windows Authentication"



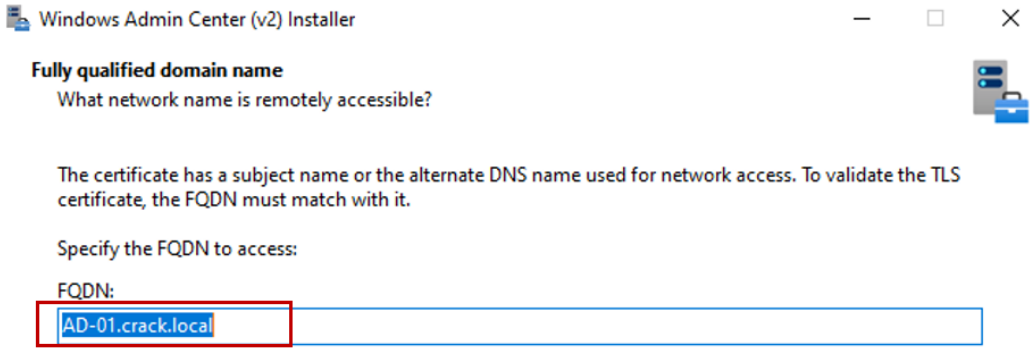
9. Port: 6516 → Next



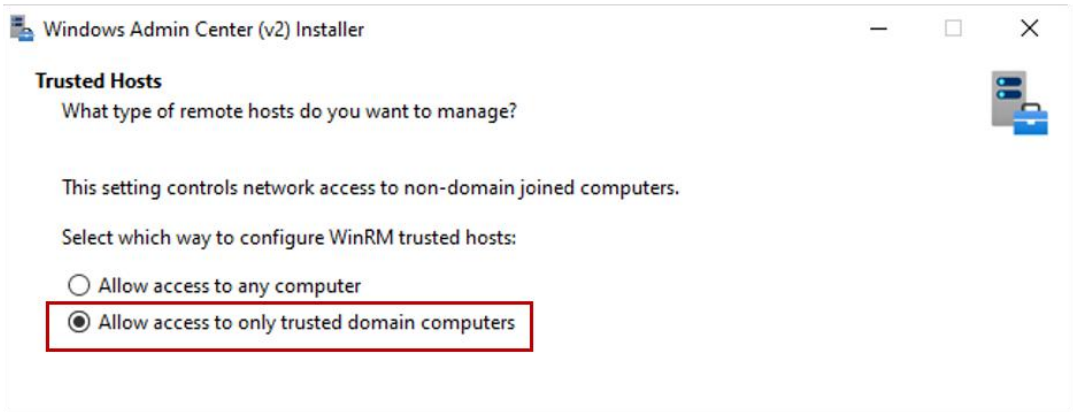
10. Select TLS Certificate



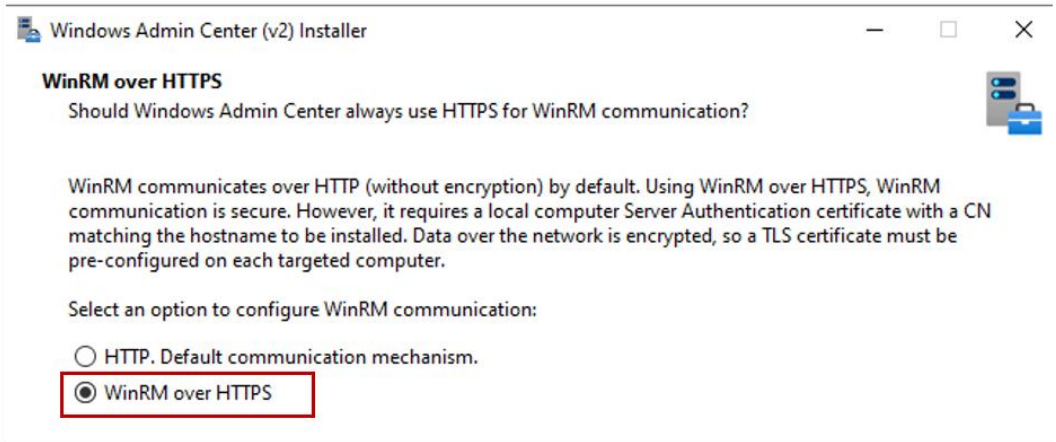
11. Fully Qualified Domain Name, Leave It As Is and Click Next



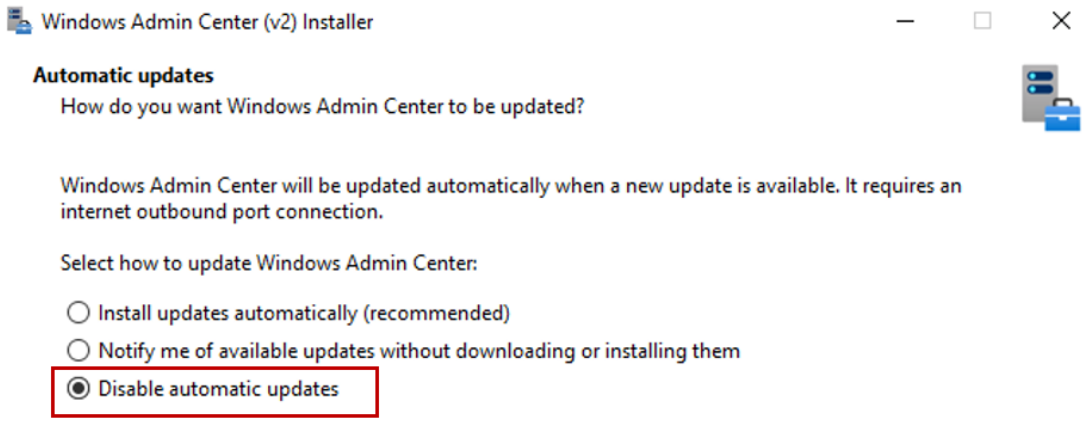
12. Select "Allow access to only trusted domain computers"

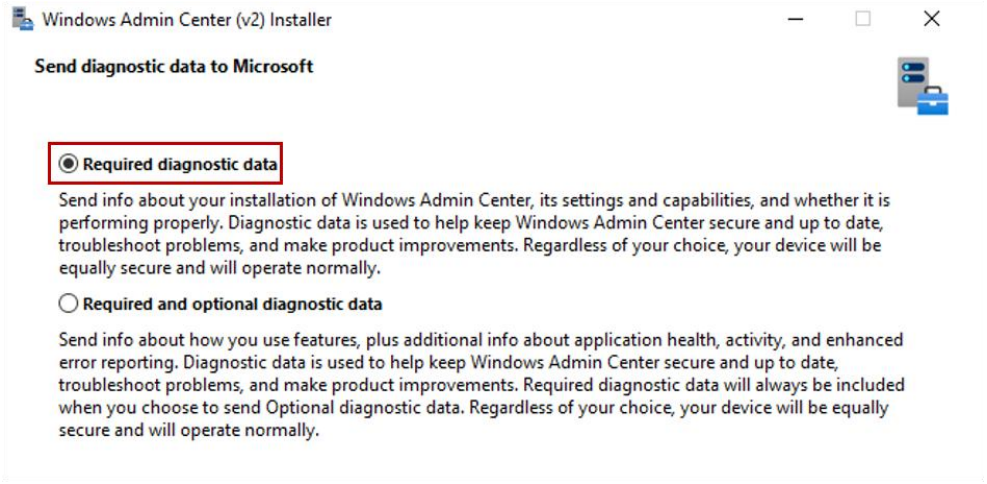


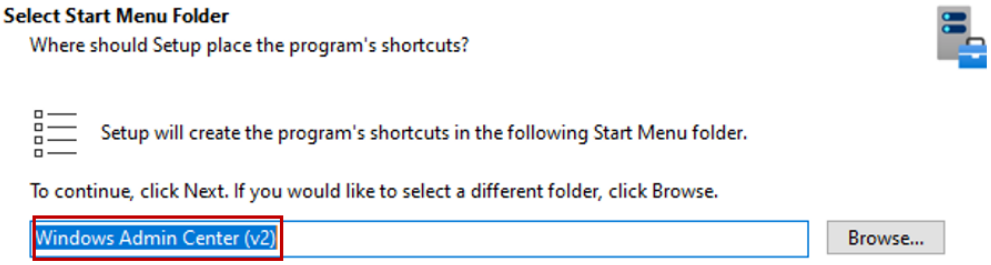
13. Select: **Use WinRM over HTTPS only** → Next



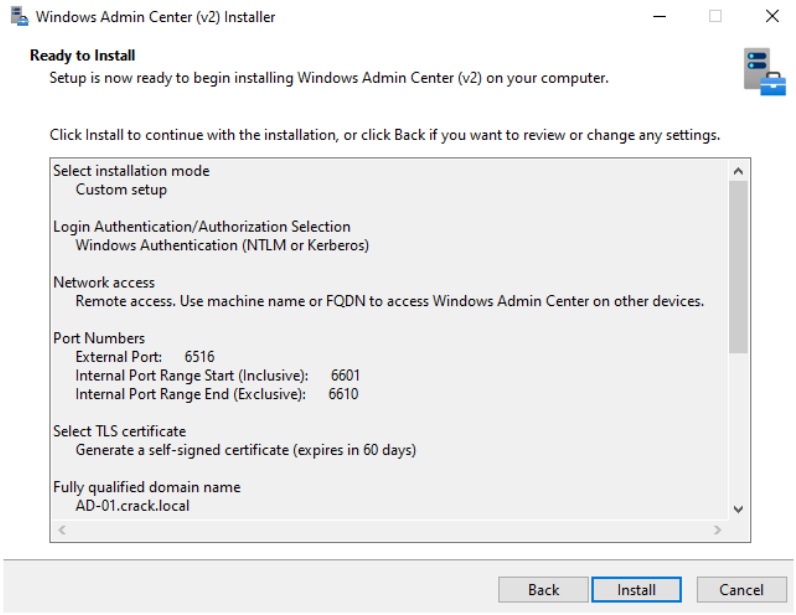
14. Select "Disable automatic updates"







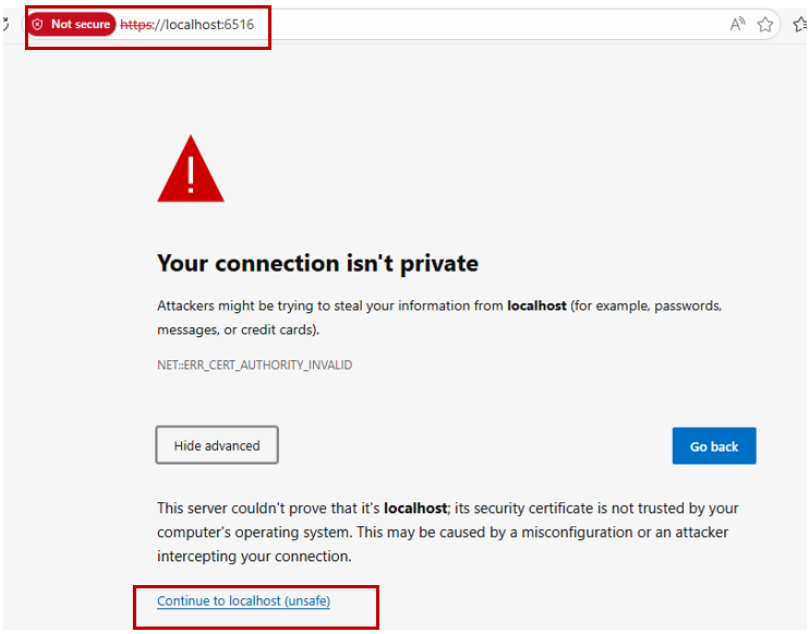
15. Click **Install** — takes about 3 minutes

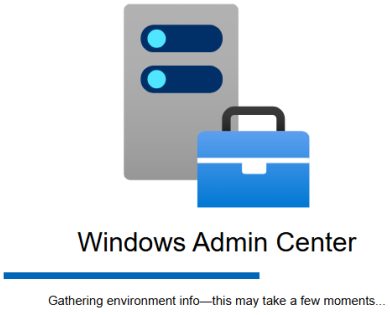


16. Click Finish

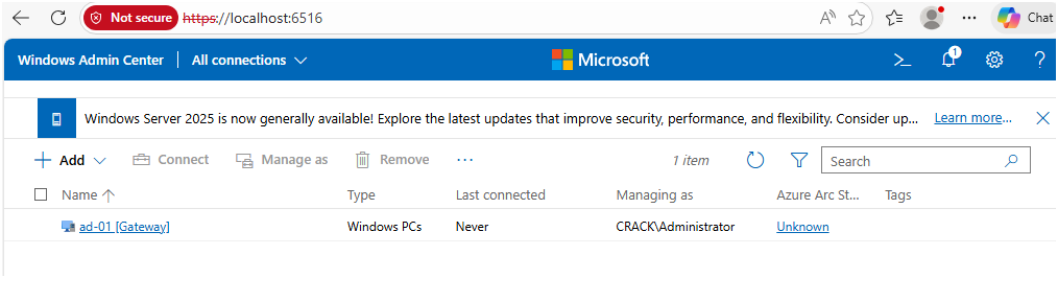
16. ACCESS WINDOWS ADMIN CENTER

1. Open Browser on ADMIN01
2. Go to: https://localhost:6516
3. A certificate warning appears — click **Advanced** → **Continue to localhost**





4. You reach the Windows Admin Center dashboard



17. ADD DC01 AS A MANAGED SERVER

It will not generate connections; we will have to enter these commands in both the virtual machines to add DC01 in Windows Admin Center.

17.1 On DC01 PowerShell:

```
# 1. Allow unencrypted WinRM

winrm set winrm/config/service '@{AllowUnencrypted="true"}'

# 2. Set trusted hosts

winrm set winrm/config/client '@{TrustedHosts="*"}'

# 3. Add WinRM firewall rules

netsh advfirewall firewall add rule name="WinRM Port 5985" dir=in
action=allow protocol=TCP localport=5985

netsh advfirewall firewall add rule name="WinRM HTTPS 5986" dir=in
action=allow protocol=TCP localport=5986

# 4. Create certificate

$cert = New-SelfSignedCertificate -DnsName "DC01","DC01.crack.local","Enter
your DC01 IP Address" -CertStoreLocation "cert:\LocalMachine\My"

# 5. Create HTTPS listener

New-Item -Path WSMan:\localhost\Listener -Transport HTTPS -Address * -
CertificateThumbprint $cert.Thumbprint -Force

# 6. Export certificate

Export-Certificate -Cert $cert -FilePath "C:\dc01cert.cer"

# 7. Restart WinRM

Restart-Service WinRM -Force
```

```
PS C:\Users\Administrator> winrm set winrm/config/service '@{AllowUnencrypted="true"}'
Service
    RootSDDL = O:NSG:BAD:P(A;;GA;;;BA)(A;;GR;;;IU)S:P(AU;FA;GA;;;WD)(AU;SA;GXGW;;;WD)
    MaxConcurrentOperations = 4294967295
    MaxConcurrentOperationsPerUser = 1500
    EnumerationTimeoutms = 240000
    MaxConnections = 300
    MaxPacketRetrievalTimeSeconds = 120
    AllowUnencrypted = true
    Auth
        Basic = false
        Kerberos = true
        Negotiate = true
        Certificate = false
        CredSSP = false
        CbtHardeningLevel = Relaxed
    DefaultPorts
        HTTP = 5985
        HTTPS = 5986
    IPv4Filter = *
    IPv6Filter = *
    EnableCompatibilityHttpListener = false
    EnableCompatibilityHttpsListener = false
    CertificateThumbprint
    AllowRemoteAccess = true

PS C:\Users\Administrator> winrm set winrm/config/client '@{TrustedHosts="*"}'
Client
    NetworkDelayms = 5000
    URLPrefix = wsman
    AllowUnencrypted = false
    Auth
        Basic = true
        Digest = true
        Kerberos = true
        Negotiate = true
        Certificate = true
        CredSSP = false
    DefaultPorts
        HTTP = 5985
        HTTPS = 5986
    TrustedHosts = *
```

```
PS C:\Users\Administrator> netsh advfirewall firewall add rule name="WinRM Port 5985" dir=in action=allow protocol=TCP localport=5985
Ok.

PS C:\Users\Administrator> netsh advfirewall firewall add rule name="WinRM HTTPS 5986" dir=in action=allow protocol=TCP localport=5986
Ok.

PS C:\Users\Administrator> $cert = New-SelfSignedCertificate -DnsName "DC01","DC01.crack.local","192.168.100.100" -CertStoreLocation "cert:\Loc
alMachine\My"
```

```
PS C:\Users\Administrator> New-Item -Path WSMan:\localhost\Listener -Transport HTTPS -Address * -CertificateThumbprint $cert.Thumbprint -Force


   WSManConfig: Microsoft.WSMan.Management\WSMan::localhost\Listener

Type            Keys                                Name
----            ----                                ----
Container       {Transport=HTTPS, Address=*}        Listener_1305953032


PS C:\Users\Administrator>
PS C:\Users\Administrator> Export-Certificate -Cert $cert -FilePath "C:\dc01cert.cer"


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        4/22/2026   1:19 AM            818 dc01cert.cer


PS C:\Users\Administrator> Restart-Service WinRM -Force
WARNING: Waiting for service 'Windows Remote Management (WS-Management) (WinRM)' to stop...
WARNING: Waiting for service 'Windows Remote Management (WS-Management) (WinRM)' to stop...
PS C:\Users\Administrator>
```

17.2 On ADMIN01 PowerShell:

```
# 1. Allow unencrypted

winrm set winrm/config/client '@{AllowUnencrypted="true"}'

# 2. Trust all hosts

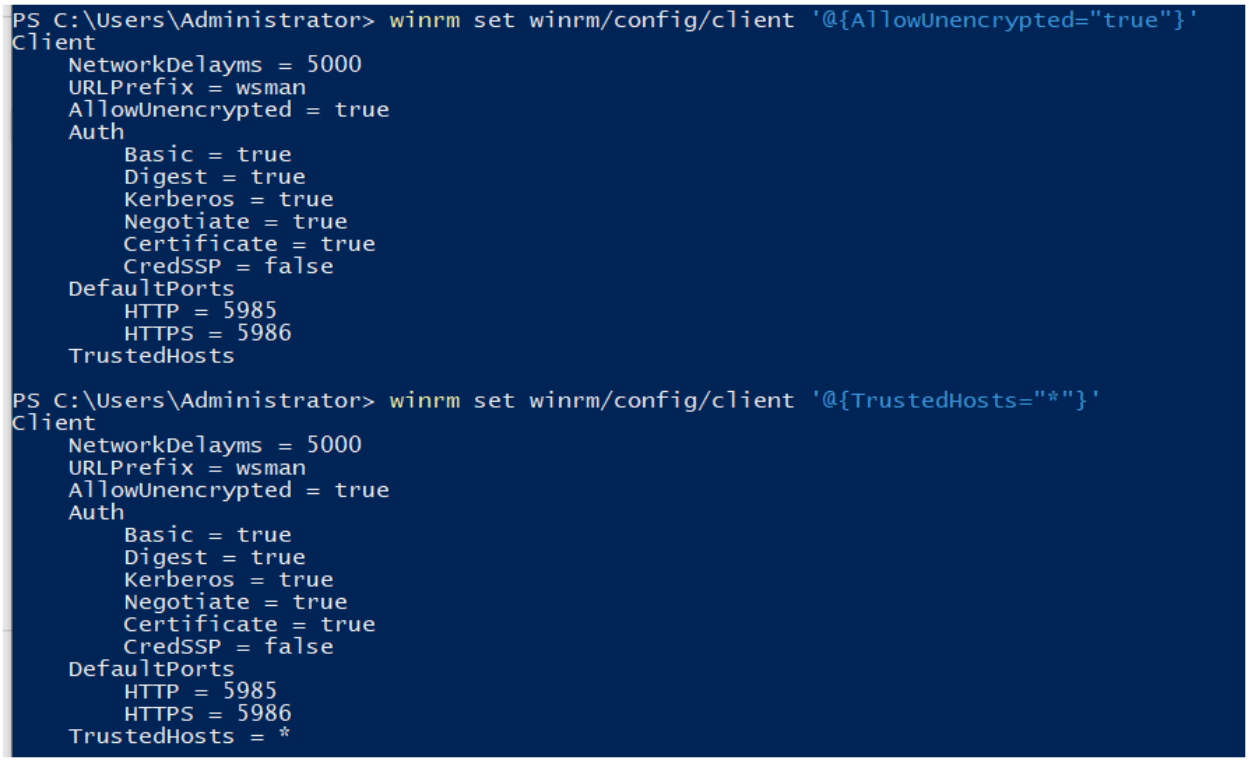
winrm set winrm/config/client '@{TrustedHosts="*"}'

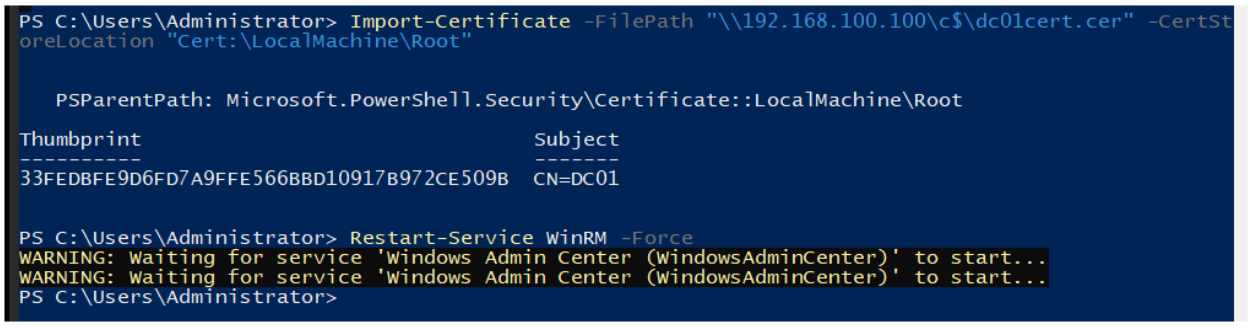
# 3. Import DC01 certificate

Import-Certificate -FilePath "\\Enter your DC01 IP Address\c$\dc01cert.cer" -
CertStoreLocation "Cert:\LocalMachine\Root"

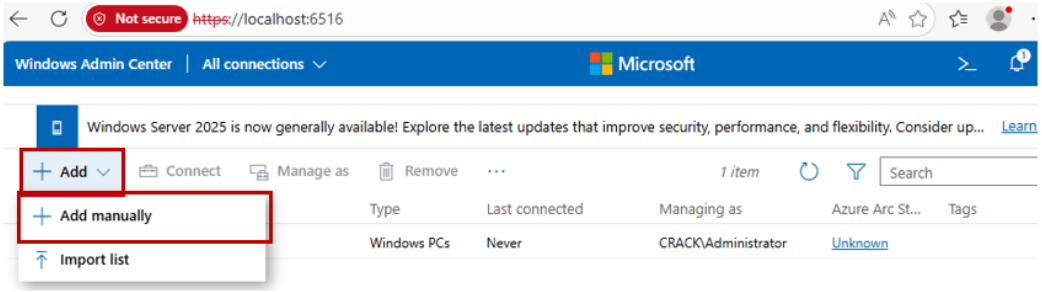
# 4. Restart WinRM

Restart-Service WinRM -Force
```

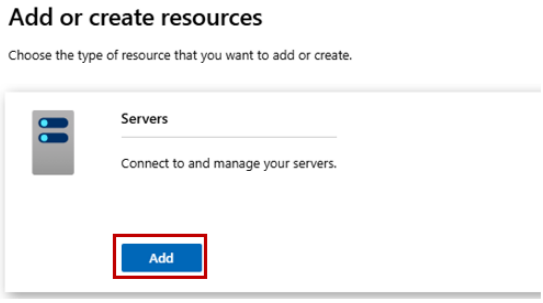




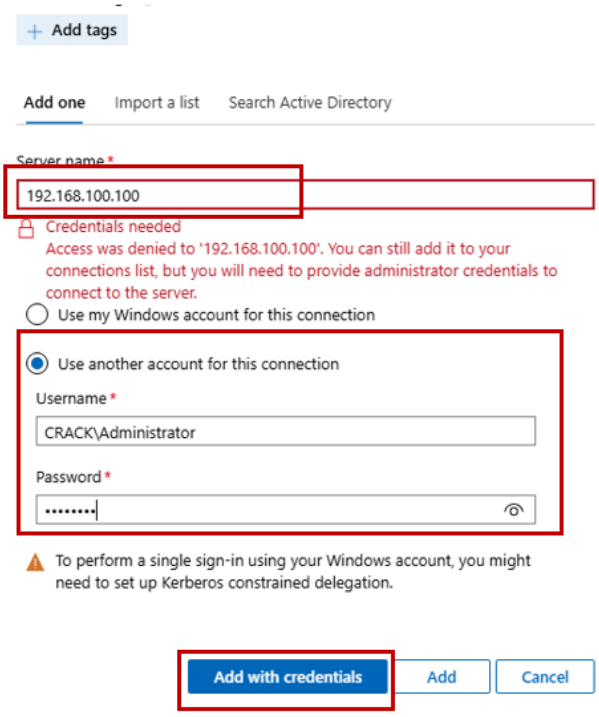
1. Click **+ Add** (top left area)

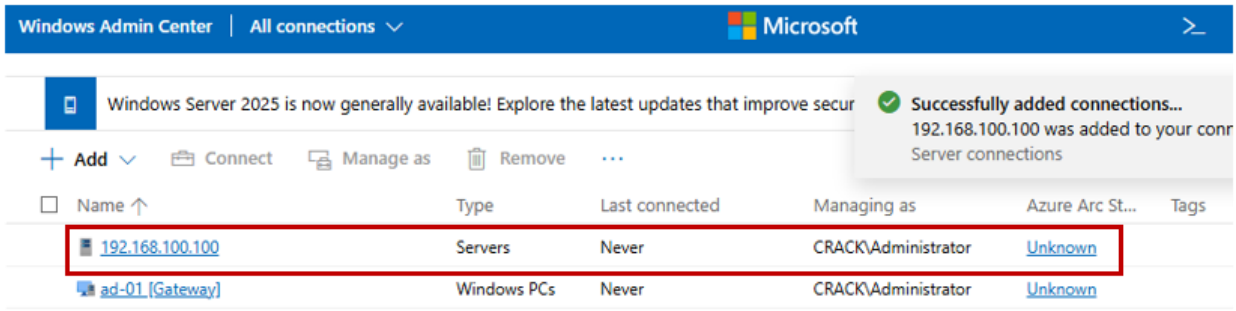


2. Click **Servers → Add**



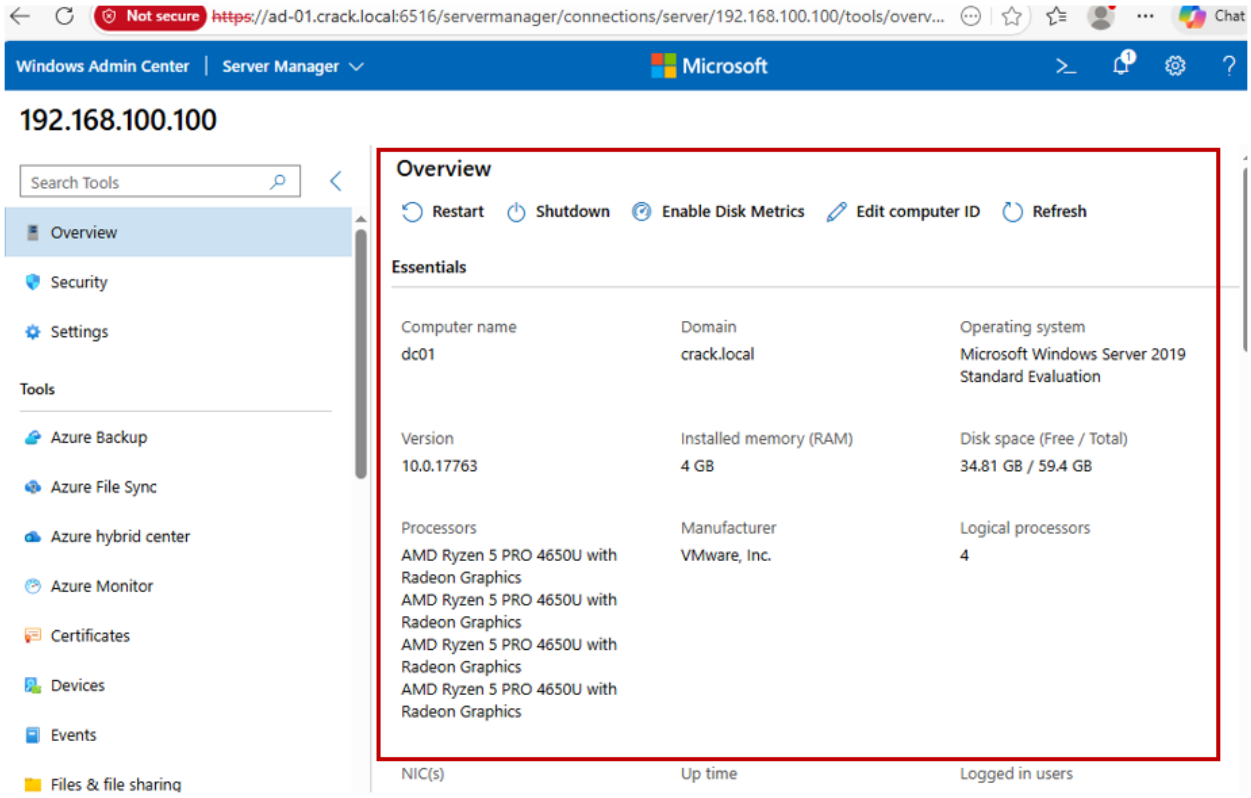
3. Server name: Enter your DC01 Ip Address and scroll down and click Add
4. Username: CRACK\Administrator
5. Password: Admin123





In WAC:

- Click Add
- Connection established



Now Add VPN01

Do the same certificate setup on **VPN01 first:**

17.3 On VPN01 PowerShell:

```
winrm quickconfig

winrm set winrm/config/service '@{AllowUnencrypted="true"}'

winrm set winrm/config/client '@{TrustedHosts="*"}'

$cert = New-SelfSignedCertificate -DnsName "VPN01","VPN01.crack.local","Enter
VPN01 IP Address" -CertStoreLocation "cert:\LocalMachine\My"

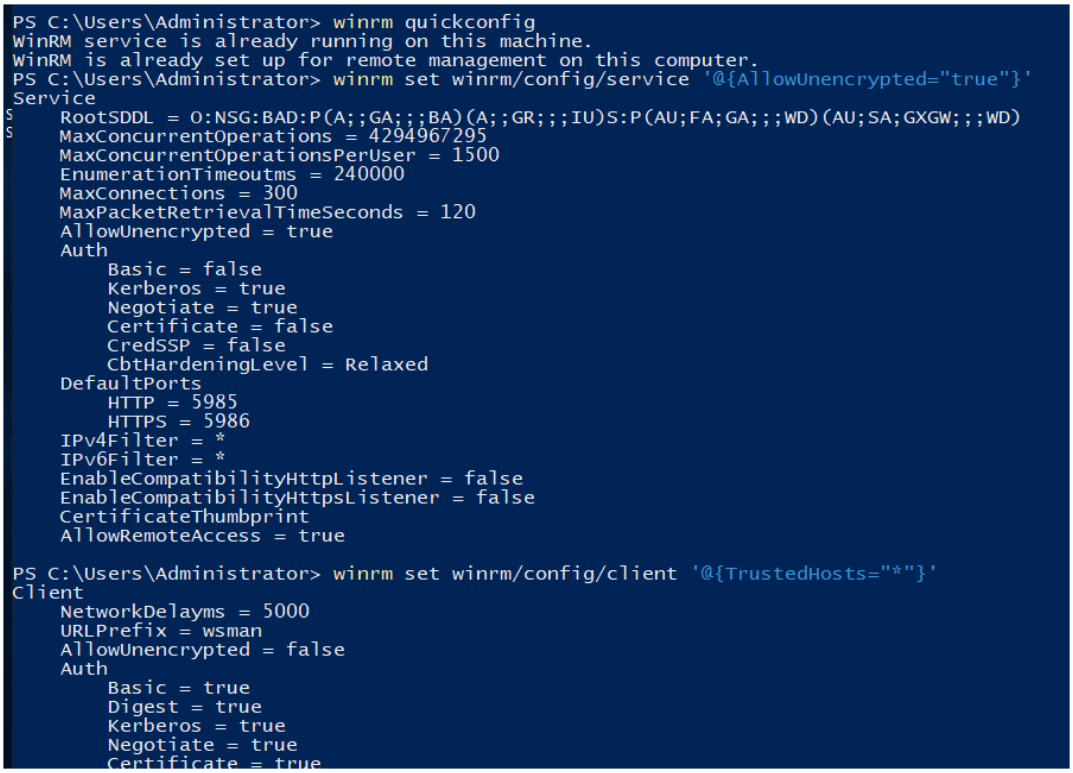
New-Item -Path WSMan:\localhost\Listener -Transport HTTPS -Address * -
CertificateThumbprint $cert.Thumbprint -Force

Export-Certificate -Cert $cert -FilePath "C:\vpn01cert.cer"

netsh advfirewall firewall add rule name="WinRM Port 5985" dir=in
action=allow protocol=TCP localport=5985

netsh advfirewall firewall add rule name="WinRM HTTPS 5986" dir=in
action=allow protocol=TCP localport=5986

Restart-Service WinRM -Force
```



```
PS C:\Users\Administrator> $cert = New-SelfSignedCertificate -DnsName "VPN-01","VPN-01.crack.local"
,"192.168.100.130" -CertStoreLocation "cert:\LocalMachine\My"
PS C:\Users\Administrator> New-Item -Path WSMan:\localhost\Listener -Transport HTTPS -Address * -Ce
rtificateThumbprint $cert.Thumbprint -Force


   WSManConfig: Microsoft.WSMan.Management\WSMan::localhost\Listener

Type            Keys                                Name
----            ----                                ----
Container       {Transport=HTTPS, Address=*}        Listener_1305953032


PS C:\Users\Administrator> Export-Certificate -Cert $cert -FilePath "C:\vpn01cert.cer"


    Directory: C:\


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        4/22/2026   2:19 AM            826 vpn01cert.cer


PS C:\Users\Administrator> netsh advfirewall firewall add rule name="WinRM Port 5985" dir=in action
=allow protocol=TCP localport=5985
Ok.

PS C:\Users\Administrator> netsh advfirewall firewall add rule name="WinRM HTTPS 5986" dir=in actio
n=allow protocol=TCP localport=5986
Ok.

PS C:\Users\Administrator> Restart-Service WinRM -Force
PS C:\Users\Administrator>
```

On ADMIN01 PowerShell:

```
Import-Certificate -FilePath "\\Enter VPN01 IP Address\c$\vpn01cert.cer" -
CertStoreLocation "Cert:\LocalMachine\Root"

Restart-Service WinRM -Force
```

```
PS C:\Users\Administrator>
PS C:\Users\Administrator> Import-Certificate -FilePath "\\192.168.10.50\c$\vpn01cert.cer" -CertStoreLocation "Cert:\LocalMachine\Root"


   PSParentPath: Microsoft.PowerShell.Security\Certificate::LocalMachine\Root

Thumbprint                                Subject
----------                                -------
1F41AE1A9F05CACDCF3AF28EAA0D31400AD80131  CN=VPN01

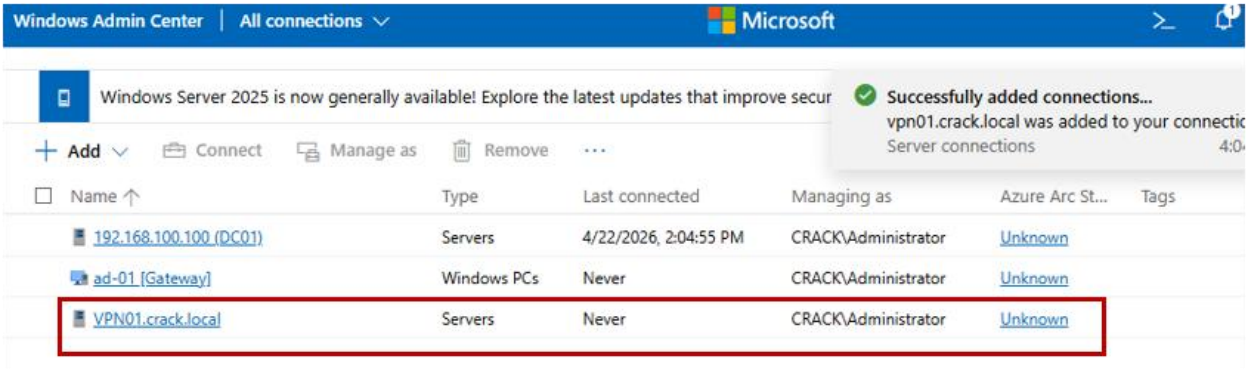

PS C:\Users\Administrator> Restart-Service WinRM -Force
WARNING: Waiting for service 'Windows Admin Center (WindowsAdminCenter)' to start...
PS C:\Users\Administrator>
```

Activate Windows
Go to Settings to activate Windows.

Then in WAC:

- Click **+ Add → Add manually → Servers → Add**
- Server name: 192.168.100.130
- Username: CRACK\Administrator
- Password: Admin123

- Click **Add** with credentials → **Continue**



NOTE!!!!

DUE to System Problem I mistakenly switched my IPS and the new ip Range includes

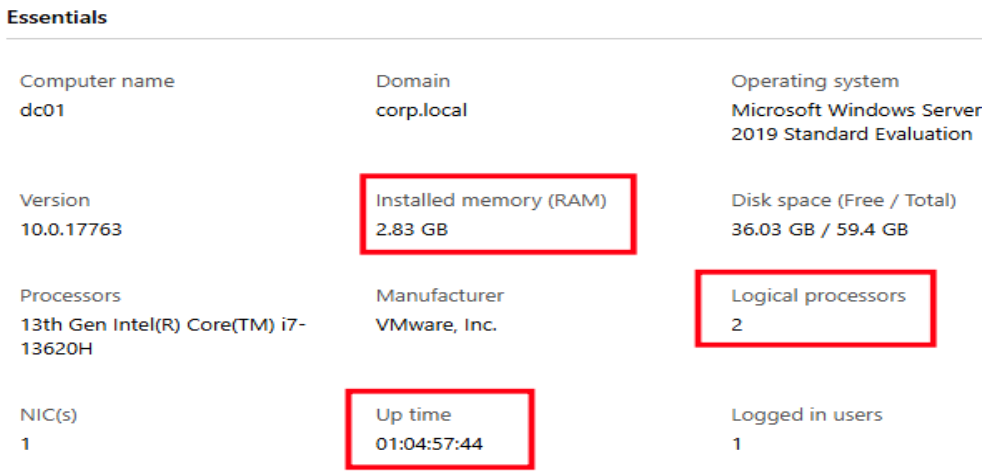
All the configurations are same but we will continue from here as domain corp with these IP's

VM	**IP Address**	**Operating System**	**Role**
DC01	192.168.10.10	Windows Server 2019	Domain Controller — from Module 2
ADMIN01	192.168.10.20	Windows 10 Pro	Management Workstation — from Module 2
CL01	192.168.10.30	Windows 10 Pro	Client Workstation — from Module 2
VPN01	192.168.10.50	Windows Server 2019	VPN Server — new in Module 3
IDS01	192.168.10.60	Kali Linux	Intrusion Detection — new in Module 3
ATTACK01	192.168.10.40	Kali Linux	Attack Simulation — new in Module 3

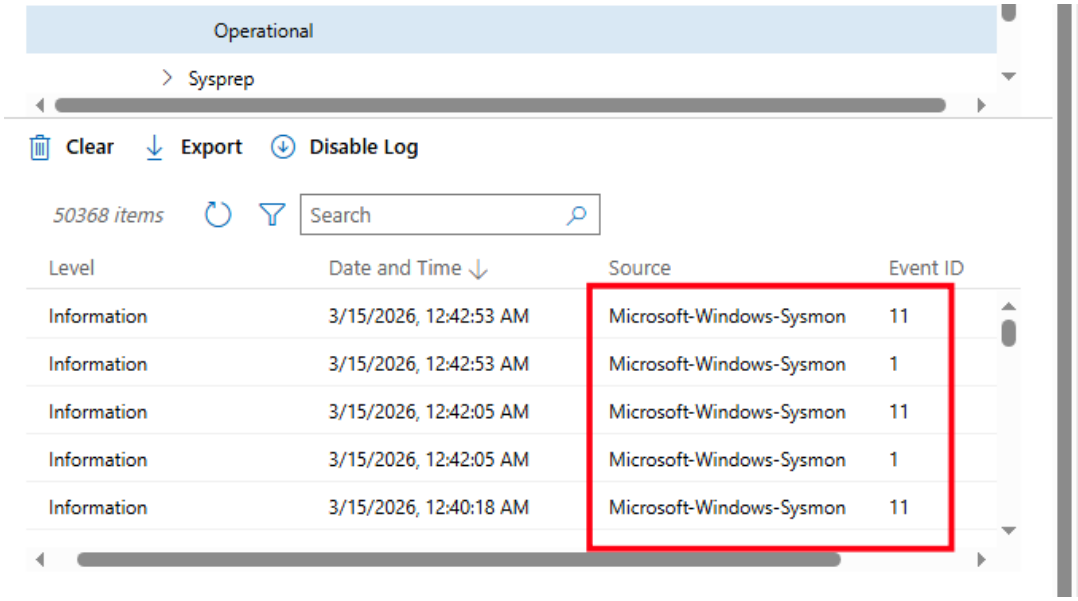
17.4 EXPLORE WAC CAPABILITIES

Click **DC01** on the server list. You can now:

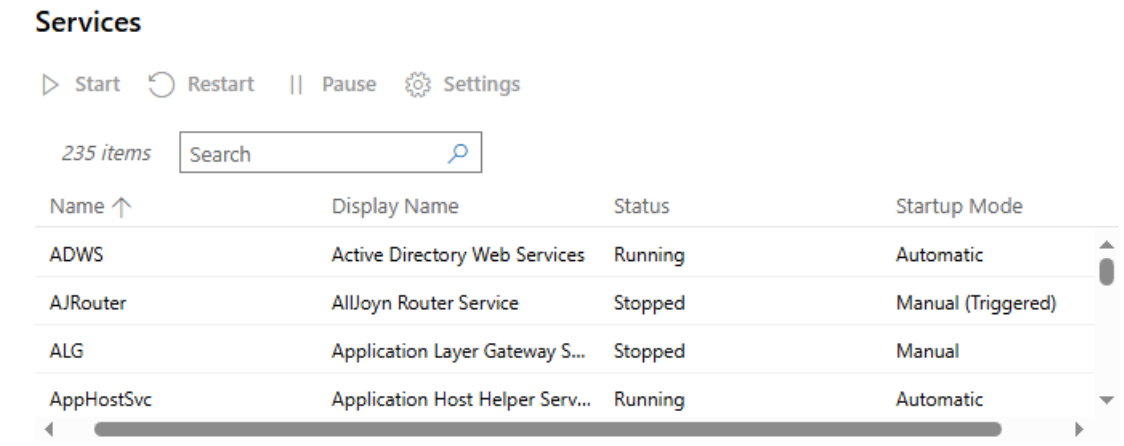
- **Overview** — see live CPU, RAM, uptime



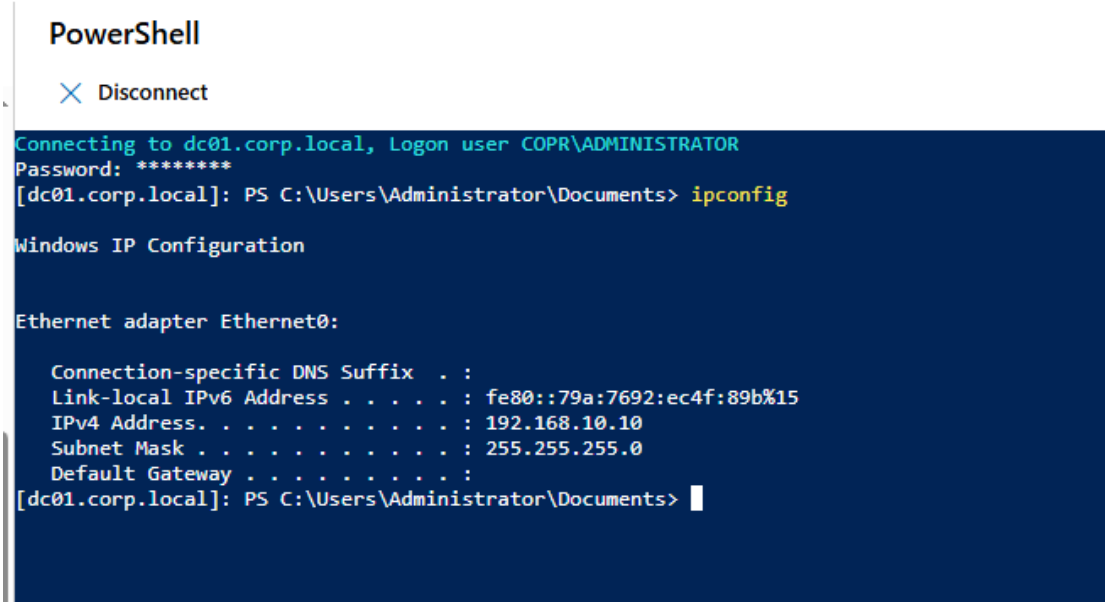
- **Events** — browse Windows Security logs remotely



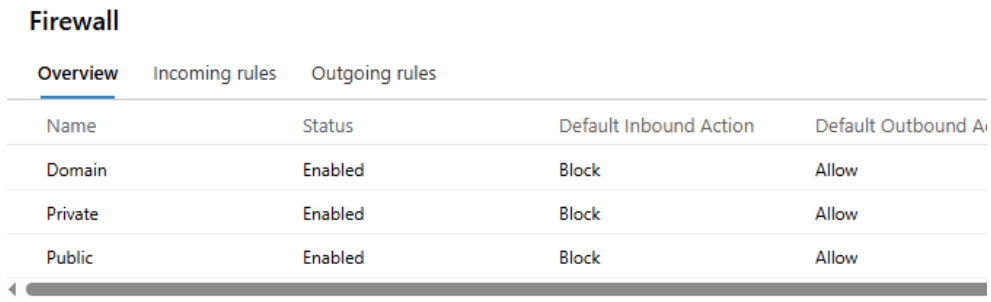
- **Services** — start/stop Windows services

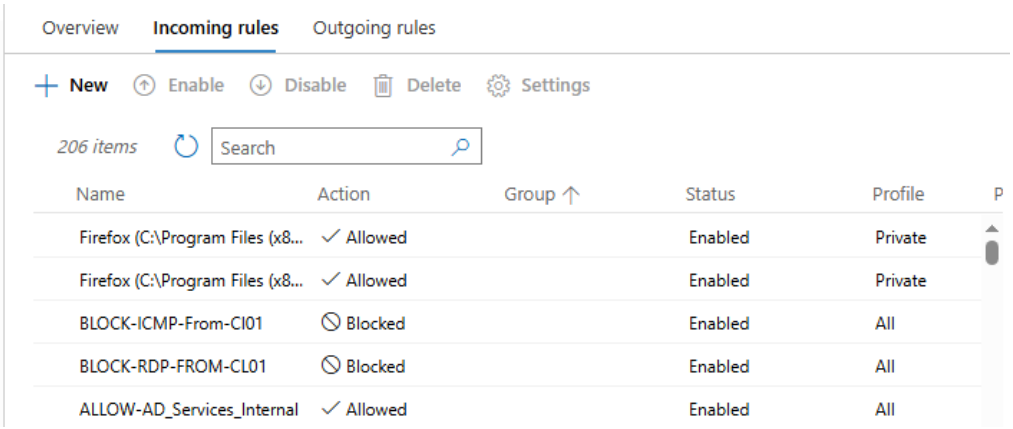


- **PowerShell** — run PowerShell commands on DC01 from your browser

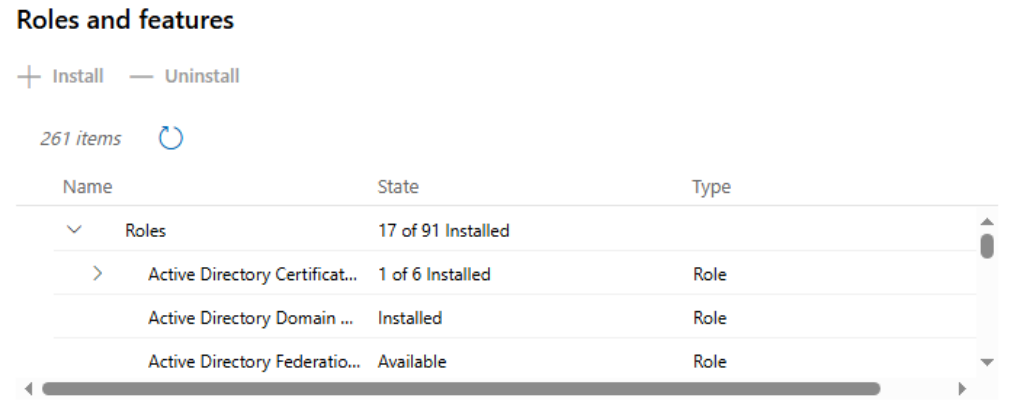


- **Firewall** — review all firewall rules





- **Roles & Features** — add/remove Windows roles



All of this goes over HTTPS on port 6516 — encrypted and logged.

18. ATTACK SIMULATION (ATTACK01 Kali Linux)

Before starting any attack:

Go to **IDS01** and make sure this is running:

```
sudo tail -f /var/log/suricata/fast.log
```



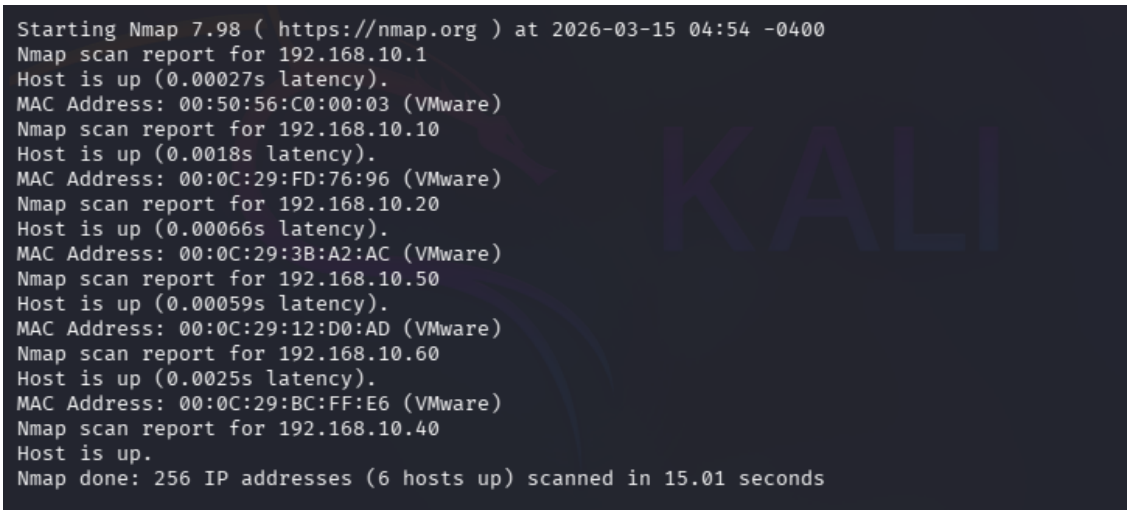
Keep this terminal visible in a side window. You will watch alerts appear here in real time.

18.1 ATTACK 1 – NETWORK RECONNAISSANCE

On **ATTACK01**, open a terminal:

```
# Discover all live hosts on the network

nmap -sn 192.168.10.0/24
```



You should see DC01, ADMIN01, CL01, VPN01, IDS01 all listed as "up."

Port scan DC01

```
nmap -sS -p 1-1024 192.168.10.10
```

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 04:58 -0400
Nmap scan report for 192.168.10.10
Host is up (0.0010s latency).
Not shown: 1014 filtered tcp ports (no-response)
PORT    STATE SERVICE
22/tcp  open  ssh
53/tcp  open  domain
80/tcp  open  http
88/tcp  open  kerberos-sec
135/tcp open  msrpc
389/tcp open  ldap
445/tcp open  microsoft-ds
464/tcp open  kpasswd5
593/tcp open  http-rpc-epmap
636/tcp open  ldapssl
MAC Address: 00:0C:29:FD:76:96 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 9.60 seconds
```

Aggressive full scan with OS and version detection

```
nmap -A -T4 192.168.10.10
```

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 05:03 -0400
Nmap scan report for 192.168.10.10
Host is up (0.0010s latency).
Not shown: 987 filtered tcp ports (no-response)
PORT     STATE SERVICE       VERSION
22/tcp   open  ssh           OpenSSH for_Windows_7.7 (protocol 2.0)
| ssh-hostkey:
|   2048 18:d4:ee:68:0b:13:75:ae:72:7b:02:32:11:f6:7b:35 (RSA)
|   256 62:09:22:06:f9:e1:84:a8:5e:1d:37:96:ee:a9:4f:78 (ECDSA)
|_  256 8a:6f:6f:be:b6:70:e8:29:b8:6c:0f:d4:be:a4:dd:61 (ED25519)
53/tcp   open  domain        Simple DNS Plus
80/tcp   open  http          Microsoft IIS httpd 10.0
|_http-server-header: Microsoft-IIS/10.0
|_http-title: IIS Windows Server
| http-methods:
|_  Potentially risky methods: TRACE
88/tcp   open  kerberos-sec  Microsoft Windows Kerberos (server time: 2026-03-15 09:04:16Z)
135/tcp  open  msrpc         Microsoft Windows RPC
389/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: corp.local, Si
te: Default-First-Site-Name)
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.corp.local
| Not valid before: 2026-03-13T23:09:23
|_Not valid after:  2027-03-13T23:09:23
|_ssl-date: 2026-03-15T09:05:25+00:00; +1s from scanner time.
445/tcp  open  microsoft-ds?
464/tcp  open  kpasswd5?
593/tcp  open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp  open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain: corp.local, Si
te: Default-First-Site-Name)
|_ssl-date: 2026-03-15T09:05:25+00:00; +1s from scanner time.
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.corp.local
| Not valid before: 2026-03-13T23:09:23
|_Not valid after:  2027-03-13T23:09:23
3268/tcp open  ldap          Microsoft Windows Active Directory LDAP (Domain: corp.local, Si
te: Default-First-Site-Name)
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.corp.local
| Not valid before: 2026-03-13T23:09:23
|_Not valid after:  2027-03-13T23:09:23
|_ssl-date: 2026-03-15T09:05:25+00:00; +1s from scanner time.
3269/tcp open  ssl/ldap      Microsoft Windows Active Directory LDAP (Domain: corp.local, Si
te: Default-First-Site-Name)
|_ssl-date: 2026-03-15T09:05:25+00:00; +1s from scanner time.
```

```
te: Default-First-Site-Name)
|_ssl-date: 2026-03-15T09:05:25+00:00; +1s from scanner time.
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.corp.local
| Not valid before: 2026-03-13T23:09:23
|_Not valid after:  2027-03-13T23:09:23
5986/tcp open  ssl/wsmans?
| ssl-cert: Subject: commonName=DC01
| Subject Alternative Name: DNS:DC01, DNS:DC01.corp.local, DNS:192.168.10.10
| Not valid before: 2026-03-14T08:04:42
|_Not valid after:  2027-03-14T08:24:42
|_ssl-date: 2026-03-15T09:05:25+00:00; +1s from scanner time.
| tls-alpn:
|   h2
|_  http/1.1
MAC Address: 00:0C:29:FD:76:96 (VMware)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 cl
osed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 2019|10|11 (97%)
OS CPE: cpe:/o:microsoft:windows_server_2019 cpe:/o:microsoft:windows_10 cpe:/o:microsoft:wi
ndows_11
Aggressive OS guesses: Windows Server 2019 (97%), Microsoft Windows 10 1803 (91%), Microsoft
 Windows 10 1903 - 21H1 (91%), Microsoft Windows Server 2019 (91%), Microsoft Windows 11 (88
%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: Host: DC01; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
| smb2-time:
|   date: 2026-03-15T09:04:49
|_  start_date: N/A

TRACEROUTE
HOP RTT     ADDRESS
1   1.03 ms 192.168.10.10

OS and Service detection performed. Please report any incorrect results at https://nmap.org/
submit/ .
Nmap done: 1 IP address (1 host up) scanned in 85.24 seconds
```

Check IDS01 — you should see alerts

```
03/15/2026-05:04:44.175849  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36360
 → 192.168.10.10:80
03/15/2026-05:04:44.175848  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36320
 → 192.168.10.10:80
03/15/2026-05:04:44.171014  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36374
 → 192.168.10.10:80
03/15/2026-05:04:44.185659  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36336
 → 192.168.10.10:80
03/15/2026-05:04:44.185657  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36304
 → 192.168.10.10:80
03/15/2026-05:04:44.175850  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36328
 → 192.168.10.10:80
03/15/2026-05:04:44.185660  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36390
 → 192.168.10.10:80
03/15/2026-05:04:44.185667  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36362
 → 192.168.10.10:80
03/15/2026-05:04:44.194443  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36402
 → 192.168.10.10:80
03/15/2026-05:04:44.340902  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36424
 → 192.168.10.10:80
03/15/2026-05:04:44.362087  [**] [1:2024364:5] ET SCAN Possible Nmap User-Agent Observed [**] [Classification: Web Application Attack] [Priority: 1] {TCP} 192.168.10.40:36436
 → 192.168.10.10:80
```

IDS is detecting the scan.

18.2 ATTACK 2 – SMBv1 AND ETERNALBLUE CHECK

```
nmap --script smb-protocols 192.168.10.10
```

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 05:10 -0400
Nmap scan report for 192.168.10.10
Host is up (0.00095s latency).
Not shown: 987 filtered tcp ports (no-response)
PORT     STATE SERVICE
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
88/tcp   open  kerberos-sec
135/tcp  open  msrpc
389/tcp  open  ldap
445/tcp  open  microsoft-ds
464/tcp  open  kpasswd5
593/tcp  open  http-rpc-epmap
636/tcp  open  ldapssl
3268/tcp open  globalcatLDAP
3269/tcp open  globalcatLDAPssl
5986/tcp open  wsmans
MAC Address: 00:0C:29:FD:76:96 (VMware)

Host script results:
| smb-protocols:
|   dialects:
|     2.0.2
|     2.1
|     3.0
|     3.0.2
|_    3.1.1

Nmap done: 1 IP address (1 host up) scanned in 15.16 seconds
```

SMB Version	**Present?**	**Meaning**
SMBv1	NOT listed	Successfully disabled
SMBv2 *(2.0.2, 2.1)*	Present	Working correctly
SMBv3 *(3.0, 3.0.2, 3.1.1)*	Present	Working correctly

```
nmap --script smb-vuln-ms17-010 192.168.10.10
```

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 05:14 -0400
Nmap scan report for 192.168.10.10
Host is up (0.0022s latency).
Not shown: 987 filtered tcp ports (no-response)
PORT     STATE SERVICE
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
88/tcp   open  kerberos-sec
135/tcp  open  msrpc
389/tcp  open  ldap
445/tcp  open  microsoft-ds
464/tcp  open  kpasswd5
593/tcp  open  http-rpc-epmap
636/tcp  open  ldapssl
3268/tcp open  globalcatLDAP
3269/tcp open  globalcatLDAPssl
5986/tcp open  wsmans
MAC Address: 00:0C:29:FD:76:96 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 10.61 seconds
```

The scan was completed and **no vulnerability section appeared** at the bottom.

```
Host script results: ← MISSING

smb-vuln-ms17-010:  ← MISSING

VULNERABLE:         ← MISSING
```

When a system **IS vulnerable** it shows:

```
Host script results:

smb-vuln-ms17-010: VULNERABLE

State: VULNERABLE
```

This proves your SMBv1 disable in Part 4 worked. EternalBlue cannot exploit a server without SMBv1.

18.3 ATTACK 3 – BRUTE FORCE ATTACK (tests your GPO lockout)

Create Wordlist on Kali

```
nano mywordlist.txt
```

Type these exactly:

```
password

123456

admin

Password1

letmein

welcome

test123

Admin123

Password123

Hruser123

qwerty

abc123

monkey

dragon master
```



Run the brute force:

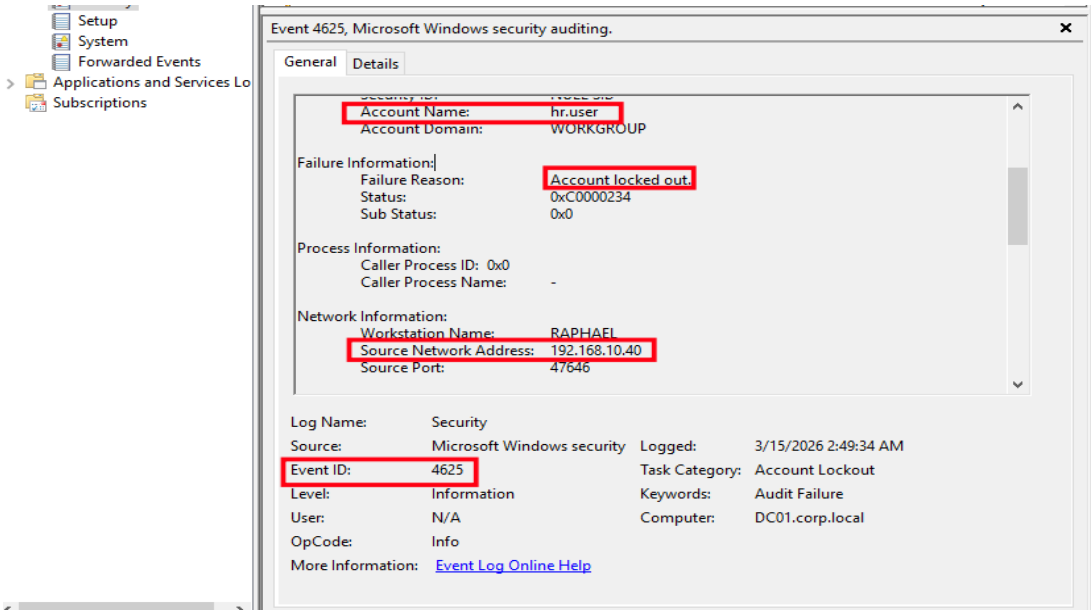
hydra -l hr.user -P mywordlist.txt -t 1 smb2://192.168.10.10

```
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service orga

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-15 05:49:39
[WARNING] Workgroup was not specified, using "WORKGROUP"
[DATA] max 1 task per 1 server, overall 1 task, 15 login tries (l:1/p:15), ~15 tries per task
[DATA] attacking smb2://192.168.10.10:445/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-15 05:49:39
```

Account got locked after 5 attempts

After 5 wrong attempts → hr.user locked

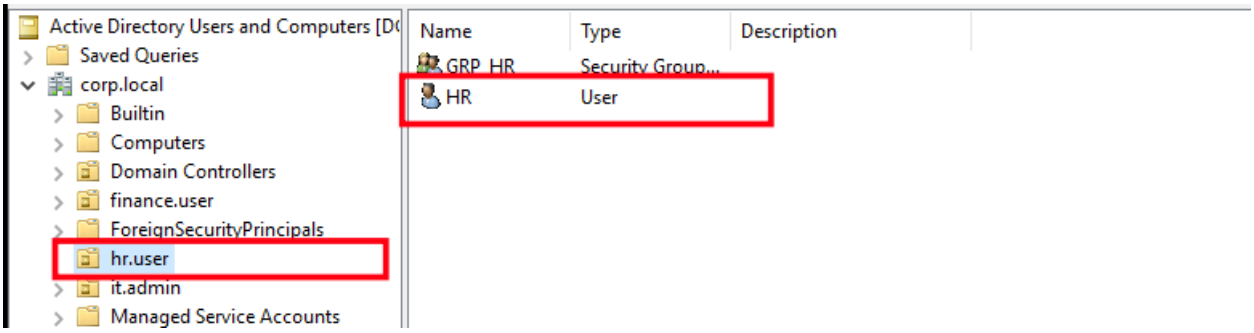


Hydra couldn't find password because account was locked

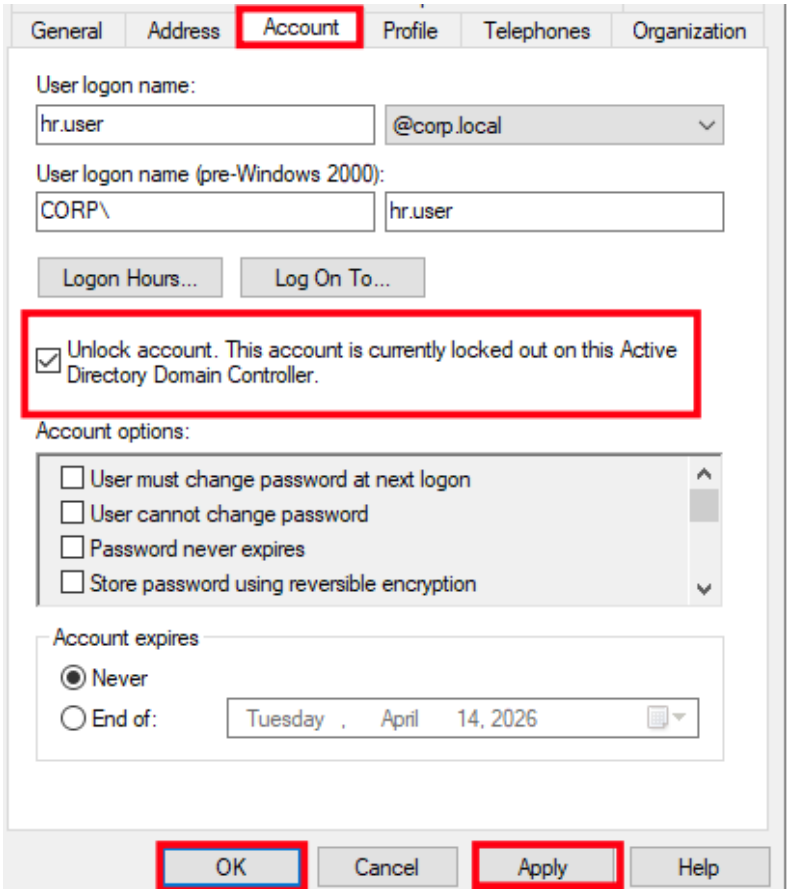


Unlock hr.user after the test:

1. DC01 → dsa.msc
2. HR OU → double-click hr.user



3. **Account** tab → check **"Unlock account"** → Apply → OK



18.4 ATTACK 4 – LLMNR POISONING TEST

Wrong credentials first: On ATTACK01 — try to connect to SoftEther VPN with wrong password using SoftEther VPN Client, or simply try:

On **ATTACK01**:

```
sudo responder -I eth0 -wv
```



Leave this running and watching.

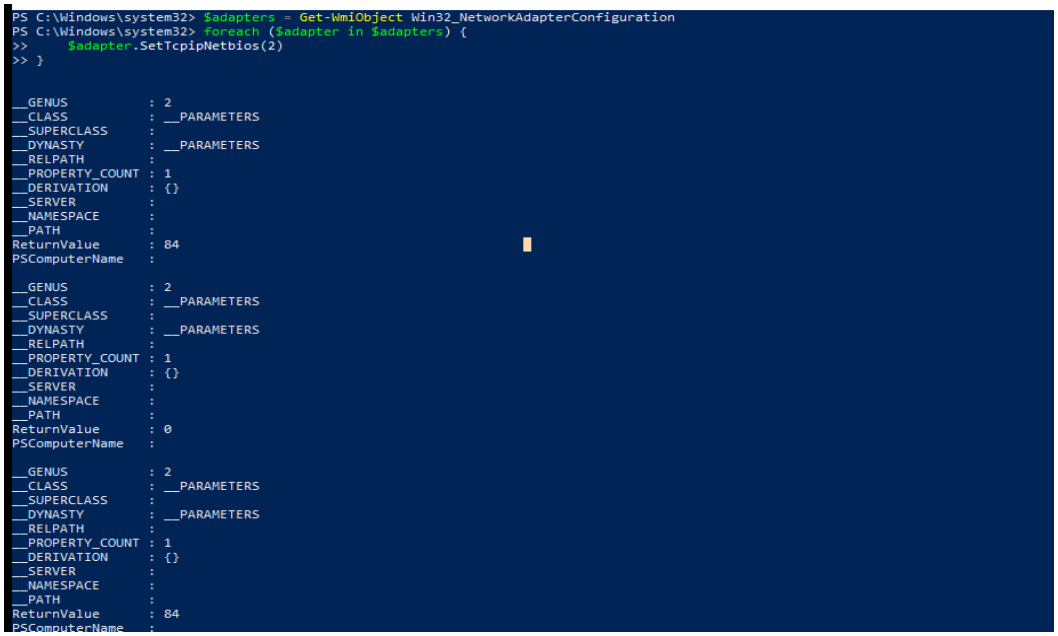
Disable NetBIOS on CL01

Open PowerShell as Administrator on CL01 and run:

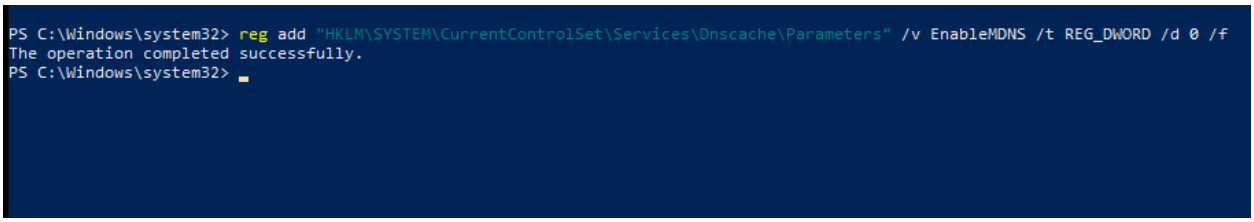
```
$adapters = Get-WmiObject Win32_NetworkAdapterConfiguration

foreach ($adapter in $adapters) {

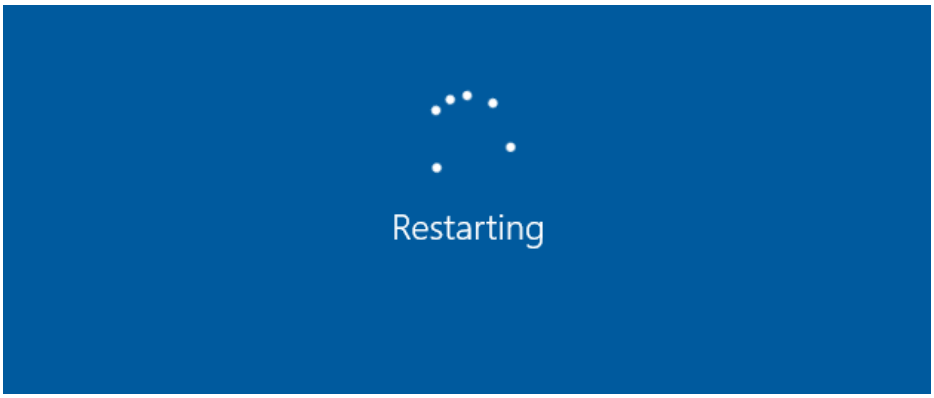
    $adapter.SetTcpipNetbios(2)}
```



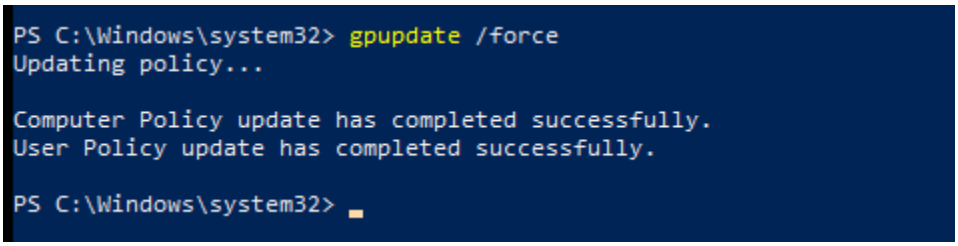

```
reg add "HKLM\SYSTEM\CurrentControlSet\Services\Dnscache\Parameters" /v
EnableMDNS /t REG_DWORD /d 0 /f
```



```
Restart-Computer
```

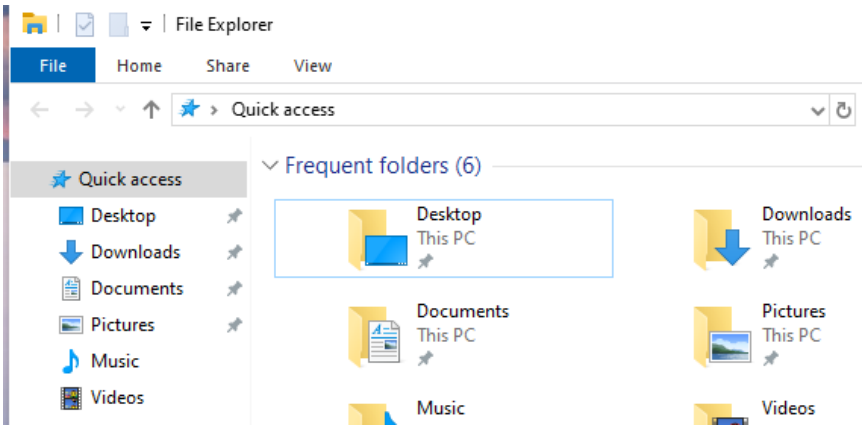



```
gpupdate /force
```

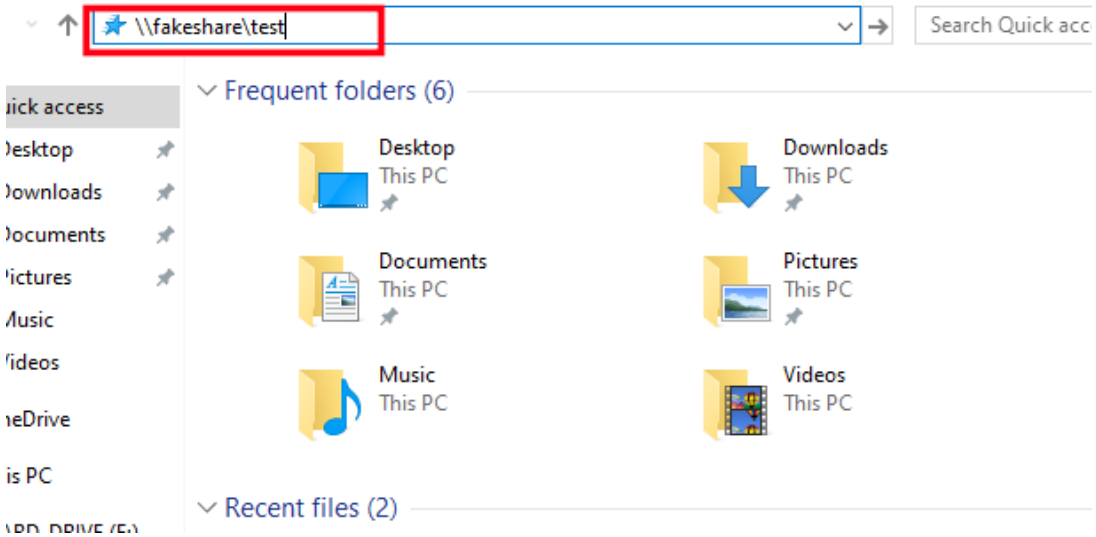


Go to **CL01**:

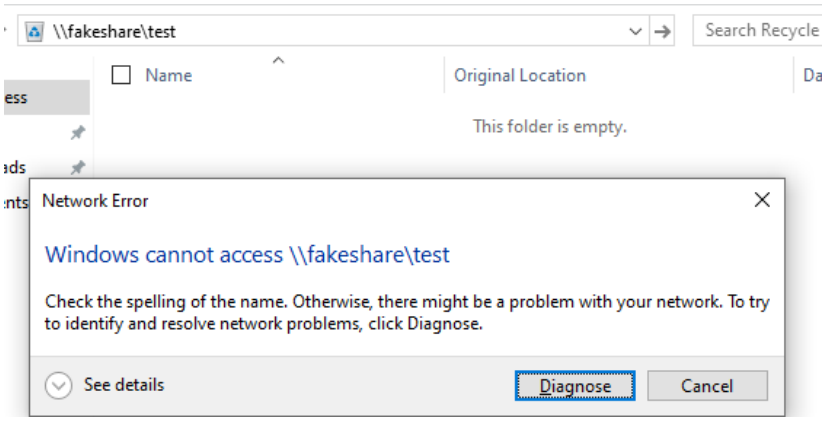
1. Open File Explorer



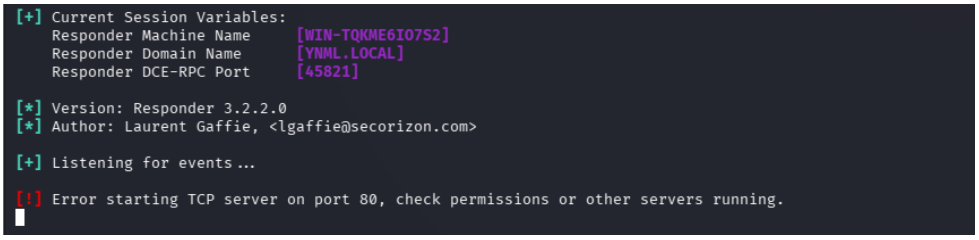
2. In the address bar at the top, type: \\fakeshare\test → press Enter



3. It tries to find this fake server



Now Check Kali Responder Terminal



This proves LLMNR is disabled. If it was still on, Responder would have captured the NTLM hash of whoever is logged into CL01 — and you could crack it to get their password.

18.5 ATTACK 5 – VPN ACCESS TEST

Test if VPN port is open

```
nmap -p 443 192.168.10.50
```

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 07:36 -0400
Nmap scan report for 192.168.10.50
Host is up (0.0021s latency).

PORT    STATE SERVICE
443/tcp open  https
MAC Address: 00:0C:29:12:D0:AD (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds
```

Correct credential test:

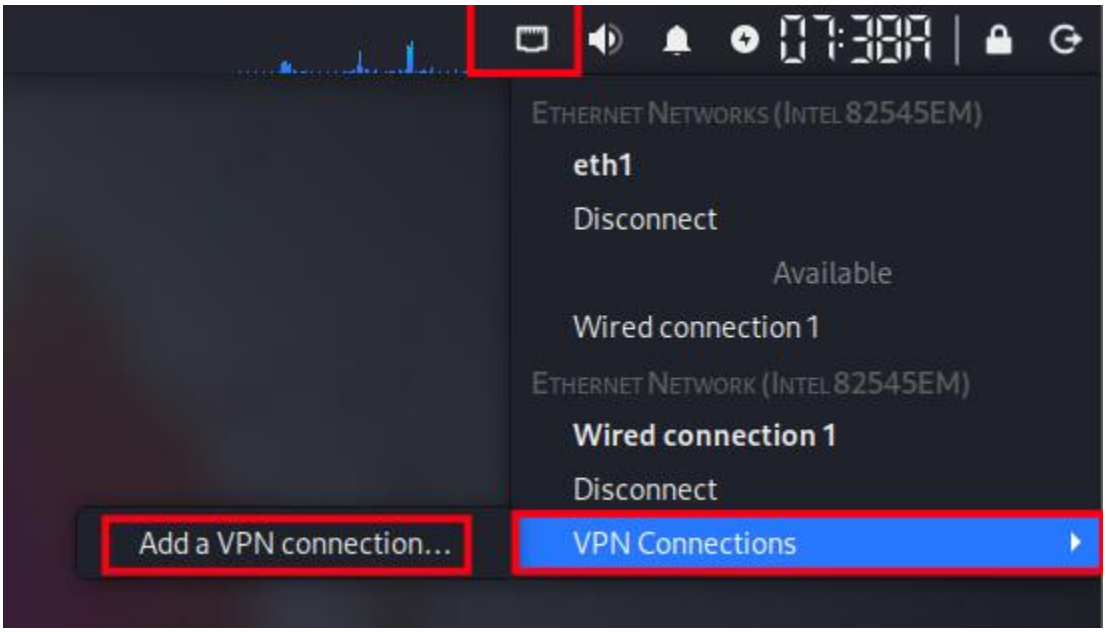
On ATTACK01, install VPN client:

```
sudo apt install network-manager-l2tp network-manager-l2tp-gnome -y
```

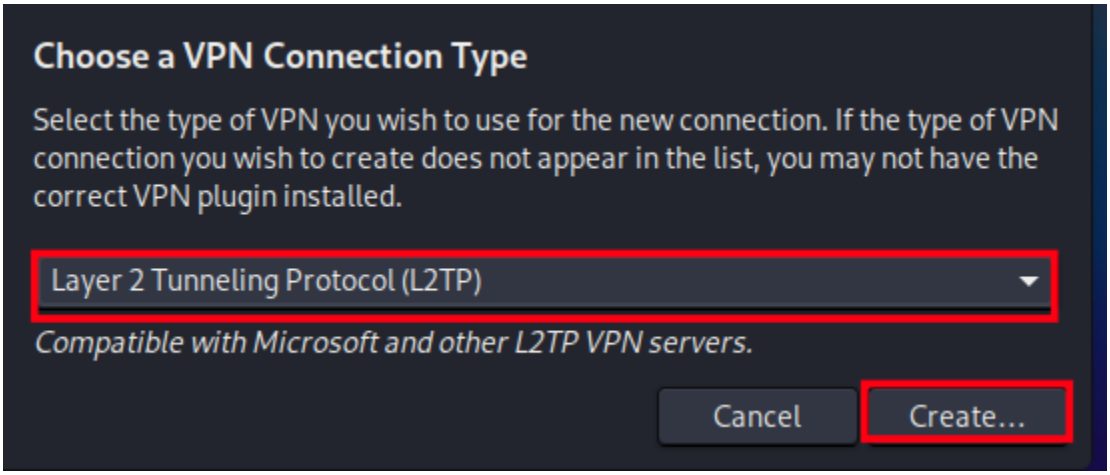
```
┌──(raphael㉿raphael)-[~]
└─$ sudo apt install network-manager-l2tp network-manager-l2tp-gnome -y
network-manager-l2tp is already the newest version (1.52.0-2).
network-manager-l2tp set to manually installed.
network-manager-l2tp-gnome is already the newest version (1.52.0-2).
network-manager-l2tp-gnome set to manually installed.
The following packages were automatically installed and are no longer required:
  bloodhound.py  libaudio2      libavformat61            libfuse2t64  libmjpegutils-2.1-0t6
  curlftpfs      libavfilter10  libconfig-inifiles-perl  libgav1-1    libmpeg2encpp-2.1-0t6
Use 'sudo apt autoremove' to remove them.
```

Connect through GNOME network manager:

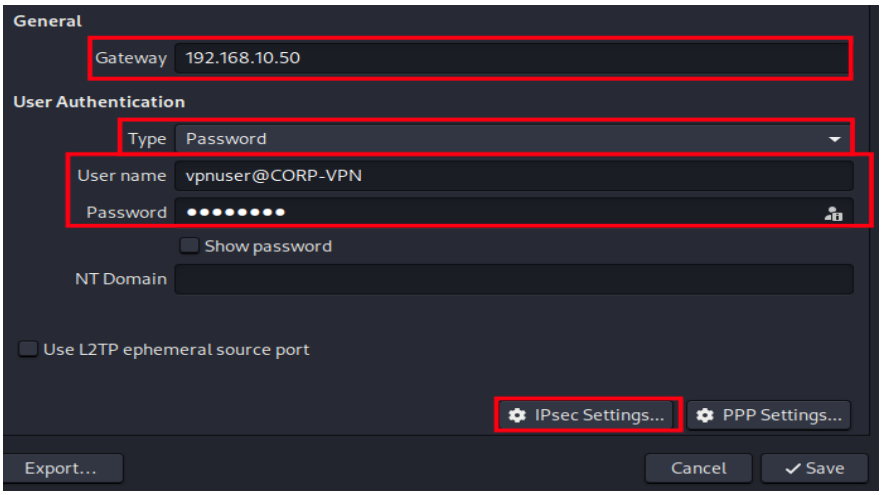
1. Click network icon → **VPN Connections** → **Add a VPN connection**



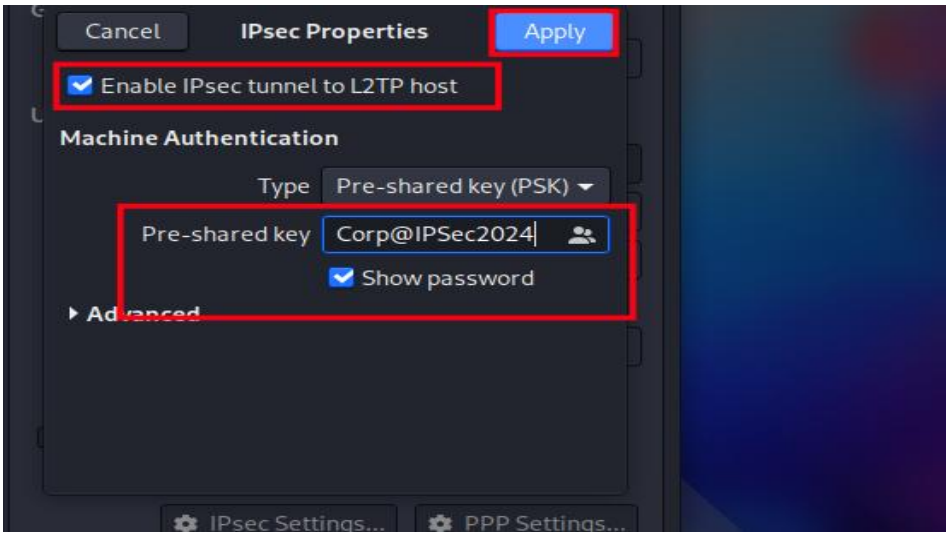
2. Select **Layer 2 Tunneling Protocol (L2TP)**



3. Gateway: 192.168.10.50
4. Type: Password
5. Username: vpnuser@CORP-VPN
6. Password: Admin123
7. NT Domain: Leave Blank

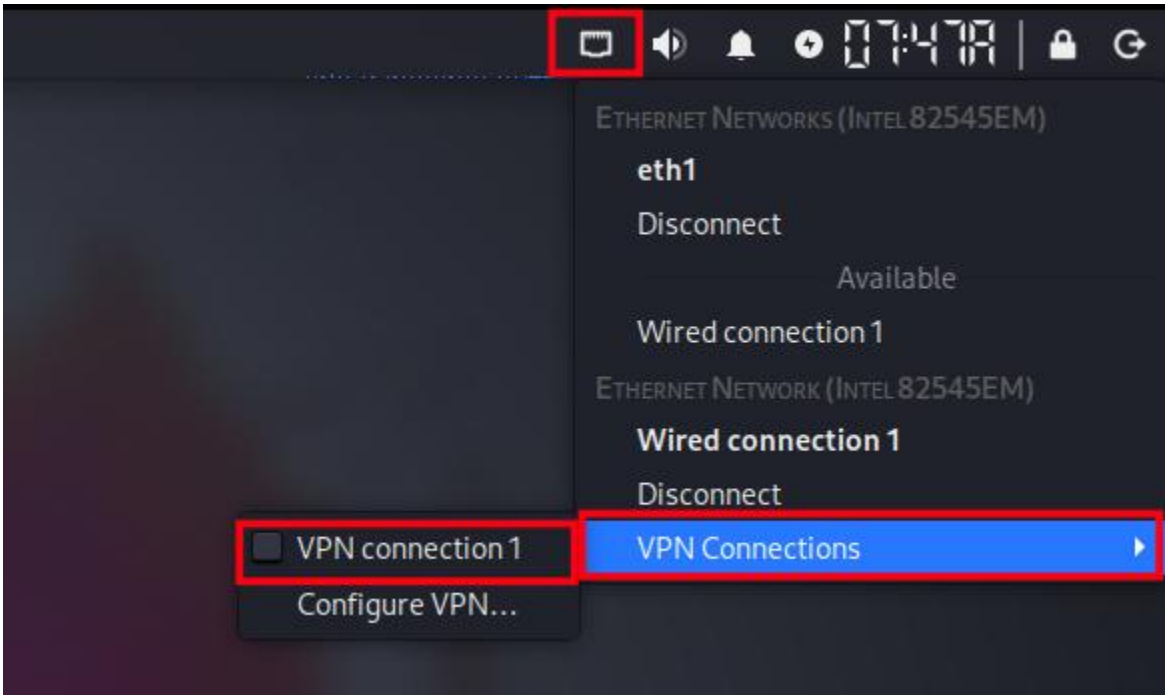


8. **Then Click IPsec Settings and fill in:**
   - Check **"Enable IPsec tunnel to L2TP host"**
   - Pre-shared key: Corp@IPSec2024



Click Save

9. After saving → click the VPN toggle to connect



After connecting:

```
ip addr show
```

```
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:81:d7:ae brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.40/24 brd 192.168.10.255 scope global noprefixroute eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::ac0c:3d36:e59f:b1a0/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:81:d7:b8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.148/24 brd 192.168.50.255 scope global dynamic noprefixroute eth1
       valid_lft 969sec preferred_lft 744sec
    inet6 fe80::b661:318a:ff51:586f/64 scope link
       valid_lft forever preferred_lft forever
15: ppp0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1400 qdisc noqueue state UNKNOWN group default qlen 3
    link/ppp
    inet 192.168.30.10 peer 1.0.0.1/32 scope global ppp0
       valid_lft forever preferred_lft forever

┌──(raphael㉿raphael)-[~]
```

Test internal access through the VPN tunnel:

```
ping 192.168.10.10
```

```
┌──(raphael㉿raphael)-[~]
└─$ ping 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=128 time=3.39 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=128 time=0.677 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=128 time=0.849 ms
^C
── 192.168.10.10 ping statistics ──
3 packets transmitted, 3 received, 0% packet loss, time 2024ms
rtt min/avg/max/mdev = 0.677/1.639/3.391/1.240 ms
```

18.6 ATTACK 6 – TLS VERSION VERIFICATION

```
nmap --script ssl-enum-ciphers -p 636 192.168.10.10
```

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-15 09:14 -0400
Nmap scan report for 192.168.10.10
Host is up (0.00060s latency).

PORT    STATE SERVICE
636/tcp open  ldapssl
| ssl-enum-ciphers:
|   TLSv1.2:
|     ciphers:
|       TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (secp384r1) - A
|       TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
|       TLS_DHE_RSA_WITH_AES_256_GCM_SHA384 (dh 2048) - A
|       TLS_DHE_RSA_WITH_AES_128_GCM_SHA256 (dh 2048) - A
|       TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384 (secp384r1) - A
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 (ecdh_x25519) - A
|       TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (secp384r1) - A
|       TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (ecdh_x25519) - A
|       TLS_RSA_WITH_AES_256_GCM_SHA384 (rsa 2048) - A
|       TLS_RSA_WITH_AES_128_GCM_SHA256 (rsa 2048) - A
|       TLS_RSA_WITH_AES_256_CBC_SHA256 (rsa 2048) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA256 (rsa 2048) - A
|       TLS_RSA_WITH_AES_256_CBC_SHA (rsa 2048) - A
|       TLS_RSA_WITH_AES_128_CBC_SHA (rsa 2048) - A
|       TLS_RSA_WITH_3DES_EDE_CBC_SHA (rsa 2048) - C
|     compressors:
|       NULL
|     cipher preference: server
|     warnings:
|       64-bit block cipher 3DES vulnerable to SWEET32 attack
|_  least strength: C
MAC Address: 00:0C:29:FD:76:96 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.53 seconds
```

What The Results Show:

TLS Version	Status	Expected
TLS 1.0	NOT listed	Blocked successfully
TLS 1.1	NOT listed	Blocked successfully
TLS 1.2	Only version listed	Correct

TLS 1.0 and **TLS 1.1** are completely gone — only **TLS 1.2** is availab

19. CONCLUSION

This lab successfully implemented and validated all technical network security controls across the corp.local environment. Network segmentation, protocol hardening, TLS enforcement, VPN access, intrusion detection and secure remote management were all configured and confirmed working through live attack simulation from ATTACK01.

The most significant finding during testing was that security controls must be applied to every machine on the network. During the LLMNR poisoning test, Responder captured credential hashes from CL01 because NetBIOS and mDNS were still active on that machine despite DC01 being fully hardened. This demonstrated that a single unprotected endpoint is sufficient for an attack to succeed.

Module 2 and Module 3 together form a complete security framework. The group policies, domain users and Active Directory infrastructure built in Module 2 were essential to the technical controls implemented here. Administrative and technical security must work together — neither is effective without the other.